How the Financing of Balance-Sheet Policy Shapes Its Effects

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September 6, 2026

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Abstract

I ask whether the financing of Federal Reserve balance-sheet policy shapes its effects on real GDP and the price level. In a monthly U.S. SVAR I identify two balance-sheet shocks, one financed by reserves and one by overnight reverse repurchases (ON RRP). Identification combines sign, zero, and narrative restrictions with shifts in the volatility of the structural errors, and imposing the zero restrictions requires a new sampler that I develop. Both balance-sheet contractions lower the price level, and only the reserve-financed contraction lowers GDP. Over the early quantitative easings, the COVID easing, and the first quantitative tightening, the reserve-financed shock is the most important driver of the unexpected change in GDP among the labeled shocks. Over the ON RRP drain of the second quantitative tightening, the ON RRP-financed shock is the most important driver of the unexpected price-level change.

Keywords: quantitative tightening, quantitative easing, central bank liabilities, bank funding, zero restrictions, shock heteroskedasticity

JEL codes: C11, C32, E51, E52, E58, G21

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1 Introduction

A balance-sheet operation of the Federal Reserve moves the securities of the System Open Market Account (SOMA) on the asset side. An equal amount of overnight claims on the Federal Reserve moves on the liability side, and two claims carry that amount. Reserves are a deposit at the Federal Reserve that banks hold. Overnight reverse repurchases (ON RRP) are an overnight loan to the Federal Reserve that money market funds (MMFs) hold. A purchase settles in reserves, and the balance stays in reserves or moves into ON RRP as the holders of the cash decide (Figure 1). The two claims move different balance sheets. Reserve financing moves the bank deposits of the public, and banks lend to firms and households. ON RRP financing moves the Treasury bills that MMFs hold and the cash they lend in the repurchase agreement (repo) market, where hedge funds finance Treasury positions. This paper asks whether the transmission of a balance-sheet operation depends on which of the two claims finances the operation.

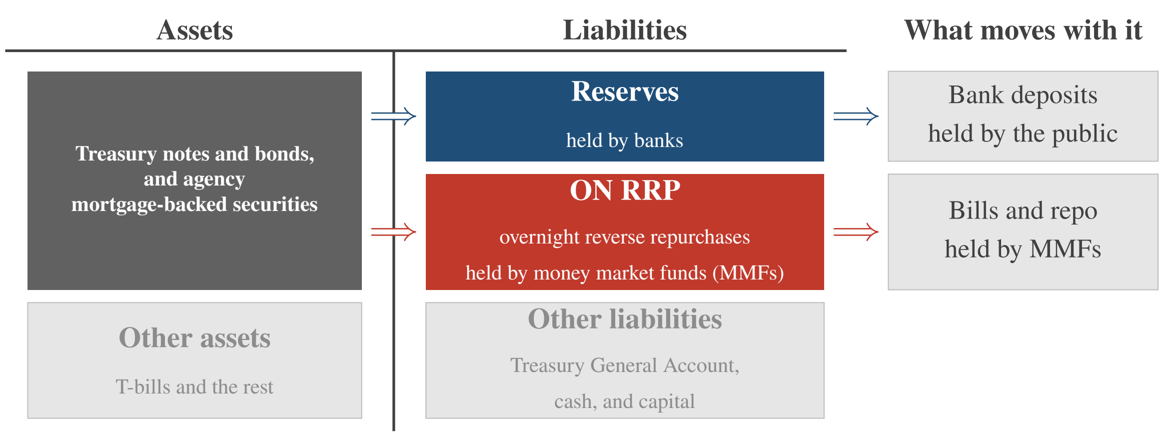


Figure 1: Two ways to finance the same operation.

Note. A stylized Federal Reserve balance sheet. A purchase of securities on the asset side creates reserves, and the balance stays in reserves or moves into ON RRP as the holders of the cash decide. The right column names what moves with each claim. Reserve financing moves the bank deposits of the public. ON RRP financing moves the Treasury bills that money market funds hold and the cash they lend in the repo market to hedge funds, which hold Treasury coupons. Section 2 documents the two routes.

The Federal Reserve financed its two quantitative tightening (QT) episodes with different overnight liabilities. QT1 drained the reserves that the earlier quantitative easing (QE) purchases had filled. QT2 left the reserves that the COVID purchases had filled nearly intact. It drained the ON RRP balances of MMFs instead (Figure 2). The two episodes are the same operation financed by the two claims of Figure 1. The transmission of balance-sheet policy to real GDP and the price level may differ with the claim. The question motivates two balance-sheet shocks, a reserve-financed shock (RFBS) and an ON RRP-financed shock (OFBS).

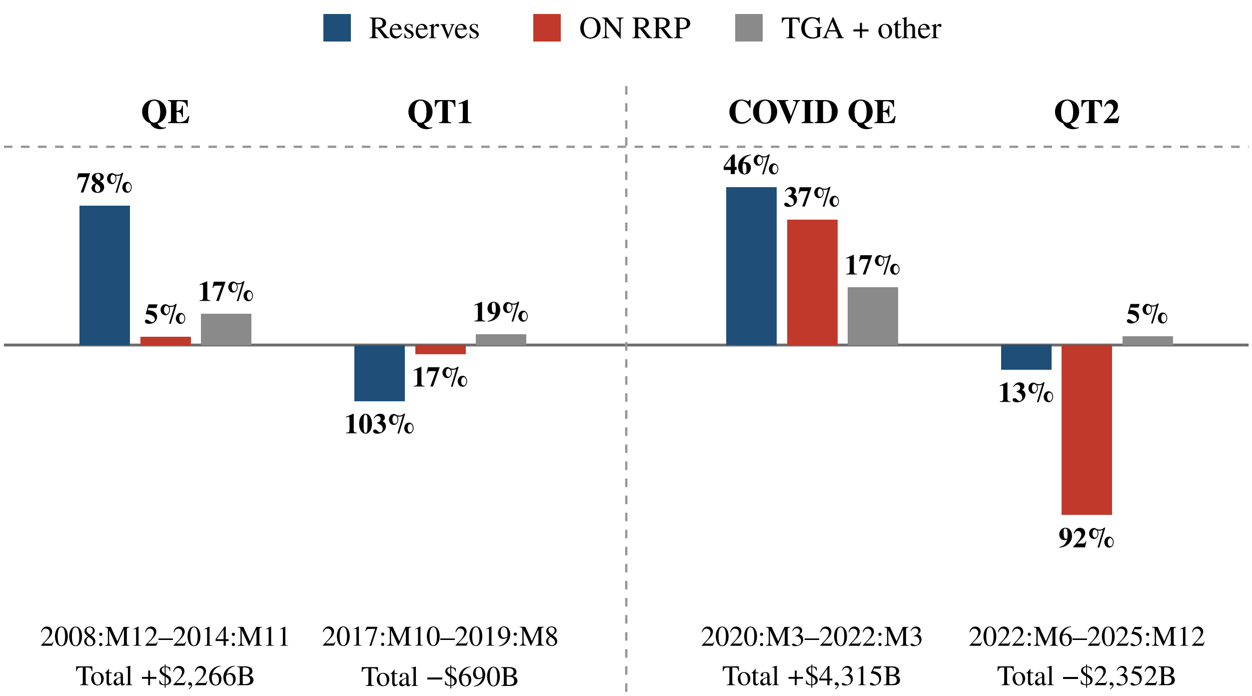


Figure 2: Liability transition across balance-sheet policy.

Note. Bars show the change in each Federal Reserve liability over the episode windows of Table A.1, in USD billions, with QE episodes plotted upward and QT episodes downward. Shares are percent of each episode's total asset change. Gray bars sum the TGA, currency, and residual liabilities and are net positive in every episode. Inside the COVID QE window the ON RRP buildup broadly matched the contemporaneous TGA drawdown, as Section 2 documents. Source: Federal Reserve H.4.1 and NY Fed SOMA holdings.

I identify the two shocks in a seven-variable monthly U.S. SVAR, alongside demand, supply, and monetary policy (MP) shocks. I normalize each shock to a one-percent impact decline in SOMA holdings and display both in the contractionary direction. Both contractions lower the deflator, and only the RFBS contraction lowers GDP.

The forecast-error variance decomposition gives the share of the forecast variance that each shock accounts for at a given horizon. At three years the two balance-sheet shocks together account for 13% of the forecast variance of GDP and 10% of the forecast variance of the deflator. The shares repeat the pattern of the impulse responses. The RFBS share of the GDP forecast variance is 11% against 1.5% for the OFBS shock, and the RFBS share of the deflator forecast variance is 7% against

3%.

The historical decomposition gives the contribution of each shock to the unexpected change in a variable over an episode. The contributions below are posterior medians among the five labeled shocks. Over the 2008 to 2014 easing the RFBS shock is the most important driver of the unexpected changes in both variables. It is also the most important driver of the unexpected change in GDP over QT1 and over the first year of the COVID expansion, before the ON RRP buildup. Over the buildup and the QT2 drain the OFBS shock is the most important driver of the unexpected change in the deflator.

These estimates require two things. The first is a model that fits a heavy-tailed sample. Every QE and QT episode falls after the onset of the Global Financial Crisis, and the period supplies almost two decades of monthly U.S. data. The two decades contain the financial crisis, the COVID collapse, and the post-2021 inflation surge, and the macroeconomic data of the period are heavy-tailed. The emergency rate cuts of March 2020 and the emergency purchases of April 2020 reach 5.7 and 8.6 standard deviations relative to their regime volatilities (Section 3.2). As Brunnermeier et al. (2021) show, fitting a homoskedastic model pushes the shock-scale variation into the uncertainty of the impulse responses. The credible bands become wide and uninformative.

The second is the restriction that identifies the responses of GDP and the price level. Sign and narrative restrictions fix the direction of the balance-sheet movements and leave those responses unrestricted. A timing assumption in the tradition of Leeper et al. (1996) and Christiano et al. (2005) adds identifying content. In that tradition real activity and prices do not respond to a policy shock within the month. The balance sheet changes gradually after a policy announcement (Nakashima et al., 2024), and its real effects plausibly arrive after the quantities accumulate.

Both requirements have to hold in one sampler that imposes zero restrictions on a heteroskedastic model, and none does. This paper develops the heteroskedastic absorb-and-rotate sampler with zero restrictions (HARS-Z). HARS-Z extends the heteroskedastic absorb-and-rotate sampler (HARS) of Kim and Zha (2026), which combines the heteroskedastic likelihood of Brunnermeier et al. (2021) with sign restrictions and with the narrative restrictions of Antolín-Díaz and Rubio-Ramírez

(2018).[1] A zero restriction confines the rotation to a manifold of Haar measure zero, where every step of HARS fails with probability one. I replace the QR map of HARS with a map γ_z built on the null-space construction of Arias et al. (2018), which holds the zeros at every draw. I characterize the measure the construction induces on that set and reweight the rotations to recover the uniform measure. The column-wise elliptical slice sampling update of Kim and Zha (2026) (ESS; Murray et al., 2010) stays valid there, and I prove it.

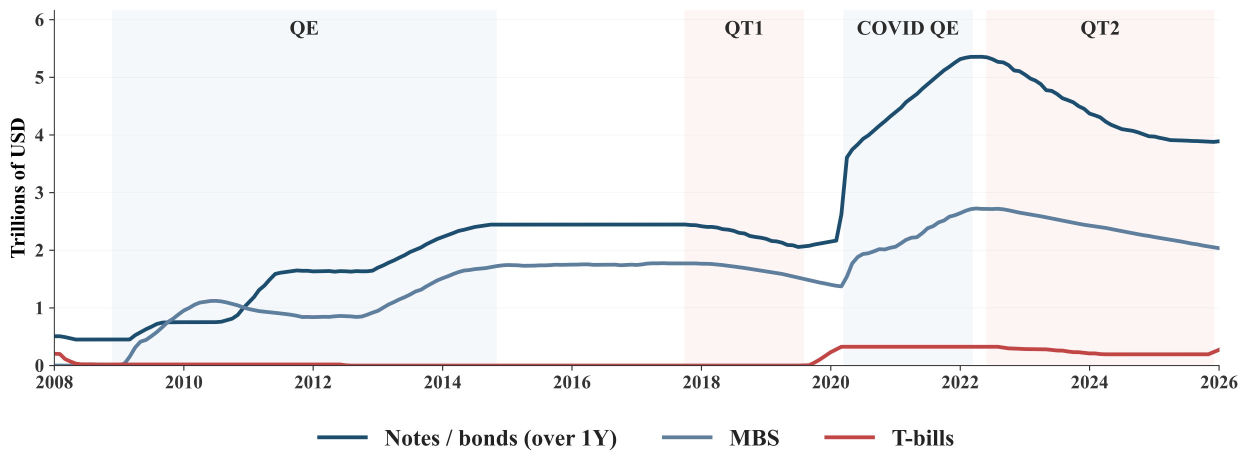
To my knowledge, no existing paper estimates the real effects of balance-sheet policy separately by the liability that finances it. Most of the empirical literature measures the financial-market effects of balance-sheet policy in event studies around announcements, for QE (Krishnamurthy and Vissing-Jorgensen, 2012; Swanson, 2021) and for QT (Smith and Valcarcel, 2023; Lloyd and Ostry, 2024; Lu and Valcarcel, 2024). Theoretical work models the transmission of balance-sheet policy (Chen et al., 2012; d'Avernas et al., 2024; Kumhof and Salgado-Moreno, 2024; Adrian et al., 2025). In these models reserves, or the short-term debt of a consolidated government, finance the purchases. Nakashima et al. (2024) estimate the real effects of the Bank of Japan's balance-sheet policy with a different identification. Their shocks separate the size of the balance sheet from its asset composition, not the liability side.

The rest of the paper proceeds as follows. Section 2 documents the balance-sheet evidence. Section 3 presents the empirical model, the data, and the identification. Section 4 develops HARS-Z and establishes its validity. Section 5 reports the impulse responses, their robustness, and the decompositions. Section 6 concludes. The appendices contain the balance-sheet data, the estimation diagnostics, the formal statement of the narrative restrictions, the review of HARS and the theory of HARS-Z, the proofs, and the robustness exercises.

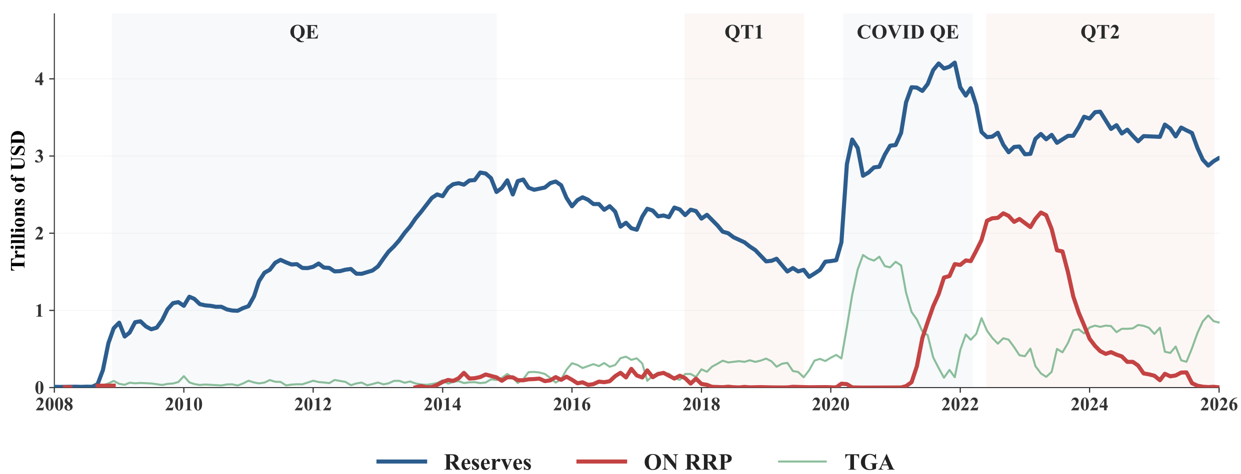
[1] Carriero et al. (2024) combine the same heteroskedastic device with sign, narrative, and proxy restrictions. Neither sampler imposes zero restrictions.

2 The Liability Side of QE and QT

Quantitative tightening (QT) shrinks the Federal Reserve's balance sheet by letting securities run off at maturity without reinvestment. Figure 3 plots the assets and liabilities. Every QE raises both sides and every QT lowers them together. The balance-sheet identity fixes the totals and leaves open which liabilities absorb the change.



(a) Federal Reserve asset holdings.



(b) Federal Reserve liabilities.

Figure 3: The Federal Reserve's balance sheet.

Note. Panel (a) plots the Treasury and MBS holdings and panel (b) plots the reserve, ON RRP, TGA, and currency balances, in USD billions. The episode windows follow Table A.1. Source: Federal Reserve H.4.1 and NY Fed SOMA holdings.

The asset side. The two expansions differ only in size. Both expansions raised Treasury and MBS holdings together, and both contractions lowered them together (Figure 3a). QT2 ran longer and larger than QT1, with coupon runoff of \$1.47 trillion against \$0.38 trillion. The larger contraction followed the larger expansion. The COVID QE raised total assets by \$4.3 trillion, roughly twice the 2008 to 2014 expansion (Appendix A).

The liability side. The episodes diverge in which liability they moved. Each share is percent of the episode's total asset change. Reserves absorbed 78% of the 2008 to 2014 expansion. QT1 reversed less than a third of that expansion and drained essentially the entire contraction from reserves. The COVID QE split between reserves (46%) and ON RRP (37%), with the remainder in the Treasury General Account (TGA) and currency. QT2 drained ON RRP, 92% of the contraction, and left reserves nearly untouched at 13%. Reserves and ON RRP together fell by more than the contraction, and the Treasury General Account and currency rose to make up the difference (Appendix A).

The ON RRP buildup. The ON RRP balance was near zero when QT1 began and near \$2 trillion when QT2 began. The COVID purchases settled in reserves, and reserves kept rising into late 2021 as the buying continued (Figure 3b). The balance grew after March 2021, when the Treasury cut bill issuance and drew down the TGA and the released cash flowed through money market funds (MMFs) into ON RRP (Wong, 2021). The expiry of the temporary Supplementary Leverage Ratio (SLR) relief in March 2021 added to it.[2]

The QT2 drain and the ON RRP facility. QT2 drained the ON RRP facility. QT1 ran when the facility held almost nothing, and it drained reserves instead. The Committee's 2022 principles state that the runoff ends when reserves reach the level it judges consistent with an ample reserves regime (Federal Open Market Committee, 2022b). The drain ran from mid-2023, a year after the runoff

[2]The SLR requires large banks to hold tier 1 capital against total leverage exposure without risk weights. In April 2020 the Federal Reserve temporarily excluded Treasuries and reserve balances from the exposure measure to ease strains in the Treasury market. The exclusion expired as scheduled on March 31, 2021, and reserves and Treasuries again counted toward bank leverage. Banks discouraged large deposits, the cash moved to MMFs, and the funds found few bills to buy (Afonso et al., 2025). Wong (2021) attributes the largest share of the ON RRP buildup to the fall in bill supply during the TGA drawdown, with the SLR expiry and the June 2021 rate adjustment adding to it.

began. Heavy Treasury bill issuance pushed T-bill yields above the ON RRP rate, and MMFs pulled their cash out of the facility and bought bills. The shift lowered ON RRP balances and left reserves unchanged.

Where the ON RRP cash went. A reader may take the ON RRP drain as MMFs exchanging repo with the Federal Reserve for bills, with no duration leaving the Federal Reserve. The drain did more than that. Over QT2 the Federal Reserve let $1.5 trillion of Treasury coupons run off (Table A.1). Private investors took on that duration under the ON RRP drain, as under the reserve drain of QT1. Hedge fund holdings of U.S. government debt rose by $2.5 trillion from 2022:Q1 to 2025:Q4, against $4.1 trillion of net coupon issuance over QT2, and MMFs supplied the repo cash that financed the purchases. MMFs lent $2.3 trillion more in repo over QT2, 54 percent of the increase through the Fixed Income Clearing Corporation (FICC), where hedge funds borrow. They also held $2.0 trillion more in Treasury securities (Figure 4).[3] What the ON RRP drain lacks is a bank. Neither MMFs nor hedge funds lend to firms and households.

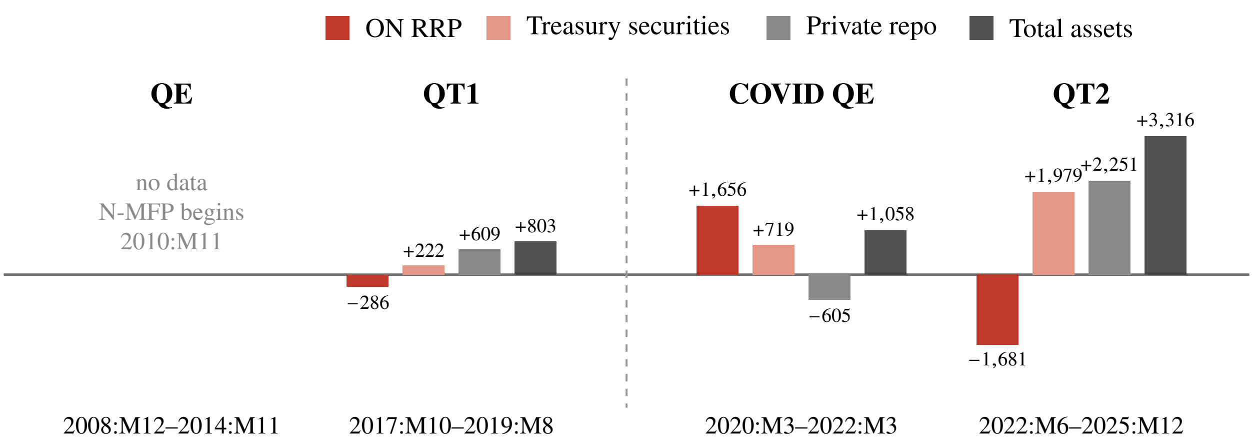


Figure 4: The money market fund balance sheet by episode.

Note. Bars show the change in each item of the money market fund balance sheet over the episode windows of Table A.1, in USD billions, at month-end. ON RRP is the funds' repo with the Federal Reserve, private repo is repo with every other counterparty, and total assets sums every holding. Form N-MFP begins in 2010:M11, after the QE window opens. Table A.3 in Appendix A reports the same changes with the hedge fund positions and Treasury issuance. Source: SEC Form N-MFP through the OFR Money Market Fund Monitor.

[3] The figure and the tables of Appendix A are sector aggregates and show neither whom MMFs bought from nor the path of any dollar. MMFs may hold Treasury coupons with remaining maturity under thirteen months, and the data do not split Treasury holdings by class.

I read the drain this way. Duration leaves the Federal Reserve under both financings, and the term-spread restriction of Section 3 applies to both balance-sheet shocks. What differs is whether the claim passes through a bank balance sheet on the way to private investors. Reserve financing moves the deposit side of bank balance sheets, and Table A.2 and Figure A.1 in Appendix A report the four episodes. The asymmetry in Section 5, where only the reserve-financed contraction lowers GDP, is the difference a bank makes.

Members of the Federal Open Market Committee watched the ON RRP facility as they set the pace of the runoff. Waller (2024) read the funds remaining in the facility as excess liquidity and as room to reduce securities holdings further. Logan (2024) traced the decline to Treasury issuance and argued for slowing the runoff as the facility approaches a low level, when further runoff would fall on reserves. The Committee lowered the monthly Treasury redemption cap from $60 billion to $25 billion in June 2024. It gave the risk of a sharp fall in reserves as the reason (Federal Open Market Committee, 2024).

The gap in existing work. Existing work prices a runoff without separating the liability that finances it. Crawley et al. (2022) in an FRB/US simulation and Wei (2022) in a preferred-habitat model convert a runoff into an equivalent policy-rate increase. Both price it through the duration risk that private investors absorb. Wei (2022) traces which liability a runoff drains, reserves when banks buy the reissued coupon securities and ON RRP when MMFs buy bills. He does not estimate the two cases. In Kumhof and Salgado-Moreno (2024) and d'Avernas et al. (2024) reserves finance balance-sheet policy and banks transmit it, and neither model includes an ON RRP facility.

The terms and the realized split. The Federal Reserve set the terms that govern the split between reserves and ON RRP. It set the operating framework, the ON RRP award rate, the per-counterparty caps, and the runoff caps (Federal Open Market Committee, 2022b,a). Within those terms the composition of Treasury issuance, bank regulation, and the portfolio choices of MMFs affected the realized split (Afonso et al., 2025), and the realized split moved within the sample.[4] If the transmission differs with the financing, the caps and the issuance composition affect what a runoff of a given size does to GDP and the price level. I take the financing as given and identify two balance-sheet shocks, one financed by reserves and one financed by ON RRP.

3 Empirical Model

Let y_t be the $n \times 1$ vector of endogenous variables and x_t the $r \times 1$ vector of predetermined terms, including the constant and p lags of y_t. The structural equation is

$$y_t' A_0 = x_t' A_+ + \eta_t', \tag{1}$$

where A_0 is the $n \times n$ nonsingular structural impact matrix and A_+ is the $r \times n$ matrix stacking the constant and lag coefficients. The vector η_t collects the model shocks, independent across equations and time conditional on the regime.[5]

Following the deterministic variance-regime specification of Brunnermeier et al. (2021), let

$$s_t : \{1, \ldots, T\} \to \{1, \ldots, h\}$$

be a known date-to-regime map that assigns each date to one of h volatility regimes. The shocks are heteroskedastic across regimes and heavy-tailed within them,

$$\eta_{i,t} \mid \nu_{i,t},\ s_t = k \ \sim\ \mathcal{N}\big(0,\ \lambda_{i,k}\, \nu_{i,t}\big), \qquad i = 1, \ldots, n, \tag{2}$$

where $\lambda_{i,k} > 0$ is the relative variance of shock i in regime k and $\nu_{i,t}$ is a latent scale with an

[4] The Federal Reserve raised the per-counterparty caps from \$30 billion to \$160 billion over 2021, and the higher caps accommodated the ON RRP buildup. From mid-2023 the shift of Treasury issuance toward bills drained the QT2 contraction from ON RRP rather than from reserves. Over the first three years of the runoff, securities holdings fell by \$1.3 trillion while bank reserves rose by \$350 billion (Logan, 2024).

[5] Under this right-multiplication convention, which (1) shares with Kim and Zha (2026) and Arias et al. (2018), columns of A_0 correspond to equations.

inverse-gamma distribution, as in Brunnermeier et al. (2021). The inverse-gamma parameters make each standardized shock a Student-t with unit scale, and the variance of shock i in regime k is proportional to $\lambda_{i,k}$. I normalize the regime-specific variances to satisfy

$$\frac{1}{h}\sum_{k=1}^{h}\lambda_{i,k} = 1, \qquad i = 1,\ldots,n,$$

which fixes the scale of each shock variance path across regimes.

Post-multiplying (1) by A_0^{-1} gives the reduced form

$$y_t' = x_t'B + u_t', \qquad B = A_+A_0^{-1}, \qquad u_t' = \eta_t'A_0^{-1}, \tag{3}$$

where the regime-specific reduced-form covariances

$$\Sigma_k \;=\; \mathrm{Var}(u_t \mid s_t = k) \;=\; (A_0^{-1})'\,\Lambda_k\,A_0^{-1}, \qquad \Lambda_k := \mathrm{diag}(\lambda_{1,k},\ldots,\lambda_{n,k}), \tag{4}$$

identify the columns of A_0 up to signed permutation whenever the variance ratios across regimes are not mutually proportional (Rigobon, 2003; Lanne et al., 2010). The covariances identify (A_0, Λ) but do not label the shocks. In this paper, the economic labeling comes from the restricted rotation, and the two roles stay separate throughout.

3.1 Data and Priors

I estimate the model of Section 4 on a monthly U.S. system with seven variables ($n = 7$) and $p = 6$ lags. The sample is November 2008 through March 2026.[6]

The system contains seven variables. The first four are the federal funds rate (FFR), the log of SOMA holdings, log(Reserves + ON RRP), and the ON RRP ratio. The ratio is (ON RRP + c)/(Reserves + ON RRP + c) with c = \$250 billion.[7] SOMA holdings cover Treasury notes and

[6]The November 2008 start coincides with the financial-crisis interventions that preceded QE1. It drops the pre-crisis regime, where reserves stayed near zero and reserve variance was orders of magnitude smaller.

[7]The constant keeps the ratio from moving on the small ON RRP balances of the years before 2021. The composition

bonds plus agency MBS and exclude bills.[8] The last three are the log of real GDP, the log of the GDP deflator (GDPDEF), and the term spread (the 10-year Treasury yield less the 3-month bill rate). I map quarterly GDP and the deflator to monthly frequency by Chow–Lin interpolation and aggregate every series to monthly means.

The design uses two liability variables. The size variable log(Reserves + ON RRP) sums the two liabilities that absorbed the QE and QT episodes of Section 2. It treats reserves and ON RRP as close substitutes, two forms of overnight central-bank money that commercial banks and money market funds hold, respectively (Diamond et al., 2024; Copeland et al., 2021). The ON RRP ratio (ON RRP + c)/(Reserves + ON RRP + c) measures the sector split of that total. The ratio distinguishes the two contractions. A reserve drain and an ON RRP drain reduce the size variable by the same amount, but they move the ratio in opposite directions. The tables abbreviate the two variables as log(R + RRP) and ON RRP/(R + RRP).[9]

The constant c keeps the ratio from moving on a small base. With ON RRP balances near zero, the raw ratio ON RRP/(Reserves + ON RRP) does not respond to reserves and tracks ON RRP dollar for dollar. Balances never exceeded \$243 billion before the 2021 buildup. Over QT1 they fell from \$154 billion to \$7 billion while reserves fell by \$770 billion, and the raw ratio fell from 6.1 to 0.4 points. A reserve drain alone would have raised it. A raw-ratio design reads that reserve-financed drain as an ON RRP drain. I set c to the largest monthly balance before 2021, rounded up to \$250 billion. With that constant the ratio moves 0.7 points over QT1 and 36 points over the QT2 drain, and the pre-2021 balances read as noise.[10]

The priors follow Sims and Zha (1998) and Brunnermeier et al. (2021). Each element of A_0 has an independent Gaussian prior with mean 100 on the diagonal, 0 off the diagonal, and standard

paragraph below states its role and its value.

[8]The Federal Reserve treats bills separately from the programs. It purchased bills from October 2019 into 2020 to keep reserves ample. It described those purchases as technical measures that do not represent a change in the stance of monetary policy (Federal Open Market Committee, 2019). Under the 2022 runoff the decline in Treasury holdings covers coupon securities, and bills fall only to the extent that coupon maturities run below the monthly cap (Federal Open Market Committee, 2022a). Bill holdings track the maturity schedule and the reserve-management purchases rather than the purchase and runoff decisions.

[9]The ratio lies between $c/(\text{Reserves} + c)$ and one and equals $c/(\text{Reserves} + c)$ before the ON RRP facility became active. It enters in levels, with no log or inverse-hyperbolic transform.

[10]Section 5.2 reports the estimates for $c = \$100$ billion and $c = \$500$ billion.

deviation 200. The centering at $A_0 = 100I$ scales each equation to an expected prior standard deviation of 0.01, and the standard deviation of 200 keeps the prior diffuse. A Dirichlet($\alpha = 2$) prior on the normalized variance shares centers the prior on equal variances across regimes. It enforces the normalization that the relative variances of each shock average to one.

Sims–Zha dummy observations center each equation on an independent random walk, except the ON RRP ratio and the term spread, which I center on white noise.[11] Following Brunnermeier et al. (2021), the latent scales are i.i.d. inverse-gamma and make each standardized shock Student-t with 4.99 degrees of freedom. Section 3.2 reports the residual diagnostics that motivate the regime structure and the Student-t errors, including the two-step fit of the degrees of freedom.

3.2 Volatility Regimes and Heavy-Tailed Shocks

For the heteroskedasticity identification I use $h = 5$ volatility regimes aligned with Federal Reserve policy phases. Table 1 reports the dates. The breaks fall at January 2011, July 2015, April 2020, and March 2022. The first two breaks follow Brunnermeier et al. (2021). I add the last two, and the COVID QE and QT2 periods each form their own regime.

Table 1: Volatility regimes.

Regime	Start	End	Description
1	2008:M11	2010:M12	QE1 and the crisis response.
2	2011:M1	2015:M6	QE2 through the taper.
3	2015:M7	2020:M3	Liftoff, QT1, and the 2019 repo episode.
4	2020:M4	2022:M2	COVID-19 QE.
5	2022:M3	2026:M3	QT2.

Note: End dates are inclusive. The first two breaks (2011:M1 and 2015:M7) follow Brunnermeier et al. (2021). I add the last two (2020:M4 and 2022:M3), and the COVID QE and QT2 periods each form their own volatility regime.

The sample spans the financial crisis, COVID, and the post-2021 tightening. Figure 5 plots the reduced-form residuals u_t in (3) from estimating (1) with a single variance regime and Gaussian

[11] The hyperparameters are overall tightness 3, lag decay 0.5, sum-of-coefficients (μ) weight 1, and co-persistence (λ) weight 5. The dummy scales use each variable's univariate AR(p) residual standard deviation, and the prior precision stays invariant to units.

errors, without the identification restrictions, under the prior of Section 3.1. Each panel shows the posterior median residual of one equation in its own units. The five subsamples follow the dates of Table 1 and partition the residual sample for description only. The number above each subsample is its residual standard deviation. The standard deviations vary across subsamples, and the peak subsample differs across equations. Non-proportional patterns of this kind motivate the regime variances and are the condition under which (4) identifies the columns of A_0.[12]

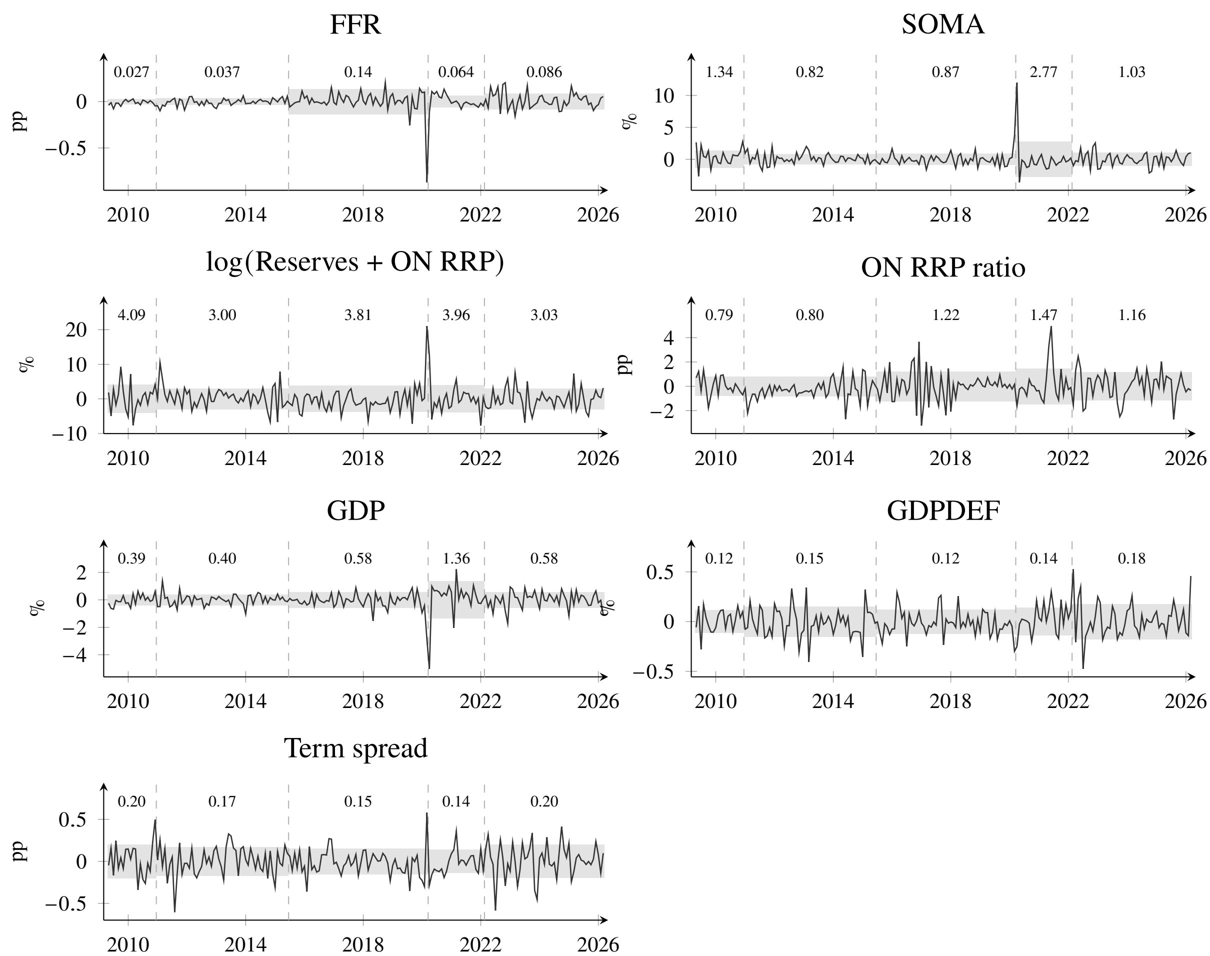


Figure 5: Reduced-form residuals under a single variance regime.

Note. Each panel plots the posterior median of the reduced-form residual of one equation from the estimation of (1) with a single variance regime, Gaussian errors, no identification restrictions, and the prior of Section 3.1. Units are percent for the log levels (SOMA, log(Reserves + ON RRP), GDP, GDPDEF) and percentage points for the rates and the ratio (FFR, ON RRP ratio, term spread). Dashed lines mark the regime breaks of Table 1. The shaded band in each subsample is plus and minus one subsample standard deviation of the series, and the number above each subsample is that standard deviation in the units of the panel. The residual sample runs from 2009:M5 to 2026:M3.

[12]At the posterior mode of the full model and across the retained draws, the estimated variance ratios are far from mutually proportional (Table E.1 in Appendix E).

Table 2: Regime-standardized residuals under five variance regimes.

	Excess kurtosis	max $|z|$
FFR	7.4	5.7
SOMA	20.7	8.6
log(Reserves + ON RRP)	1.0	3.7
ON RRP ratio	1.6	4.1
GDP	1.5	3.6
GDPDEF	1.2	3.8
Term spread	0.6	3.5
Pooled	5.1	8.6

Note: The table estimates equation (1) with the five variance regimes of Table 1, Gaussian errors, no identification restrictions, and the prior of Section 3.1. It reports the excess kurtosis and the largest absolute value of the structural residuals standardized by $\sqrt{\lambda_{i,s_t}}$. Entries are posterior medians. The pooled row stacks all equations over the full sample.

The heteroskedastic model with five variance regimes absorbs the shifts in volatility but not the heavy-tailed shocks in the data. Table 2 repeats the estimation with the five regimes in place of the single regime. It standardizes each structural residual by the square root of its estimated regime variance λ_{i,s_t}. Under Gaussian errors, with no heavy-tailed shocks, these standardized residuals would show excess kurtosis near zero.[13]

Pooled across the seven equations and the full sample, the excess kurtosis is 5.1. The SOMA residual reaches 8.6 standard deviations in April 2020. That month contains the emergency purchases at the onset of COVID, the largest balance-sheet action in the sample. The FFR residual reaches 5.7 standard deviations in March 2020, the month of the emergency cuts to the zero lower bound. Without the latent scales, these residuals would contaminate the estimated variances of their regimes, and the contamination would weaken the heteroskedasticity identification.

I fit the Student-t degrees of freedom to the standardized residuals by the two-step procedure of Brunnermeier et al. (2021), and the fit gives 4.68. The estimation uses 4.99, the value fitted before

[13] Kurtosis is the expectation of the fourth power of a standardized random variable and equals three for a Gaussian. Excess kurtosis subtracts this benchmark. Positive values indicate that extreme observations occur more often than under a Gaussian. A Student-t with $\delta > 4$ degrees of freedom has excess kurtosis $6/(\delta - 4)$.

the constant entered the ON RRP ratio, and the likelihood at 4.99 lies 0.13 log points below the fitted value.[14] The fitted distribution matches the residuals on the frequency of extreme residuals. In the 1,421 observations a t with 4.68 degrees of freedom predicts 1.0 residual beyond six standard deviations, and the sample contains one, the April 2020 SOMA residual.[15] A t with 4.68 degrees of freedom has excess kurtosis of 8.8, and the pooled residuals show 5.1.

3.3 Identification

I identify the shocks with three sets of restrictions. Sign restrictions label five of the seven shocks. Zero restrictions hold at impact. The zeros bind under the average transmission, the responses evaluated at the average of the relative variances across regimes, and need not hold in the regime-specific responses (Appendix C.1). Narrative restrictions tie RFBS and OFBS to the episodes that generated them. All sign restrictions bind from impact through horizon five and must hold both under the average transmission and in every regime. I describe the two balance-sheet shocks first and the three remaining labeled shocks second. Table 3 collects the restrictions on the balance-sheet shocks, and Table 4 collects the restrictions on the demand, supply, and monetary policy (MP) shocks.

The balance-sheet shocks. Table 3 collects the restrictions on RFBS and OFBS. The two shocks correspond to the two financing channels. I normalize the positive direction of both shocks as an expansion. A positive RFBS shock raises SOMA, raises log(Reserves + ON RRP), and compresses the term spread, and the expansion settles in reserves. A positive OFBS shock moves the same three variables in the same direction, and the expansion settles in ON RRP. The ON RRP ratio separates the two columns. A reserve-financed balance-sheet expansion lowers the ratio (reserves raise the total while ON RRP is unchanged). An ON RRP-financed expansion raises the ratio directly. Without the ratio restriction the two columns would satisfy identical restrictions. The restrictions

[14] Brunnermeier et al. (2021) obtain 5.7 on their sample. On this sample the likelihood at 5.7 lies 2.0 log points below the fitted value, and the Gaussian likelihood lies 89 log points below.

[15] At five standard deviations the expected count is 2.2, and the sample contains two, the March 2020 FFR residual and the April 2020 SOMA residual.

Table 3: Restrictions on the two balance-sheet shocks.

Response variable	RFBS	OFBS
Panel A: sign and zero restrictions		
FFR		
SOMA	+	+
log(Reserves + ON RRP)	+	+
ON RRP/(Reserves + ON RRP)	−	+
GDP	Z	Z
GDPDEF	Z	Z
Term spread	−	−
Panel B: narrative restrictions (window W, regime average)		
Window W	2020:M4–2021:M2	2023:M9–2024:M7
Mean shock realization, eq. (B.1)	+	−
|HD| on log(R + RRP), eq. (B.2)	dominant	dominant

Note: A + or − is a sign restriction on the response of the row variable to the column shock, binding from impact through horizon five $[0, 5]$. Z is a zero at impact. Blank cells are unrestricted. Panel B ties each balance-sheet shock to its window. Over the window the mean realization of RFBS must be positive and the mean realization of OFBS must be negative, and the shock's historical-decomposition contribution to log(Reserves + ON RRP) must exceed the sum of the contributions of all other labeled shocks.

carry no episode label. QE and QT1 enter as positive and negative realizations of RFBS, and the QT2 drain enters as negative OFBS realizations.

One economic assumption generates the zero restrictions in Panel A of Table 3.

Assumption 1 (Balance-sheet shocks do not reach real activity and prices within the month)**.** *The two balance-sheet shocks have no impact effect on real GDP or the GDP deflator.*

Assumption 1 follows the timing of Leeper et al. (1996), whose sluggish private sector responds to financial variables only with a delay. Production and final-goods pricing involve planning processes, and movements in interest rates and in the balance sheet reach output and prices smoothly over time rather than within the month. Christiano et al. (2005) impose the same timing on the MP shock. Assumption 1 applies it to the two balance-sheet shocks instead, and the MP shock stays sign-restricted at impact. The balance sheet changes gradually after each program announcement (Nakashima et al., 2024), and its real effects plausibly arrive after the quantities accumulate. The timing argument binds more tightly for the quantity channel than for the rate channel.

Beyond impact, the GDP and deflator responses to both balance-sheet shocks stay unrestricted. The zeros bind on the average transmission, the reported object, and the regime-specific impact responses stay unrestricted (Section 4). The macroeconomic transmission is data-determined and can differ across the two channels.

I impose two narrative restrictions, one for each balance-sheet shock, and Panel B of Table 3 summarizes them.[16] Each adapts the single-date narrative sign restrictions of Antolín-Díaz and Rubio-Ramírez (2018) to a window average and ties the shock to the episode that generated it through two requirements. The mean realization of the shock over the window must carry the sign of its episode. The shock's contribution to the unexpected movement of log(Reserves + ON RRP) over the window must dominate the contributions of the other labeled shocks.[17] The baseline measures the contribution by the multi-horizon interval historical decomposition of Antolín-Díaz and Rubio-Ramírez (2018). Appendix B states the two requirements formally.

I tie the RFBS shock to the first phase of the COVID purchases, April 2020 through February 2021, when the expansion settled predominantly in bank reserves. The window carries a positive sign. From March 2021 the inflows settled increasingly in money market funds and the ON RRP facility (Section 2). Including those months would risk misallocating the TGA-driven movement in log(Reserves + ON RRP) to the RFBS shock.

I tie the OFBS shock to the sharp ON RRP drain of September 2023 through July 2024. The window contains eleven months, the same length as the RFBS window. A positive OFBS shock is an expansion, the QT2 drain corresponds to negative OFBS realizations, and the window carries a negative sign. Over that window the ON RRP facility ran down steadily while reserves stayed roughly flat. This window starts more than a year after QT2 began in mid-2022, as Section 2

[16] I also estimated the model with four narrative restrictions. The first addition anchors a negative mean RFBS realization over the QT1 window. The second anchors a positive mean OFBS realization over the ON RRP buildup, April 2021 through February 2022. The estimates stay almost identical, and the two restrictions below are enough in this sample.

[17] I impose the dominance on log(Reserves + ON RRP) rather than on SOMA for two reasons. Section 2 defines the two episodes by their effects on the liability side, and log(Reserves + ON RRP) carries no zero restriction. The requirement discriminates the balance-sheet shock against every labeled shock, including the demand, supply, and MP shocks. On SOMA the zero restrictions would leave only the other balance-sheet shock as a competitor.

documents.[18] Tying the restrictions to these phases, rather than to full episodes, keeps each inside the window where its channel was the active margin. Averaging over windows rather than single dates makes the restriction robust to the timing noise of individual months.

The demand, supply, and MP shocks. Table 4 collects the restrictions on the three remaining labeled shocks. Labeling them serves the narrative restrictions. The dominance requirement compares each balance-sheet shock against the other labeled shocks, and the labels put demand, supply, and MP among the competitors it must dominate over its window. Without the labels the comparison would run against the other balance-sheet shock alone. Demand, supply, and a conventional MP shock follow standard conventions. A demand shock raises the FFR, GDP, and the GDP deflator. A supply shock raises GDP and lowers the deflator. An MP shock raises the FFR and lowers GDP and the deflator. One economic assumption generates the zero restrictions on SOMA.

Table 4: Restrictions on the demand, supply, and MP shocks.

Response variable	Demand	Supply	MP
FFR	+		+
SOMA	Z	Z	Z
log(Reserves + ON RRP)			
ON RRP/(Reserves + ON RRP)			
GDP	+	+	−
GDPDEF	+	−	−
Term spread			

Note: A + or − is a sign restriction on the response of the row variable to the column shock, binding from impact through horizon five $[0, 5]$. Z is a zero at impact. Blank cells are unrestricted. The regime covariances identify (A_0, Λ) and do not label the shocks, and the two unlabeled shocks carry no restriction.

[18] Part of the ON RRP decline over the window coincides with the post-debt-ceiling rebuild of the Treasury General Account (TGA). The Treasury refilled the TGA through heavy bill issuance from June through October 2023, and the rebuild drew down ON RRP rather than reserves. This raises the question of whether to attribute the drain to QT2 or to the fiscal rebuild. The two are far apart in magnitude. From September 2023 onward the monthly decline in ON RRP is roughly three times the contemporaneous rise in the TGA. The rebuild does not account for the bulk of the drain, and among the five labeled shocks OFBS is the dominant driver of log(Reserves + ON RRP) over the window. I read the restriction as identifying the dominant balance-sheet shock, while noting that the bill issuance associated with the rebuild plausibly amplified the same channel. Section 5.2 reports specifications that control the TGA and Treasury issuance directly.

Assumption 2 (Balance-sheet policy drives SOMA within the month)**.** *Within the month the Federal Reserve moves its SOMA holdings only through its balance-sheet policy. The demand, supply, and MP shocks have no impact effect on SOMA.*

Assumption 2 sets three zeros on SOMA, and the grounds differ by shock. For the demand and supply shocks the zero follows the policy-equation exclusions of Leeper et al. (1996). In their system the policy instrument does not respond within the month to output and final-goods prices, the counterparts of GDP and the GDP deflator here. They give two grounds. The data for those series arrive only after the month ends, and a shift in the policy stance requires consultation and consensus building. The preannounced schedules and caps of asset purchases and runoff strengthen both grounds, and the within-month path of the balance sheet does not react to macroeconomic news. The restriction constrains the impact responses of SOMA only, and GDP and the GDP deflator carry zeros under Assumption 1 instead.

For the MP shock the zero reflects the separation of the two instruments. Since October 2008 the Federal Reserve has implemented the policy rate through administered rates on reserves and the ON RRP facility rather than through open market operations. A rate surprise requires no change in asset holdings within the month. An MP shock that moved SOMA contemporaneously would also conflate the two instruments. The sample begins in November 2008 and lies entirely within this operating framework.

Leeper et al. (1996) find that movements in reserves and M1 in their sample arise mostly from policy accommodating shifts in private demand. An identification that treats reserves or M1 as a policy instrument mislabels those demand shifts as policy shocks. They call such identifications unreliable. The critique applies to any quantity that private demand can move. SOMA is not such a quantity. Only the Federal Reserve trades in its own portfolio, and under administered rates the accommodation of demand runs through the liability side rather than through asset holdings.

4 HARS-Z

This section presents the sampler that draws from the posterior of the heteroskedastic model of Section 3 conditional on the sign, zero, and narrative restrictions of Section 3.3. I develop HARS-Z by extending HARS, the sampler of Kim and Zha (2026, hereafter KZ) that combines shock heteroskedasticity with sign and narrative restrictions, to include zero restrictions. I adopt the notation of KZ throughout. Appendices C and D contain the full theoretical analysis. Readers focused on the estimates can note Algorithm 1 and continue to Section 5.

HARS (KZ, Algorithm 1) combines likelihood-preserving posterior draws of the model parameters with rotations drawn from a well-defined probability distribution restricted by the sign and narrative restrictions, adopting a column-wise elliptical slice sampler (ESS; Murray et al., 2010). Appendix C.1 reviews the sampler and the repair steps in detail.

One may be tempted to impose the zeros by adding them to $SR(\theta, Q)$, the indicator that the sign and narrative restrictions hold, and running HARS unchanged. This approach fails, and the failure originates in the measure rather than in the algorithm. Throughout, $\theta = (A_0, \Lambda, A_+)$ collects the structural objects with $\Lambda = \{\Lambda_k\}_{k=1}^{h}$, and $Q \in O(n)$ denotes the rotation.

A zero restriction is not a truncation. A zero restriction on the impulse responses is a linear equality constraint on the j-th column q_j of Q. Following the notation of Arias et al. (2018, hereafter ARRW), the $z_j \times n$ matrix $Z_j(\theta)$ stacks the z_j zero restrictions on shock j as rows. The restrictions on shock j read $Z_j(\theta)\, q_j = 0$, the rows depend on the draw through A_0 and the moving-average coefficients, and the sampler must rebuild them at every iteration. The zeros bind under the average transmission only and need not hold in the regime-specific responses, where $\Lambda_k^{1/2}$ reweights the constrained column. Define the zero manifold

$$\mathcal{Q}(\theta) \;=\; \left\{ Q \in O(n) : Z_j(\theta)\, q_j = 0 \text{ for every restricted } j \right\}. \tag{5}$$

Whenever some $z_j \geq 1$, $\mathcal{Q}(\theta)$ is a lower-dimensional smooth manifold in $O(n)$ and $\mu(\mathcal{Q}(\theta)) = 0$, where μ is Haar measure on $O(n)$, as ARRW note in motivating their Algorithm 2.

Sign and narrative restrictions without the zeros truncate the conditional distribution of Q to a subset of positive Haar measure, and the posterior target in HARS (Appendix C.1) remains well defined. The zeros lower the dimension of the admissible set $R(\theta) = \{Q \in \mathcal{O}(n) : SR(\theta, Q) > 0\}$ instead, and the condition $\mu(R(\theta)) > 0$ fails at every θ with at least one zero restriction. The restricted posterior target is not defined, the Gaussian coordinatization never reaches the zeros, and every step of KZ's Algorithm 1 fails with probability one, including its initialization. Appendix C.2 shows formally why HARS fails under zero restrictions. The sampler must parameterize the zero-restricted set directly rather than reach it by sampling. It must build the rotation on $\mathcal{Q}(\theta)$, and its slice-sampling updates must never leave it. The rest of this section constructs exactly such a parameterization.

HARS-Z replaces the QR map γ with a map $\gamma_z(\cdot\,;\theta)$, built on the null-space construction of ARRW's Algorithm 2, that sends a Gaussian matrix to a rotation on $\mathcal{Q}(\theta)$ by construction. The state of the chain is the pair (θ, X), where X is an $n \times n$ Gaussian matrix, the coordinates in which HARS runs its column-wise ESS update. The sampler recomputes the rotation $Q = \gamma_z(X; \theta)$ whenever it needs one. A structural update moves the manifold $\mathcal{Q}(\theta)$, the recomputed rotation lies on the new manifold, and the zeros hold after every move. Appendix C.3 builds the map γ_z and establishes its properties.

HARS-Z targets the analogue of the posterior target in HARS on the zero-restricted set, with the uniform measure on $\mathcal{Q}(\theta)$ in place of Haar measure on $\mathcal{O}(n)$,

$$\pi_\nu(d\theta, dQ \mid y, SR) \;=\; \pi(\theta \mid y)\, \frac{\mathbf{1}\{Q \in R(\theta)\}\, \mu\big(dQ \mid \mathcal{Q}(\theta)\big)}{\mu\big(R(\theta) \mid \mathcal{Q}(\theta)\big)}\, d\theta, \tag{6}$$

where $\mu(\cdot \mid \mathcal{Q}(\theta))$ denotes the uniform measure on $\mathcal{Q}(\theta)$ and $R(\theta) = \{Q \in \mathcal{Q}(\theta) : SR(\theta, Q) > 0\}$ now denotes the admissible set on the zero-restricted set. Generating the rotation through γ_z induces the null-space construction measure on $\mathcal{Q}(\theta)$, not the uniform measure. Weighting each retained rotation by the volume element $v(Q; \theta)$ of the construction, normalized within each retained draw of θ, recovers the target π_ν. The column-wise ESS update remains valid on $\mathcal{Q}(\theta)$ and leaves the conditional target invariant. Appendix C.4 establishes these results.

The HARS-Z sampler, stated below as Algorithm 1, alternates a posterior draw of θ with a rotation draw of X. The posterior draw is the sampler of Brunnermeier et al. (2021, hereafter BPSS), taken from KZ unchanged. Following KZ's Theorem 1, partition the parameter collection as $z_0 = (A_0, \Lambda)$ and $z_+ = A_+$, and let s_0 collect everything held fixed in the block update. The first block moves by Metropolis–Hastings (MH) on the collapsed density with A_+ integrated out, and the companion block moves by a draw from its Gibbs conditional $p_+(A_+ \mid z_0, s_0)$. The rejection counts generated by the repair steps below are diagnostics of computational efficiency, not density terms (KZ, Proposition 5). Each repair returns a draw from the correct conditional distribution. The counts require no reweighting and enter no posterior computation. Appendices C.3 and C.4 state the lemma, propositions, and corollaries that Algorithm 1 cites.

Algorithm 1. (Heteroskedastic absorb-and-rotate sampler with zero restrictions (HARS-Z).) Let the current admissible state be (θ, X) with $Q = \gamma_z(X; \theta)$ and $SR(\theta, Q) > 0$, and fix the tuning parameters N_{ess}, N_θ, N_Q. For the initialization, construct an admissible rotation by a numerical search at the initial θ and convert it into Gaussian coordinates by Lemma 1(iii). One iteration proceeds as follows. Each step matches its counterpart in Algorithm 1 of KZ.

1. **Posterior draw with admissibility check.** Propose $z_0^{\text{prop}} = (A_0^{\text{prop}}, \Lambda^{\text{prop}})$ by the first-block MH step and set $z_0^\star = z_0^{\text{prop}}$ if accepted and $z_0^\star = z_0$ otherwise. Draw the companion block $z_+^\star \sim p_+(A_+ \mid z_0^\star, s_0)$ together with the latent Student-t scales, and form the completed draw $\theta^\star = (z_0^\star, z_+^\star)$. Rebuild the rotation at the new draw, $Q^\star = \gamma_z(X; \theta^\star)$. The rebuilt rotation satisfies every zero restriction by construction. Check the sign and narrative restrictions. If $SR(\theta^\star, Q^\star) > 0$, set $\theta \leftarrow \theta^\star$ and go to Step 3. Otherwise the rotation is stale. Go to Step 2a.

2a. **Posterior-draw repair by the joint procedure.** Repair on the completed draw, as required by KZ's Theorem 1. For at most N_θ attempts, draw a completed candidate $\theta^{\text{cand}} = (z_0^{\text{cand}}, z_+^{\text{cand}})$ and retain it only if $SR\big(\theta^{\text{cand}}, \gamma_z(X; \theta^{\text{cand}})\big) > 0$. The first-block MH step, anchored at the last admissible state θ (not at the stale draw $\theta^\star$), proposes the candidate block z_0^{cand}, and

$z_+^{\text{cand}} \sim p_+(A_+ \mid z_0^{\text{cand}}, s_0)$. On success, set $\theta \leftarrow \theta^{\text{cand}}$ and go to Step 3. If the N_θ attempts retain no candidate, discard all candidates and go to Step 2b, carrying the draw $\theta^\star$ that triggered the repair. Failed candidates, and the stale draw $\theta^\star$ itself, are auxiliary proposals. The chain never retains them.

2b. **Global rotation repair.** Hold $\theta^\star$ fixed. For $j = 1, \ldots, N_Q$, draw a fresh Gaussian matrix $G^{(j)}$ and propose $Q^{(j)} = \gamma_z(G^{(j)}; \theta^\star)$, the analogue on $\mathcal{Q}(\theta^\star)$ of the independent Haar draws in Step 2b of KZ's Algorithm 1. Let $j^\star$ be the first index with $SR(\theta^\star, Q^{(j^\star)}) > 0$. If such an index exists, set $(\theta, X) \leftarrow (\theta^\star, G^{(j^\star)})$ and go to Step 3. If no index succeeds after N_Q attempts, declare sampler failure and proceed to Step 5.

3. **Rotation draw.** From the admissible state, run N_{ess} column-wise ESS updates of X at fixed θ (Proposition 2), producing Gaussian states $X^{(1)}, \ldots, X^{(N_{\text{ess}})}$ and rotations $Q^{(r)} = \gamma_z(X^{(r)}; \theta)$ with $SR(\theta, Q^{(r)}) > 0$ for every r. Retain all N_{ess} rotations together with θ for posterior impulse-response computation.

4. **Advance the state.** Set the next state to $(\theta, X^{(N_{\text{ess}})})$ and return to Step 1.

5. **Sampler failure.** If a posterior move produces a draw $\theta^\star$ for which neither the joint posterior-draw repair nor the global rotation repair restores sign admissibility, the sampler cannot advance. Increase N_θ or N_Q, improve the repair search, or revisit the restriction set.

Retaining N_{ess} rotations per structural draw divides the cost of the structural update across several rotation draws. The ESS updates evaluate only the admissibility indicator, not the likelihood. Appendix C.4 establishes the validity of Algorithm 1 step by step. Posterior inference weights each retained rotation by its volume element, normalized within each retained draw of θ, and reports weighted medians and bands. Appendix C.5 states the estimator. HARS-Z nests HARS. With an empty zero set, γ_z reduces to the QR map γ, $\mathcal{Q}(\theta) = O(n)$, and the construction measure $\mu_z(\cdot\,; \theta)$ is Haar measure. The volume weight is identically one, the weighted summaries reduce to the unweighted ones, and the sampler reduces to KZ's Algorithm 1.

5 Results

I estimate the baseline posterior with HARS-Z, and a second chain with a different random seed checks convergence. Appendix E reports the sampler settings and the convergence diagnostics.

5.1 Baseline Impulse Responses

Figures 6 and 7 report the impulse responses to the two balance-sheet shocks under the baseline identification of Section 3.3, with the relative variances set to their average of one across regimes. The sign and narrative restrictions of Section 3.3 define each shock in its expansionary direction, and the identification is unchanged. Each figure shows one shock in its contractionary direction, the direction of the QT episodes. To make the two channels directly comparable, I normalize each shock to a one-percent impact decline in SOMA holdings.[19] The decline continues after impact. The median SOMA response reaches 5.6 percent after thirty-six months for the RFBS shock and 4.3 percent after twelve months for the OFBS shock. At the QT2 average level those declines are about \$360 billion and \$280 billion. Figure F.1 in Appendix F reports the responses to the demand, supply, and MP shocks.

Figure 6 plots the responses to the RFBS shock. The shaded regions are the 68% credible bands, and the dashed lines are the 90% bands. The restricted responses hold by construction. Panel A of Table 3 restricts the expansionary direction. Under the contractionary display, SOMA, log(Reserves + ON RRP), the ON RRP ratio, and the term spread move with the signs of Panel A reversed, and the zero restrictions hold GDP and the GDP deflator at zero. The economic content lies in the GDP and GDP-deflator responses beyond impact, where no restriction binds.

A negative RFBS shock lowers GDP persistently. The GDP response turns negative around three months, stays negative through thirty-six months, and the median decline reaches about 0.6 percent at thirty-six months. The deflator response trends downward over the same horizons, and its

[19]The dollar value of the normalization depends on the level of SOMA holdings. Over the QT2 window SOMA holdings averaged \$6,482 billion, and one percent of that level is \$64.8 billion. That amount equals 1.08 months of Treasury redemptions at the \$60 billion monthly cap in place before June 2024. Over the full sample one percent of SOMA ranges from \$4.1 billion in 2008:M11 to \$76.9 billion at the COVID peak.

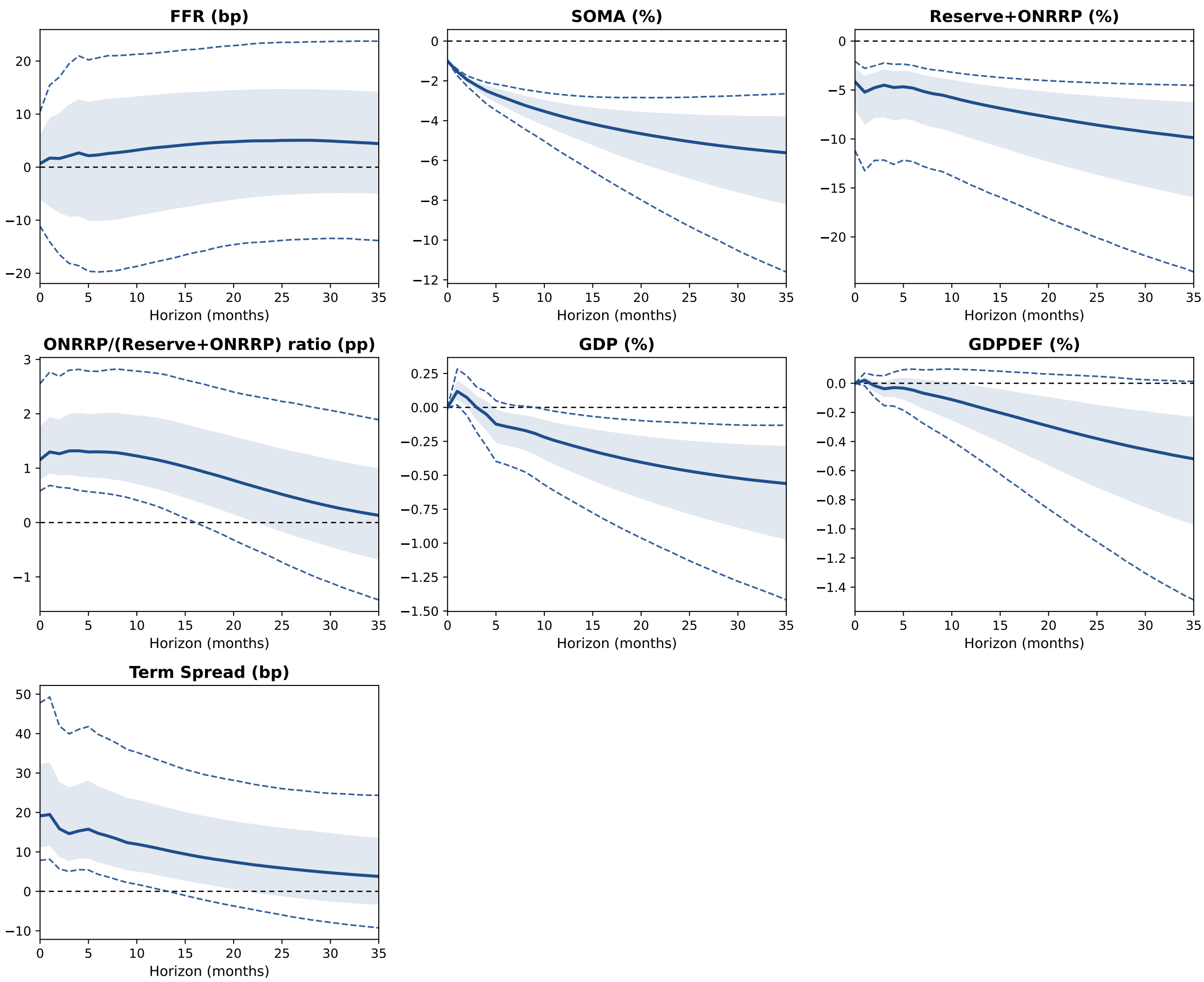


Figure 6: Impulse responses to the RFBS shock.

Note. Impulse responses to the RFBS shock, normalized to a one-percent impact decline in SOMA holdings. I evaluate the responses at the regime average of the relative variances. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. Vertical axes are in basis points for the FFR and the term spread, percent for SOMA, log(Reserves + ON RRP), GDP, and GDPDEF, and percentage points for the ON RRP ratio.

median decline reaches 0.5 percent at thirty-six months. The 68% bands support both responses, excluding zero from five months onward for GDP and from eleven months onward for the deflator. At the 90% level the GDP bands exclude zero from ten months onward. The 90% deflator bands include zero at every horizon, with an upper bound within 0.1 percentage points of zero from twelve months onward.

Figure 7 plots the responses to the OFBS shock. The restricted responses again hold by construction and reverse the signs of the OFBS column in Panel A of Table 3. The economic content

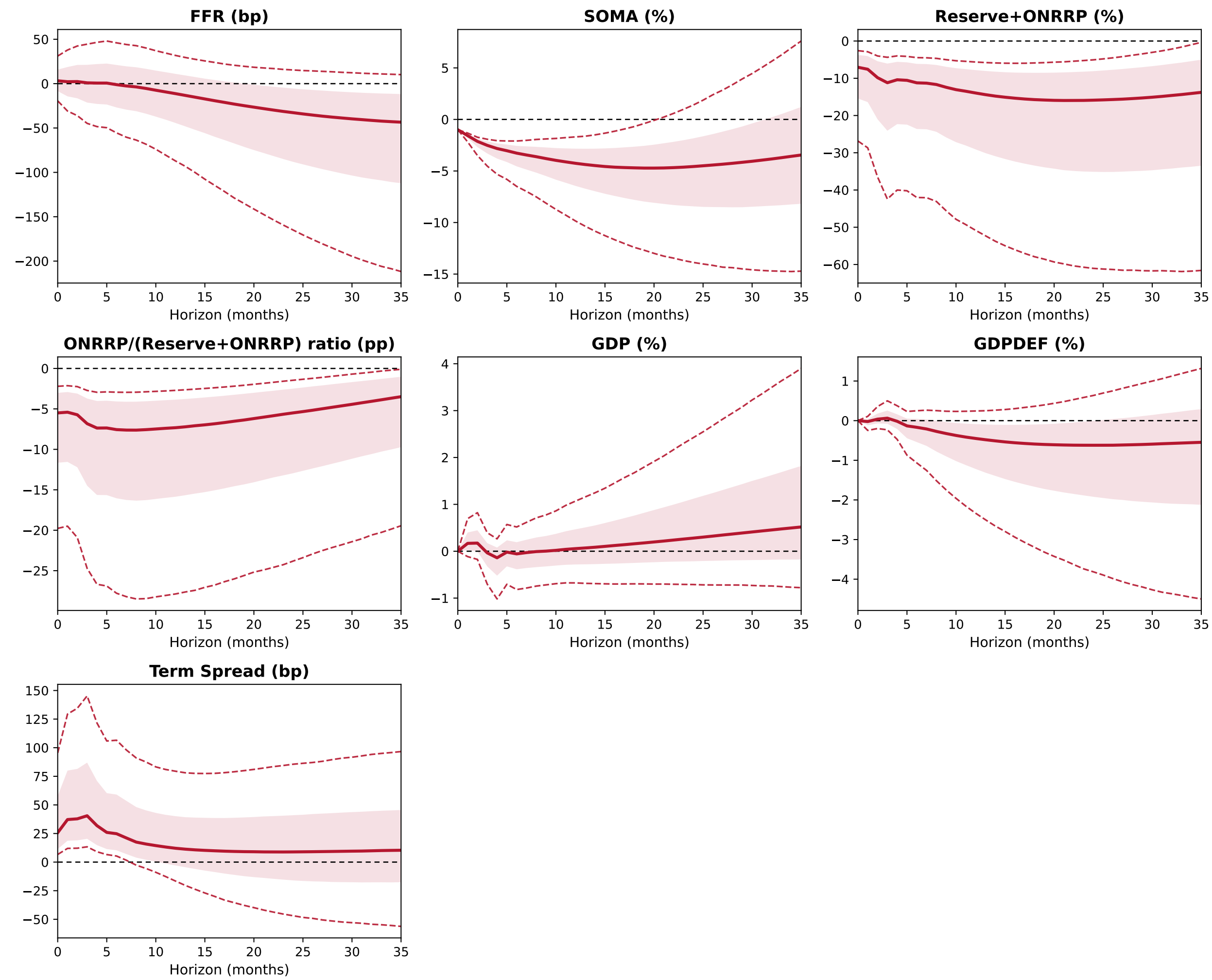


Figure 7: Impulse responses to the OFBS shock.

Note. Impulse responses to the OFBS shock, normalized to a one-percent impact decline in SOMA holdings. The figure shows the shock in the direction of the QT2 drain. I evaluate the responses at the regime average of the relative variances. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. Vertical axes are in basis points for the FFR and the term spread, percent for SOMA, log(Reserves + ON RRP), GDP, and GDPDEF, and percentage points for the ON RRP ratio.

again lies in the GDP and GDP-deflator responses beyond impact, where no restriction binds.

A negative OFBS shock lowers the deflator over the first two years. The deflator response turns negative after three months, and the median decline reaches 0.6 percent at twenty-four months and stays near 0.5 percent through thirty-six months. The GDP response stays close to zero through the first year and drifts upward afterward, to 0.5 percent at thirty-six months. The 68% bands support the deflator decline from eight to twenty-four months and include zero afterward. The GDP bands include zero at every horizon. At the 90% level both bands include zero at every horizon.

Interpreting the two shocks. Both contractions lower the deflator, and only the RFBS contraction lowers GDP. The 68% bands support both statements, through thirty-six months for the RFBS contraction and through twenty-four months for the OFBS contraction. At the 90% level the GDP bands of an RFBS contraction exclude zero. The other bands include zero.

The RFBS responses do not correspond to a single episode. QE and QT1 enter as the positive and negative realizations of the shock (Section 3.3), and realizations of both signs also fall outside those episodes. The linear model has one response function for the shock, and realizations of both signs enter its estimation. The contraction display is a sign convention, not an estimate from the negative realizations alone. If the responses to the negative and positive realizations differ, the estimate averages the two. The figure alone cannot attribute the estimated responses to QE or to QT1.

The OFBS responses do not correspond to a single episode either. The narrative restriction ties the shock to the ON RRP drain of the QT2 window, and the drain enters as negative realizations. The ON RRP buildup of 2021 to 2022 enters as positive realizations, and the OFBS shock ranks first for the deflator over the buildup as well as over the drain (Section 5.3). The linear model has one response function for the shock, and realizations of both signs enter the estimation of that function. The contraction display is a sign convention. I read the plotted responses as the regime average over both episodes and not as the responses to the QT2 drain alone.

5.2 Robustness

The robustness exercises fall into four groups. The first varies the constant in the ON RRP ratio. The second alters the variance regimes and the error distribution. The third alters the zero restrictions, and the fourth replaces the interpolated GDP and GDP deflator with monthly industrial production and CPI series.

The constant in the ON RRP ratio. The RFBS GDP response does not move with the constant, and the two deflator responses move in opposite directions. I reran the model with c at 0, \$100 billion, and \$500 billion, keeping every regime, sign, zero, and narrative restriction. The RFBS GDP decline keeps its size and its 68% bands under all four values. As c rises the RFBS deflator decline strengthens and the OFBS deflator decline shrinks, and its 68% upper bound crosses zero between \$100 billion and \$250 billion. Both movements come from QT1. At $c = 0$ the OFBS shock accounts for half of the unexpected fall in $\log(\text{Reserves} + \text{ON RRP})$ over QT1, a window that reserves financed. That half returns to the RFBS shock as c rises, and the deflator responses follow it. At $c = 0$ the model reads the fall of the ON RRP balances from \$154 billion to near zero over 2017 to 2019 as a sequence of OFBS contractions. At \$500 billion the constant is large enough to dampen the QT2 drain in the ratio, and the OFBS shock moves the ratio less. The baseline value of \$250 billion sits between the two. Figures G.1 and G.2 in Appendix G report the responses.

A single variance regime. The homoskedastic model is infeasible on this sample. I reran the model with a single variance regime, keeping every sign, zero, and narrative restriction. The initialization fails. The numerical search at the initial θ finds no admissible rotation, and the chain never starts. The sample spans the financial crisis, COVID, and the post-2021 inflation, and the estimated variance ratios across regimes are large (Table E.1). A single-regime model forces one variance onto all of them, and no rotation satisfies the restrictions at the homoskedastic estimates. On this sample the regime structure is essential to the identification, not a refinement of it.

Three variance regimes. The baseline responses of Section 5.1 are robust to reducing the variance regimes from five to three. Two specifications reduce the regimes, and each keeps every sign, zero, and narrative restriction. The first partitions the sample at 2015:M7 and 2022:M3, and the second at 2011:M1 and 2020:M4. Under both partitions the RFBS decline in GDP keeps its 90% bands. The medians at three years are 0.74 and 0.61 percent against 0.56 in the baseline. The RFBS deflator decline keeps its 68% bands from the second year. The OFBS deflator decline keeps its 68% bands through twenty-four months under both partitions, and the OFBS GDP bands include zero. The bands are wider under the first partition. Appendix G reports the responses, Figure G.3 under the first partition and Figure G.4 under the second.

Gaussian errors. Removing the latent scales shrinks the RFBS responses and narrows the bands, and the pattern across the two shocks is unchanged. The baseline draws each standardized shock from a Student-t with 4.99 degrees of freedom (Section 3.2), and the latent scales give each shock in each month its own scale. The Gaussian specification removes the latent scales and keeps every regime, sign, zero, and narrative restriction. The RFBS decline in GDP has a median of 0.38 percent at three years against 0.56 in the baseline. Its 90% bands exclude zero from month twelve, with an upper bound within 0.01 percentage points of zero. The RFBS deflator decline excludes zero at 90% from twenty-four months, where the baseline bands include zero. The OFBS deflator decline keeps its 68% bands through twenty-four months, and the OFBS GDP bands include zero.

The two largest standardized residuals are the April 2020 SOMA residual and the March 2020 FFR residual, and the latent scales absorb them in the baseline (Section 3.2). Under Gaussian errors these months carry full weight in the likelihood. I read the smaller RFBS medians as the weight of those months in the coefficient estimates. Figure G.5 in Appendix G reports the responses.

The zero restrictions. Assumptions 1 and 2 generate the zeros, and I remove them one at a time to separate their roles. The first specification removes Assumption 1 and keeps the three SOMA zeros of Assumption 2. The GDP and GDP-deflator responses to the two balance-sheet shocks are unrestricted at impact. The second specification removes Assumption 2 and keeps the four zeros of Assumption 1. SOMA can respond within the month to the demand, supply, and MP shocks.

The two macroeconomic results depend on Assumption 1, and Assumption 2 sharpens their bands. Without Assumption 1 the RFBS response of GDP is positive at impact, 0.27 percent with 68% bands above zero. The decline at three years shrinks to 0.22 percent, with 68% bands that reach zero. The RFBS deflator bands include zero at every horizon. The OFBS deflator bands include zero at every horizon, where the baseline bands exclude zero at 68% from eight to twenty-four months. The OFBS GDP response is positive at three years at the 68% level. Without Assumption 2 the responses stay close to the baseline. The RFBS decline in GDP has a median of 0.44 percent with 90% bands that exclude zero, and the RFBS deflator decline keeps its 68% bands. The OFBS deflator decline keeps its 68% bands through twelve months.

Removing both assumptions leaves the RFBS decline in GDP and removes the OFBS decline in the deflator. The RFBS decline in GDP has a median of 0.34 percent at three years, its 68% bands exclude zero, and its 90% bands include zero. The RFBS deflator bands include zero. The OFBS GDP response is positive at the 68% level from six months, and its 90% bands include zero. The OFBS deflator bands include zero at every horizon. The SOMA, log(Reserves + ON RRP), ON RRP-ratio, and term-spread responses keep their baseline shapes with wider credible bands.

I read the positive RFBS responses at impact without Assumption 1 as reflecting the state of the economy around the balance-sheet actions rather than the transmission of the shock. The largest RFBS realizations fall in the COVID months. The latent scales downweight the extreme residuals of those months in the likelihood, but they leave the state of those months in the estimates. On this reading the estimates without the impact zeros mix the transmission with the state, and Assumption 1 separates them. The sign and narrative restrictions and the heteroskedasticity identify the four sign-restricted responses of Panel A of Table 3 on their own. The zero restrictions carry the GDP and deflator results. Appendix G reports the responses, Figures G.6 and G.7 for the single removals and Figure G.8 for the joint removal.

Exogenous fiscal and issuance controls. The baseline responses survive direct controls for the two forces outside the system that move reserves and ON RRP. The first specification adds the Treasury General Account, the ratio of the TGA to the sum of reserves, ON RRP, and the TGA, at lags zero to two. The second adds Treasury issuance, the bill share of marketable debt and the log change of coupon securities outstanding, each at lags zero to two. The exogenous coefficients enter with no shrinkage, and every regime, sign, zero, and narrative restriction stays. Under the TGA control both balance-sheet responses match the baseline at both band levels. Under the issuance controls the RFBS decline in GDP keeps its 68% bands, with a median of 0.34 percent at three years against 0.56 in the baseline. The RFBS deflator decline strengthens to 0.60 percent, and its 90% bands exclude zero from twenty-four months.

The OFBS term spread keeps its rise under the issuance controls, and the OFBS deflator bands include zero at every horizon at the 68% level. I read the TGA specification as showing that the estimated transmission does not reflect the TGA. I read the issuance specification with Section 2. Bill issuance is the route by which the ON RRP drain reaches the Treasury, and a control that absorbs that route absorbs part of the OFBS shock. Figures G.10 and G.11 in Appendix G report the responses.

The one-year yield as the policy rate. The baseline responses survive with the one-year Treasury yield in place of the funds rate. At the zero lower bound the funds rate carries no innovation, and announcements about future policy move short-maturity yields. A specification replaces the funds rate with the one-year yield and keeps everything else. The RFBS decline in GDP keeps its shape, its 90% bands exclude zero from month twelve, and its median at three years is 0.72 percent. The OFBS deflator decline keeps its 68% bands through three years, where the baseline bands include zero after twenty-four months, and the OFBS term spread keeps its rise. The OFBS GDP bands include zero, as in the baseline. Figure G.12 in Appendix G reports the responses.

Monthly outcome variables. A specification replaces the two interpolated outcome variables with the log of industrial production and the log of the CPI. At the 68% level the RFBS decline in industrial production matches the baseline GDP decline from the second year, and its 90% bands include zero. The RFBS decline in the CPI excludes zero at 68% from twenty-four months, and its 90% bands include zero. The OFBS decline in the CPI excludes zero at 68% through twelve months, and its 90% bands include zero. Its median at twelve months is 0.73 percent against 0.45 for the deflator. The OFBS response of industrial production is positive at 68% at six months and includes zero afterward, and its 90% bands include zero at every horizon. Figure G.9 in Appendix G reports the responses.

5.3 Variance and Historical Decompositions

The impulse responses report an elasticity for a one-percent decline in SOMA holdings. They do not say how much of the observed variation in GDP and the deflator the two shocks account for. This subsection reports two decompositions that answer that question. The forecast-error variance decomposition gives the share of the forecast variance at a given horizon. The historical decomposition gives the contribution of each shock to the unexpected change in a variable over an episode. Both use the baseline draws of Section 5.1 and weight the draws as the impulse responses do.

Forecast-error variance decomposition. Table 5 reports the variance decomposition under the average transmission. The shares below are posterior medians at three years, and the table gives the 68% intervals around them. The two balance-sheet shocks together account for 13% of the forecast variance of GDP and 10% of the forecast variance of the deflator. Of the GDP forecast variance the RFBS shock accounts for 11% and the OFBS shock for 1.5%. Of the deflator forecast variance the RFBS shock accounts for 7% and the OFBS shock for 3%. The same decomposition covers the two liability variables, which Table 5 omits. The two balance-sheet shocks together account for 46% of the forecast variance of log(Reserves + ON RRP). The OFBS shock accounts for 40% of the forecast variance of the ON RRP ratio.

Historical decomposition. The historical decomposition splits the accumulated forecast error of a variable over a window among the shocks that the sample realizes inside it. Let $W = \{t_0 + 1, \ldots, t_1\}$ and let $\mathcal{I}_{t_0}$ collect the observables through t_0. The contribution of labeled shock j to variable i is

$$\mathrm{HD}_{j\to i}(W) = \sum_{\ell=0}^{t_1-t_0-1} \left[\Theta_{s_{t_1-\ell},\ell}(Q)\right]_{ji} \tilde{\varepsilon}_{j,t_1-\ell}, \qquad \sum_{j=1}^{n} \mathrm{HD}_{j\to i}(W) = y_{i,t_1} - \mathbb{E}\left[y_{i,t_1} \mid \mathcal{I}_{t_0}\right], \tag{7}$$

Table 5: Forecast-error variance decomposition.

	Horizon	Share of forecast-error variance				
		Demand	Supply	MP	RFBS	OFBS
GDP	12	0.122	0.172	0.174	0.029	0.005
		[0.059, 0.235]	[0.082, 0.295]	[0.088, 0.287]	[0.011, 0.065]	[0.002, 0.016]
	24	0.108	0.189	0.106	0.080	0.009
		[0.041, 0.241]	[0.086, 0.335]	[0.053, 0.204]	[0.028, 0.157]	[0.003, 0.032]
	36	0.089	0.199	0.073	0.114	0.015
		[0.031, 0.235]	[0.078, 0.369]	[0.038, 0.155]	[0.041, 0.213]	[0.003, 0.055]
GDPDEF	12	0.122	0.407	0.146	0.010	0.013
		[0.031, 0.329]	[0.171, 0.647]	[0.044, 0.376]	[0.002, 0.039]	[0.003, 0.040]
	24	0.117	0.375	0.122	0.037	0.026
		[0.025, 0.323]	[0.162, 0.601]	[0.030, 0.337]	[0.006, 0.108]	[0.004, 0.078]
	36	0.113	0.329	0.100	0.072	0.029
		[0.025, 0.323]	[0.139, 0.546]	[0.026, 0.298]	[0.014, 0.182]	[0.006, 0.094]

Notes: Shares of the forecast-error variance under the average transmission, with the relative variances at their average of one across regimes. Horizons are in months. The first row of each pair is the posterior median and the second is the 68% credible interval, both over the retained draws. The five labeled shocks appear here, and the two unlabeled shocks account for the remainder.

where the regime index follows the date of the shock. Figure 8 reports (7) over six windows, as posterior medians with 68% intervals, for the five labeled shocks.[20] The six windows split two of the four episodes of Table A.1 at the month the financing changed. The COVID expansion splits at 2021:M2, the last month before the ON RRP buildup. QT2 splits at 2023:M5, the last month before the ON RRP balances began to fall. The narrative window of the OFBS shock lies inside the QT2 drain (Section 3.3).

[20] A contribution accumulates only the realizations of a shock inside the window. Reserves financed the balance sheet across the QE years, QT1, and the COVID expansion, and the ON RRP balances financed it over the months of the QT2 drain (Table A.1). A shock that the sample realizes over fewer months contributes less whatever its response function, and these entries do not separate a weak channel from one the sample rarely exercised.

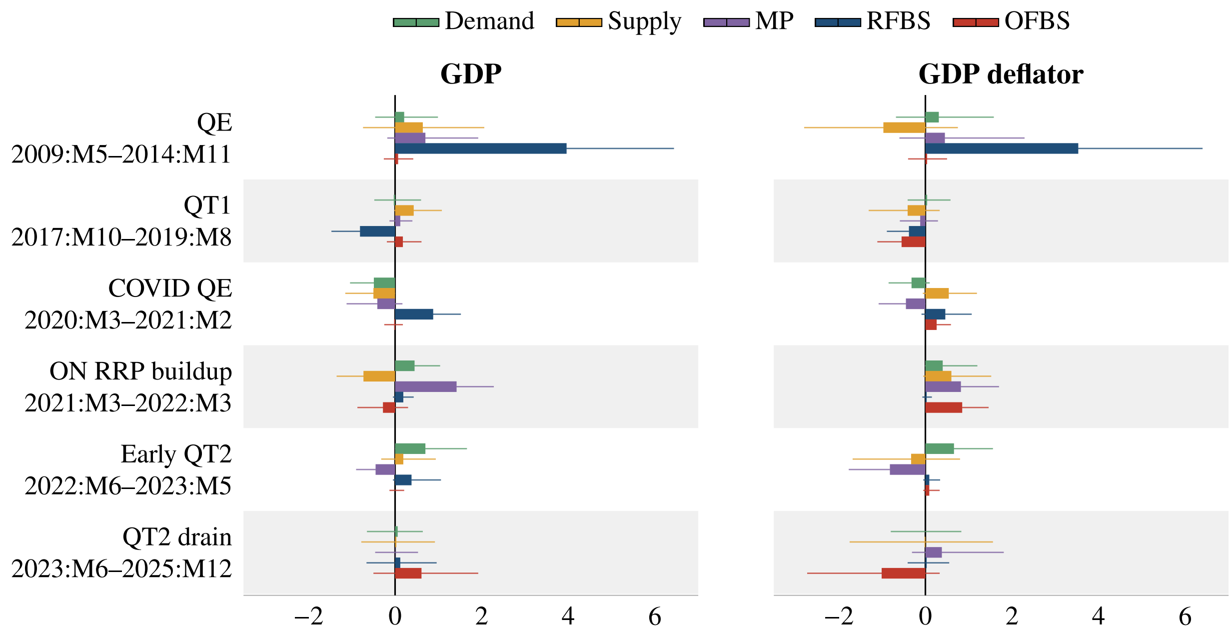


Figure 8: Historical decomposition by episode.

Note. Contribution of each labeled shock to the accumulated forecast error of the variable over the window, in percent of the variable's level relative to its within-window forecast. Entries are contributions to the level of the variable and not shares of its observed change, which also contains the path implied by the initial conditions. Bars are posterior medians and whiskers are 68% credible intervals, both over the retained draws. The QE window opens at 2009:M5, the first estimable month of the 2008:M12–2014:M11 episode under the six estimation lags. The two unlabeled shocks account for the remainder. Appendix H reports the numbers, the two unlabeled shocks, and the two liability variables.

The RFBS shock ranks first among the five labeled shocks for GDP over QE, QT1, and the first COVID window, the three windows that reserves financed. It ranks first for the deflator over QE and second over the first COVID window. Once the ON RRP balances move it drops out of the top three. It ranks last for both variables over the buildup and second for GDP over the QT2 drain.

The OFBS shock ranks last for both variables over QE and the first COVID window. It ranks first for the deflator over QT1, the buildup, and the QT2 drain, and first for GDP only over the drain.[21] Over early QT2 neither shock ranks above third.

The two shocks contribute to the two liabilities in different windows. Over QE and the first COVID window the RFBS shock accounts for the unexpected rise in log(Reserves + ON RRP), and

[21] QT1 is the one window in which the OFBS shock leads while reserves financed the drain. Section 5.2 traces the entry to the ON RRP balances that ran to zero over those months and reports how it falls as the constant c rises.

over the ON RRP buildup the OFBS shock does. Over the QT2 drain the OFBS shock accounts for the unexpected fall. Over QT1 the two shocks split the unexpected fall evenly, and the OFBS shock carries the fall in the ON RRP ratio against its forecast. Table H.3 in Appendix H reports the two liability variables.

Interpreting the decompositions. The decompositions repeat the pattern of the impulse responses in the sample. The RFBS share of the GDP forecast variance is 11% against 1.5% for the OFBS shock, and the RFBS share of the deflator forecast variance is 7% against 3%. The historical contributions locate each shock in the windows its liability financed. The RFBS shock ranks first for GDP over the three reserve-financed windows and for the deflator over QE. The OFBS shock ranks first for the deflator over the buildup and the QT2 drain, and its GDP contribution ranks first in no window but the drain.

6 Conclusion

The Federal Reserve financed its two QT episodes with different overnight liabilities. QT1 drained reserves, the liability that the 2008 to 2014 purchases had filled. QT2 drained ON RRP instead and left the reserves that the COVID purchases had filled nearly intact. Both liabilities are overnight claims on the Federal Reserve. Banks hold reserves, and money market funds hold most ON RRP balances. The two claims sit with different holders, and the transmission of balance-sheet policy to real GDP and the price level may differ with the liability that finances it. This paper asks whether it does.

The impulse responses answer the question. I normalize each shock to a one-percent impact decline in SOMA holdings and display both in the contractionary direction. Both contractions lower the deflator, and only the RFBS contraction lowers GDP.

The forecast-error variance decomposition gives the share of the forecast variance that each shock accounts for at a given horizon. At three years the two balance-sheet shocks together account for 13% of the forecast variance of GDP and 10% of the forecast variance of the deflator. The shares repeat the pattern of the impulse responses. The RFBS share of the GDP forecast variance is 11% against 1.5% for the OFBS shock, and the RFBS share of the deflator forecast variance is 7% against 3%. The two shocks together account for 46% of the forecast variance of log(Reserves + ON RRP), and the OFBS shock accounts for 40% of the forecast variance of the ON RRP ratio.

The historical decomposition gives the contribution of each shock to the unexpected change in a variable over an episode. The contributions below are posterior medians among the five labeled shocks. Over the 2008 to 2014 easing the RFBS shock is the most important driver of the unexpected changes in both variables. It is also the most important driver of the unexpected change in GDP over QT1 and over the first year of the COVID expansion, before the ON RRP buildup. Over the buildup and the QT2 drain the OFBS shock is the most important driver of the unexpected change in the deflator.

The empirical design combines heteroskedasticity, sign restrictions, narrative restrictions, and zero restrictions. The heteroskedastic likelihood separates the model shocks, and the sign and narrative restrictions label the two balance-sheet shocks. HARS-Z imposes the impact-zero restrictions on that combination, and the rotation satisfies every zero restriction at every draw.

Which liability finances a runoff can decide what the runoff does to GDP and the price level. The Federal Reserve sets the terms that govern the split between reserves and ON RRP. Within those terms the composition of Treasury issuance, bank regulation, and the portfolio choices of MMFs affected which liability each episode drained (Section 2). The asset side of the balance sheet moves the same way in every episode, but the liability side does not. The size of the balance sheet does not summarize what a balance-sheet policy does.

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A Balance-Sheet Data

This appendix collects the episode-level balance-sheet evidence discussed in Section 2. Table A.1 reports the Federal Reserve's flows by episode. Table A.2 and Figure A.1 report the bank balance sheet over the same windows, and Table A.3 reports the money market fund and hedge fund balance sheets.

Table A.1: Federal Reserve balance-sheet flows by episode, USD billions.

	QE 2008-12 → 2014-11	QT1 2017-10 → 2019-08	COVID QE 2020-03 → 2022-03	QT2 2022-06 → 2025-12
Asset side				
Total assets	+2,266	−690	+4,315	−2,352
MBS	+1,725	−267	+1,340	−666
SOMA Treasury	+1,975	−375	+2,732	−1,560
notes/bonds	+1,993	−378	+2,732	−1,467
T-bills	−18	+3	0	−92
Liability side				
Total liabilities	+2,266	−690	+4,315	−2,352
Reserves	+1,765	−709	+1,998	−312
ON RRP	+107	−115	+1,588	−2,152
TGA	+12	−43	+242	+124
Other	+382	+177	+487	−12
Absorption ratios (% of total asset change)				
Reserves	77.9	102.8	46.3	13.3
ON RRP	4.7	16.7	36.8	91.5

Notes: Asset-side figures use the Federal Reserve's H.4.1 series (WALCL) and the New York Fed's monthly SOMA Treasury holdings by maturity. Liability-side figures use H.4.1 series (WRESBAL, WTREGEN, WCURCIR) and the daily ON RRP series (RRPONTSYD). "Other" aggregates currency in circulation and residual Federal Reserve liabilities (capital, foreign and other deposits). Total liabilities equal total assets by accounting identity. All values resampled to month-start.

Reserves are a liability of the Federal Reserve and an asset of the banks that hold them. A purchase settled in reserves adds reserves on the asset side of the banking system and a deposit on its liability side, held by whoever sold the securities. Table A.2 and Figure A.1 report the bank balance sheet over the four episodes. Reserves and deposits rise together in the two purchase episodes. In QT1 reserves fall and deposit growth slows. In QT2 reserves and deposits are both nearly flat, while the ON RRP facility loses $1.9 trillion. The loan and securities rows record what banks held on the asset side beside reserves. They are not the destination of reserves, which stay in the banking system, and the table makes no causal claim about them.

Table A.2: The bank balance sheet by episode, USD billions.

	QE 2008-12 → 2014-11	QT1 2017-10 → 2019-08	COVID QE 2020-03 → 2022-03	QT2 2022-06 → 2025-12
Assets				
Reserves	+1,765	−709	+1,998	−312
Securities in bank credit	+790	+336	+1,859	−149
Loans and leases	+611	+831	+629	+1,958
Liabilities				
Deposits	+3,164	+894	+4,383	+539

Notes: Changes over the episode windows of Table A.1, from monthly means of the weekly H.8 series (all commercial banks, seasonally adjusted), month-start labels, as in that table. The reserves row is the reserves row of Table A.1. Loans and leases and securities sum to bank credit. Two reclassifications move the loans row without any lending. The FAS 166/167 consolidation of 2010:M3 to 2010:M4 adds $359 billion to consumer loans in the QE window. The 2025 reclassification of loans to nondepository financial institutions removes $189 billion from commercial and industrial loans in the QT2 window. Nominal deposits carry trend growth, and the table does not remove it.

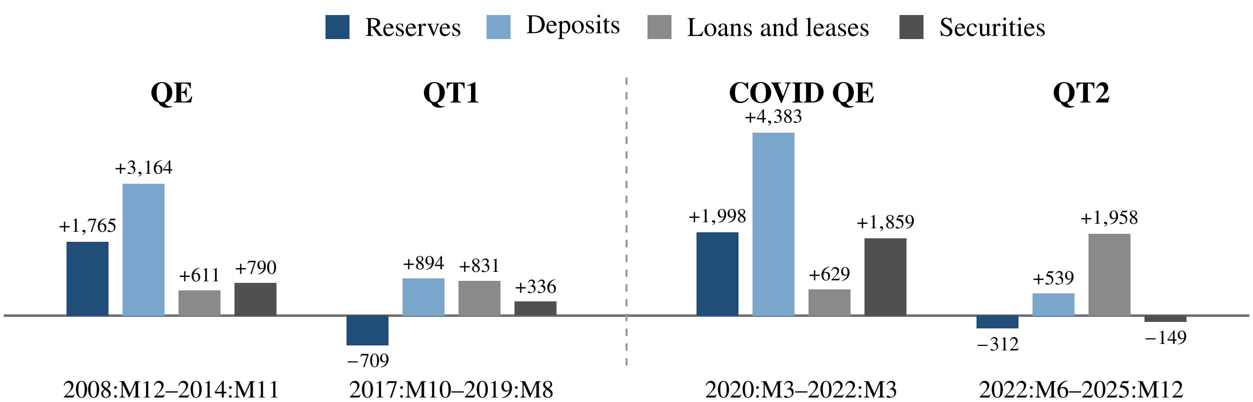


Figure A.1: The bank balance sheet by episode.

Note. Bars show the changes of Table A.2, in USD billions, with the sign of each change. Reserves are an asset of the banking system and deposits a liability. The scale is that of Figure 4. Source: Federal Reserve H.8 and H.4.1.

Table A.3 records the ON RRP route. When money market funds take cash out of the facility, the offsetting item sits on their own asset side, in Treasury securities and in repo with private borrowers. Total assets need not fall. The hedge fund rows record who borrowed the repo cash and what they held. The issuance rows record the coupons and bills that the Treasury sold over each window, which is the supply the funds and the hedge funds absorbed.

Table A.3: Money market fund and hedge fund balance sheets by episode, USD billions.

	QE 2008-12 → 2014-11	QT1 2017-10 → 2019-08	COVID QE 2020-03 → 2022-03	QT2 2022-06 → 2025-12
ON RRP facility				
All counterparties	+149	−304	+1,870	−1,873
of which MMFs		−286	+1,656	−1,681
Money market fund assets				
Treasury securities		+222	+719	+1,979
Private repo		+609	−605	+2,251
of which through FICC		+173	−106	+1,210
Total assets		+803	+1,058	+3,316
Hedge funds				
Government debt exposure		+953	−493	+2,497
Repo borrowing		+610	−212	+2,289
Treasury issuance				
Net bill issuance	−564	+530	+1,364	+2,874
Net coupon issuance	+7,163	+1,419	+5,006	+4,092

Notes: Changes over the episode windows of Table A.1. Money market fund, facility, and issuance rows are month-end levels at the last month of the window less the level at the last month-end before it. Form N-MFP is a month-end snapshot. Hedge fund rows are quarter-ends that bracket the window. Blank cells have no observation at the base date. Form N-MFP begins in 2010:M11 and Form PF in 2013, both after the QE window opens. MMFs are money market funds. Private repo is their repo with every counterparty other than the Federal Reserve. FICC is the Fixed Income Clearing Corporation, where hedge funds borrow through sponsored repo. Government debt exposure is hedge funds' gross exposure to U.S. government debt, and repo borrowing is the cash they borrow in repo, both from Form PF. Treasury securities include coupons with remaining maturity under thirteen months. Sources: Federal Reserve daily ON RRP series (RRPONTSYD), SEC Form N-MFP through the OFR Money Market Fund Monitor, SEC Form PF through the OFR Hedge Fund Monitor, and the Monthly Statement of the Public Debt.

B The Narrative Restrictions

This appendix states the narrative restrictions of Section 3.3. Each restriction ties a labeled shock to the episode that generated it through two requirements. Let W be a window of months inside a regime and j the focal shock. The first is a sign requirement on the shock's realizations. For the RFBS window the mean realization is positive,

$$\frac{1}{|W|}\sum_{t\in W}\tilde{\varepsilon}_{j,t} \;>\; 0. \tag{B.1}$$

For the OFBS window the restriction reverses the inequality in (B.1). The second restricts the historical decomposition. Let $\mathrm{HD}_{j\to L}(W)$ denote the contribution of labeled shock j to $L = \log(\text{Reserves} + \text{ON RRP})$ accumulated over the window W. The restriction requires this contribution to dominate those of the other labeled shocks,

$$\left|\mathrm{HD}_{j\to L}(W)\right| \;>\; \sum_{\substack{i \text{ labeled} \\ i\neq j}}\left|\mathrm{HD}_{i\to L}(W)\right|. \tag{B.2}$$

In the baseline, $\mathrm{HD}_{j\to L}(W)$ is the multi-horizon interval historical decomposition of Antolín-Díaz and Rubio-Ramírez (2018), the contribution of shock j to the level of $\log(\text{Reserves} + \text{ON RRP})$ at the end of the window. The contributions of the n rotated shocks add up to the unexpected movement of $\log(\text{Reserves} + \text{ON RRP})$ over the window, and each restriction assigns a share of that movement to each shock. The sum runs over the labeled shocks only, while Antolín-Díaz and Rubio-Ramírez (2018) compare the focal shock against every other shock.

C HARS and HARS-Z

This appendix expands Section 4. Appendix C.1 reviews HARS in detail. Appendix C.2 derives the zero constraint matrices and states the formal consequences of adding zero restrictions to HARS. Appendix C.3 builds the map γ_z. Appendix C.4 states the results that establish the validity of HARS-Z. Appendix C.5 states the posterior-inference estimator. The next appendix collects the proofs.

C.1 HARS in Detail

The absorbed rotation. HARS (KZ, Algorithm 1) combines likelihood-preserving posterior draws of the model parameters with rotations drawn from a well-defined probability distribution restricted by the sign and narrative restrictions. In homoskedastic SVARs, the rotations of the coefficients, $(A_0, A_+) \mapsto (A_0 Q, A_+ Q)$ with $Q \in \mathcal{O}(n)$, are observationally equivalent. Sign restrictions identify shocks by selecting, among these rotations, those that produce economically sensible impulse responses. Under heteroskedasticity this rotation invariance fails. Rotating the coefficients while holding the regime variances fixed mixes shocks whose relative variances differ across regimes and changes the heteroskedastic likelihood (KZ, Lemma 1 and Proposition 2). The rotation would change the fitted model rather than select an interpretation.

KZ's construction absorbs the regime scale before rotation. Define $\Xi_{s_t} := \Lambda_{s_t}^{-1/2}$ and

$$\varepsilon_t := \Xi_{s_t} \eta_t, \qquad E(\varepsilon_t \varepsilon_t' \mid s_t) = I_n,$$

and the shocks have unit variance in every regime. Post-multiplying (1) by Ξ_{s_t} absorbs the regime scaling into the block $(A_0 \Xi_{s_t}, A_+ \Xi_{s_t})$. Rotating the entire absorbed block by $Q \in \mathcal{O}(n)$ preserves the independent-shock representation and leaves the likelihood unchanged (KZ, Proposition 1),

$$p(y \mid \theta, Q) \;=\; p(y \mid \theta) \qquad \text{for every } Q \in \mathcal{O}(n), \tag{C.1}$$

where $\theta = (A_0, \Lambda, A_+)$ collects the structural objects and $\Lambda = \{\Lambda_k\}_{k=1}^h$. Absorbing first keeps the likelihood invariant to Q under heteroskedasticity. I refer to the rotated unit-variance shocks $\tilde{\varepsilon}_t' = \varepsilon_t' Q$ as the labeled shocks. The dynamic responses of the observables to the labeled shocks in regime k at horizon ℓ are the rows of

$$\Theta_{k,\ell}(Q) \;=\; Q' \Lambda_k^{1/2} A_0^{-1} \, \Phi_\ell(B), \tag{C.2}$$

where $\{\Phi_\ell(B)\}_{\ell \geq 0}$ denotes the moving-average coefficients of the reduced form, with $\Phi_0(B) = I_n$.[22] The regime affects the responses only through the positive diagonal matrix $\Lambda_k^{1/2}$. I also define the average transmission

$$\bar{\Theta}_\ell(Q) \;=\; Q' A_0^{-1} \, \Phi_\ell(B), \tag{C.3}$$

which evaluates the responses at the average of the relative variances across regimes, and that average equals one under the normalization. The average transmission is the object I report as the impulse response. Because Q' mixes the regime-specific scales in $\Lambda_k^{1/2}$ across shocks, the regime-specific responses are not positive rescalings of the average response, and restrictions imposed regime by regime add information.

The restricted posterior target. The likelihood invariance (C.1) separates posterior simulation from identification. Write $SR(\theta, Q)$ for the indicator that the sign and narrative restrictions are satisfied, and define the admissible set $R(\theta) = \{Q \in \mathcal{O}(n) : SR(\theta, Q) > 0\}$, as in KZ. When $\mu(R(\theta)) > 0$, where μ is Haar measure on $\mathcal{O}(n)$, KZ's Definition 1 gives the restricted posterior target

$$\pi(d\theta, dQ \mid y, SR) \;=\; p(\theta \mid y) \, \frac{\mathbf{1}\{Q \in R(\theta)\}}{\mu(R(\theta))} \, \mu(dQ). \tag{C.4}$$

[22] Under the right-multiplication convention, $y_{t+\ell}' = \cdots + u_t' \, \Phi_\ell(B)$ and $\Phi_\ell(B) = \sum_{m=1}^{\min(\ell,p)} \Phi_{\ell-m}(B) \, B_m$, where B_m is the m-th $n \times n$ lag block of B. Row j of $\Theta_{k,\ell}(Q)$ collects the horizon-ℓ responses of the n observables to labeled shock j. Unlike KZ, who write $\Theta_{k,\ell}(Q) = \Phi_\ell(B) \, Q' \Lambda_k^{1/2} A_0^{-1}$, this draft places the moving-average matrix on the right, as in (C.2), and all remaining notation is identical.

Drawing θ from its posterior and then drawing Q from Haar measure restricted to $R(\theta)$ produces exact draws from this target (KZ, Proposition 3). The restrictions truncate only the conditional distribution of the rotation. They change the support of Q, not the posterior weight on θ.

The rotation in Gaussian coordinates. HARS draws the rotation in Gaussian coordinates. An $n \times n$ matrix X with independent standard-normal entries maps to a Haar-distributed rotation $Q = \gamma(X)$ through the QR map γ. Column-wise updates of X by the elliptical slice sampler (ESS; Murray et al., 2010) leave the conditional distribution of Q in (C.4) invariant (KZ, Proposition 4). When a posterior move leaves the retained rotation stale, HARS repairs in two stages. The posterior-draw repair redraws completed parameter candidates at the fixed rotation and retains the first admissible one. This repair preserves the sign-restricted marginal target of (A_0, Λ) (KZ, Theorem 1 and Step 2a of their Algorithm 1). Only if no candidate passes does the global rotation repair draw independent rotations from Haar measure and retain the first admissible one. That draw has the restricted-Haar conditional distribution of (C.4) (KZ, Proposition 5(i) and Step 2b).

C.2 The Failure Under Zero Restrictions

Zero restrictions break this construction, and the failure originates in the measure rather than in the algorithm. The zeros invalidate the restricted target (C.4) and the Gaussian coordinatization, while the likelihood invariance (C.1) continues to hold. The obstacle is that a zero restriction is not a truncation. Sign and narrative restrictions truncate the conditional distribution of Q to a subset of positive Haar measure, and the target (C.4) remains well defined. A zero restriction instead imposes a linear equality constraint on a column of Q and lowers the dimension of the admissible set. Adding the zeros to the indicator SR fails, as Remark 1(i) states below.

Under the average transmission (C.3), the impact response of variable i_0 to shock j is $q_j' A_0^{-1} e_{i_0}$, where e_{i_0} is the i_0-th column of I_n. The response is linear in the j-th column q_j of Q, and a zero restriction on this response is a linear constraint on q_j. For a structural draw θ, I describe the zeros on shock j with two objects, following the notation of Arias et al. (2018). The scalar z_j counts the zero restrictions on shock j. The $z_j \times n$ matrix $Z_j(\theta)$ stacks those restrictions as rows. Each row is $(A_0^{-1} e_{i_0})'$ for an impact zero on variable i_0 and $(A_0^{-1} \Phi_\ell(B)\, e_{i_0})'$ for a horizon-ℓ zero, and z_j is the row count of $Z_j(\theta)$.[23] The zero restrictions on shock j read

$$Z_j(\theta)\, q_j = 0,$$

and the sampler must rebuild the constraint matrices at every iteration. The matrices depend on the draw through A_0 and $\Phi_\ell(B)$.[24] Define the zero manifold

$$\mathcal{Q}(\theta) \;=\; \big\{ Q \in \mathcal{O}(n) : Z_j(\theta)\, q_j = 0 \text{ for every restricted } j \big\}.$$

Whenever some $z_j \geq 1$, $\mathcal{Q}(\theta)$ is a lower-dimensional smooth manifold in $\mathcal{O}(n)$ and $\mu(\mathcal{Q}(\theta)) = 0$, as ARRW note in motivating their Algorithm 2.[25] The next remark states the consequences for HARS.

Remark 1 (HARS cannot impose zero restrictions)**.** Fix θ with at least one zero restriction. Then $\mu(\mathcal{Q}(\theta)) = 0$, and the following hold.

(i) The restricted posterior target (C.4) is not defined.

(ii) The Gaussian coordinatization never reaches the zeros, that is, $\Pr\{\gamma(X) \in \mathcal{Q}(\theta)\} = 0$.

(iii) Every step of KZ's Algorithm 1 fails with probability one. The failure covers the posterior draw

[23] ARRW impose the zeros on a transformation $F(A_0, A_+)$ of the structural parameters, writing $Z_j F(A_0, A_+)\, e_j = 0$ with a constant selection matrix Z_j and e_j the jth column of I_n. In their orthogonal reduced-form parameterization the rotation carries the restriction, and their construction evaluates F before the rotation, giving $Z_j F\, q_j = 0$. I fold that product into a single matrix $Z_j(\theta) := Z_j F(\theta)$, which matches KZ's convention $\tilde{\varepsilon}_t' = \varepsilon_t' Q$. The fold is notational. The transformation F appears as the rows of $Z_j(\theta)$, and the count z_j, the dimension bounds, and the null-space construction are identical to theirs.

[24] Unlike the admissibility check $SR(\theta, Q)$, in which the sign restrictions must hold in every regime and Λ enters, the zero constraints do not depend on Λ. The average transmission sets the relative variances to one, and the zeros hold exactly under (C.3). They need not hold in the regime-specific responses (C.2), where $\Lambda_k^{1/2}$ enters the product $Q' \Lambda_k^{1/2} A_0^{-1}\, \Phi_\ell(B)$ and reweights the constrained column.

[25] When $Z_j(\theta)$ has full row rank, the column q_j must lie on the unit sphere of the $(n - z_j)$-dimensional null space of $Z_j(\theta)$. The zero restrictions lower the dimension of $\mathcal{Q}(\theta)$ below the $n(n-1)/2$ dimensions of $\mathcal{O}(n)$, and any lower-dimensional subset has Haar measure zero. ARRW establish the same fact in the structural parameterization. The set of structural parameters satisfying the sign and zero restrictions is of Lebesgue measure zero in the set of all structural parameters. The zero restrictions define a lower-dimensional smooth manifold, and the volume measure of their Theorem 3 serves as the reference measure on it.

with admissibility check (Step 1), the posterior-draw repair by the joint procedure (Step 2a), the global rotation repair (Step 2b), and the rotation draw (Step 3). The column-by-column accept–reject initialization fails as well.

Proof. See Appendix D.1. □

Rotations drawn from the uniform distribution satisfy the zero restrictions with probability zero.[26] The sampler must parameterize the zero-restricted set directly rather than reach it by sampling. It must build the rotation on $Q(\theta)$, and its slice-sampling updates must never leave it. Appendix C.3 constructs exactly such a parameterization.

C.3 Extending HARS to Zero Restrictions

HARS-Z extends HARS to include zero restrictions on the rotation. Developing this extension is the paper's methodological contribution. The extension replaces the QR map γ with a map $\gamma_z(\cdot\,;\theta)$ from a Gaussian matrix to a rotation that lies on $Q(\theta)$ *by construction*. ARRW's Algorithm 2 supplies such a map, and the zero restrictions hold exactly along the chain. The subsection builds the map γ_z and sets the state of the chain and its posterior target. It identifies the measure that the target induces on $Q(\theta)$ and constructs the rotation draw.

ARRW's Algorithm 2 builds the columns of Q one at a time, following a permutation σ of the shock indices $\{1, \ldots, n\}$ that places the most-restricted shocks first. The index $\sigma(j)$ is the shock whose column the construction builds at step j. The permutation must satisfy $z_{\sigma(j)} \leq n - j$ for every j, where $z_{\sigma(j)}$ is the zero count of the shock assigned to step j. The j-th column must be orthogonal to the $j-1$ preceding columns and must satisfy its own $z_{\sigma(j)}$ zeros. A candidate direction exists only if $(j-1) + z_{\sigma(j)} < n$.[27] For every restricted shock j, the matrix $Z_j(\theta)$ must also have full row rank, that is, its z_j rows must be non-redundant.[28]

[26] The obstruction is also topological. The elliptical slice sampler of Arias et al. (2026) assumes an admissible set of positive measure (their Assumption 1). Strict-inequality sign restrictions give an open set of positive measure. Zero restrictions give a set that is neither open nor of positive measure.

[27] Later positions have more orthogonality constraints and admit fewer zeros, and the most heavily restricted shocks come first.

[28] ARRW impose this assumption on their selection matrices Z_j when they set their notation (their Section 2.2).

Given a Gaussian matrix $G = [g_1, \ldots, g_n]$, the construction proceeds one column at a time. At step j, the admissible directions for column $q_{\sigma(j)}$ form the linear subspace

$$N_j(\theta) \;:=\; \operatorname{null}\big(M_j(\theta)\big), \qquad M_j(\theta) \;:=\; \begin{bmatrix} [\, q_{\sigma(1)}, \ldots, q_{\sigma(j-1)} \,]' \\ Z_{\sigma(j)}(\theta) \end{bmatrix}, \tag{C.5}$$

where $M_j(\theta)$ stacks the preceding columns $q_{\sigma(1)}, \ldots, q_{\sigma(j-1)}$ and the zero restrictions of shock $\sigma(j)$. The null space $N_j(\theta)$ collects the vectors orthogonal to every preceding column and consistent with those zeros.

The projection at step j requires the rows of $M_j(\theta)$ to be linearly independent. The null space has the dimension $n - (j-1) - z_{\sigma(j)}$ that the ordering condition above presumes. Each block of $M_j(\theta)$ is independent on its own. The preceding columns are orthonormal, and $Z_{\sigma(j)}(\theta)$ has full row rank. The requirement binds only across the two blocks. I assume that the rows of $Z_{\sigma(j)}(\theta)$ are linearly independent of the preceding columns at every orthonormal collection $q_{\sigma(1)}, \ldots, q_{\sigma(j-1)}$ whose columns satisfy their zero restrictions.[29] ARRW derive this property from a regularity condition on their function F and the full row rank of the Z_j. See their footnote 6 in Section 2.2 and their Appendix A.3, conditions 1 and 2. I assume it directly, at each θ the chain visits rather than over the whole parameter space.

The construction projects g_j orthogonally onto $N_j(\theta)$ and normalizes,

$$q_{\sigma(j)} \;=\; \frac{P_j(\theta)\, g_j}{\big\|P_j(\theta)\, g_j\big\|}, \tag{C.6}$$

where $P_j(\theta)$ denotes the orthogonal projection onto $N_j(\theta)$ and the denominator is nonzero for almost every g_j. Write $\gamma_z(\cdot\,;\theta)$ for the map $G \mapsto Q$ that these n projection steps define. The chain below stores this Gaussian input as part of its state and denotes it by X.

[29]This independence fails only at draws of θ where some combination of the rows of $Z_{\sigma(j)}(\theta)$ falls exactly in the span of the preceding columns. The rows are functions of entries of A_0^{-1} and the moving-average coefficients. The failure requires an exact equality among these entries. A draw of θ from a continuous distribution satisfies an exact equality with probability zero. The retained draws avoid the failure set almost surely. ARRW make the same genericity argument for their set U (their Appendix A.3), and the implementation checks the condition at every construction step at no additional cost.

Two properties follow. First, each column satisfies its constraints by construction, and after n steps Q is orthogonal and lies on $\mathcal{Q}(\theta)$. The construction involves no accept–reject step. Second, the projected Gaussian column favors no direction within $N_j(\theta)$, and the normalized column is uniform over the admissible directions at step j.[30] This step-by-step distribution matches Steps 2 and 3 of ARRW's Algorithm 2.[31]

The uniform distribution of each column, conditional on the preceding columns, determines the law of the completed rotation. Write $\mu_z(\cdot\,;\theta)$ for the null-space construction measure on $\mathcal{Q}(\theta)$, the distribution of $\gamma_z(G;\theta)$ under a standard Gaussian matrix G,

$$\mu_z(C;\theta) \;:=\; \Pr\big\{\gamma_z(G;\theta) \in C\big\} \qquad \text{for every measurable } C \subseteq \mathcal{Q}(\theta). \tag{C.7}$$

Proposition 1 characterizes this measure and relates it to the conditional-Haar (uniform) measure on $\mathcal{Q}(\theta)$.

I depart from ARRW in one respect. At step j they draw $x_j \in \mathbb{R}^{n+1-j-z_j}$ from a standard normal distribution and normalize it to the unit sphere. The dimension of their coordinates varies with j and, through z_j, with the restriction pattern. I instead use a single $n \times n$ Gaussian matrix X, the coordinates in which HARS runs its ESS update. Step j takes the j-th column of X as its input. The randomness remains separate across steps, and every column keeps the fixed dimension n. The projection at each step keeps only the components in $N_j(\theta)$, and the projected column has the same distribution as the vector ARRW draw at that step. The fixed $n \times n$ matrix allows me to apply the column-wise ESS update of KZ (Proposition 4) and its validity argument without change.

Lemma 1 (Properties of the map γ_z)**.** *Fix θ such that, for every j, the matrix $Z_{\sigma(j)}(\theta)$ has full row rank and the permutation satisfies $z_{\sigma(j)} \leq n - j$. Require also that the matrix $M_j(\theta)$ in (C.5) has*

[30] The orthogonal projection of a standard Gaussian vector onto a linear subspace is a standard Gaussian within that subspace, whose distribution is invariant under rotations of the subspace. Its normalized direction is uniform on the subspace's unit sphere.

[31] At each step j, they draw a standard normal vector on $\mathbb{R}^{n+1-j-z_j}$, normalize it to the unit sphere, and map it through a matrix K_j. The columns of K_j form an orthonormal basis for the null space of M_j, the matrix that stacks the preceding columns and the zero restrictions. The orthogonal projection above induces the same uniform distribution on that null space. The two constructions draw each column from the same distribution.

full row rank at every orthonormal collection $q_{\sigma(1)}, \ldots, q_{\sigma(j-1)}$ whose columns satisfy their zero restrictions. Then the following hold for almost every G.

(i) *$\gamma_z(G;\theta) \in \mathcal{Q}(\theta)$, and every zero restriction holds exactly, with no rejection.*

(ii) *Every construction step k depends on G only through $g_1, \ldots, g_k$. Perturbing g_j leaves the columns from steps $1, \ldots, j-1$ unchanged, and the construction re-projects the perturbed and later columns into their own null spaces.*

(iii) *If $Q \in \mathcal{Q}(\theta)$ is admissible and $G_Q := [\, q_{\sigma(1)}, \ldots, q_{\sigma(n)} \,]$ collects the columns of Q in build order, then $\gamma_z(G_Q;\theta) = Q$. The construction reproduces its own output, and setting $X = G_Q$ places the chain at an admissible state.*

Proof. See Appendix D.2. □

Lemma 1 makes ARRW's Algorithm 2 usable inside a Markov chain. Part (i) restates the property of their Algorithm 2 in the fixed $n \times n$ Gaussian parameterization introduced above. Part (ii) is new. ARRW draw each rotation independently and never perturb one afterward. They never need to ask how the rotation changes when a single Gaussian column moves.[32] The column-wise ESS update moves one column of X along an ellipse and poses exactly that question. Part (ii) supplies the answer. The recomputed rotation stays on $\mathcal{Q}(\theta)$. Part (iii) reuses ARRW's support argument for a new purpose. Setting $X = G_Q$ starts the chain at any admissible rotation, whatever search produced that rotation. Part (iii) supplies the admissible state that Algorithm 1 presupposes.

The state of the chain. A sampler could store the rotation itself, keeping the matrix Q in the state and reusing it after every structural update. HARS does exactly this. KZ write the state as (θ, Q) and set $X = Q$ whenever the ESS step needs Gaussian coordinates. The choice is harmless there. The QR map γ does not depend on θ, and the stored rotation and its coordinates never disagree. On $\mathcal{Q}(\theta)$ the choice matters. The manifold $\mathcal{Q}(\theta)$ moves with θ. A stored Q lies on the old manifold and

[32] ARRW use their coordinates only for independent proposals. The QR map γ of KZ fails for the reason given in Remark 1(ii). Under γ, an elliptical perturbation of a single Gaussian column cannot restore the zeros, and the sign-only ESS never visits the manifold.

leaves the new manifold $\mathcal{Q}(\theta^\star)$ with probability one as soon as the structural draw moves. The zero restrictions fail after every structural update. I store the Gaussian coordinates instead. The state is the pair (θ, X), where X is an $n \times n$ Gaussian matrix, and the sampler recomputes the rotation from the state whenever it needs one,

$$Q \;=\; \gamma_z(X; \theta).$$

The next subsection establishes the validity of the construction. Proposition 1 derives the measure that the construction induces on the admissible set, Corollary 2 shows that the weighted draws recover conditional-Haar expectations, and Proposition 2 gives the column-wise elliptical slice sampling update that the sampler uses. Algorithm 1 collects the steps.

C.4 The Validity of HARS-Z

A structural update leaves X unchanged and changes only the rotation $\gamma_z(X; \theta)$. The construction projects the same Gaussian columns onto the null spaces of the new draw, and $\gamma_z(X; \theta^\star)$ lies on $\mathcal{Q}(\theta^\star)$ by Lemma 1(i). The zeros hold after every structural move by construction, and the admissibility check reduces to the sign and narrative restrictions. The rotation remains auxiliary, exactly as in KZ. Given (θ, X), the rotated absorbed system of (C.1) evaluated at $Q = \gamma_z(X; \theta)$ generates the data, and the conditional density $p(y \mid \theta, X)$ is well defined. The next definition states the posterior target of HARS-Z, and the corollary that follows establishes its validity.

Definition 1 (The restricted posterior target on (θ, X))**.** Let $\varphi(X)$ denote the standard Gaussian density of the $n \times n$ matrix X and define

$$\kappa(\theta) \;=\; \int \mathbf{1}\big\{SR\big(\theta, \gamma_z(X; \theta)\big) > 0\big\}\, \varphi(X)\, dX,$$

the Gaussian probability of the admissible set at θ. When $\kappa(\theta) > 0$, define the conditional distribution of X given θ,

$$\pi_X(dX \mid \theta, SR) \;:=\; \frac{\mathbf{1}\big\{SR\big(\theta, \gamma_z(X; \theta)\big) > 0\big\}\, \varphi(X)}{\kappa(\theta)}\, dX. \tag{C.8}$$

The restricted posterior target is

$$\pi(d\theta, dX \mid y, SR) \;=\; \pi(\theta \mid y)\, \pi_X(dX \mid \theta, SR)\, d\theta. \tag{C.9}$$

Definition 1 is KZ's Definition 1 in Gaussian coordinates, with $\kappa(\theta) > 0$ replacing $\mu(R(\theta)) > 0$. KZ's definition takes one fixed distribution, Haar measure on $O(n)$, and restricts it to the θ-dependent region $R(\theta)$. That structure is unavailable on rotations here. A distribution on $Q(\theta)$ assigns no mass to $Q(\theta')$ at any other draw θ', and no fixed distribution on rotations exists. The Gaussian distribution of X restores the structure. Equation (C.8) restricts the same $\varphi(X)$ to the admissible set at every draw, and θ enters only through the indicator and the map $\gamma_z(\cdot\,;\theta)$.

Corollary 1 (The Gaussian coordinates are auxiliary)**.** *Fix θ satisfying the conditions of Lemma 1 and write $Q = \gamma_z(X;\theta)$. For almost every X,*

$$p\big(y \mid \theta, X\big) = p\big(y \mid \theta, Q\big) = p\big(y \mid \theta\big).$$

If, in addition, $\kappa(\theta) > 0$, then drawing θ from its posterior and then drawing X from the conditional $\pi_X(\cdot \mid \theta, SR)$ in (C.8) produces exact draws from the joint target (C.9). Sampling X in a separate block leaves the marginal posterior of θ unchanged.

Proof. See Appendix D.3. □

Corollary 1 justifies the two-block structure of the sampler. The equalities say that the likelihood never involves X. The θ block runs the posterior sampler of BPSS exactly as in KZ. The last claim covers the rotation block. Drawing the rotation at fixed θ from the conditional (C.8) produces draws from the target (C.9) and leaves the marginal posterior of θ unchanged.

Posterior inference evaluates the impulse responses, functions of (θ, Q). The next definition states the distribution that posterior inference targets.

Definition 2 (The conditional-Haar target on (θ, Q))**.** Fix the conditions of Lemma 1 at every θ

with $\kappa(\theta) > 0$. The conditional-Haar target, restated from Section 4, is

$$\pi_v(d\theta, dQ \mid y, SR) \;=\; \pi(\theta \mid y)\, \frac{\mathbf{1}\{Q \in R(\theta)\}\, \mu\big(dQ \mid \mathcal{Q}(\theta)\big)}{\mu\big(R(\theta) \mid \mathcal{Q}(\theta)\big)}\, d\theta, \tag{6}$$

where $\mu(\cdot \mid \mathcal{Q}(\theta))$ denotes the conditional-Haar (uniform) measure on $\mathcal{Q}(\theta)$ and $R(\theta) = \{Q \in \mathcal{Q}(\theta) : SR(\theta, Q) > 0\}$ denotes the admissible set on the zero-restricted set.

Remark 2. The normalizer of (6) is positive whenever $\kappa(\theta) > 0$. The admissible set $R(\theta)$ is relatively open in $\mathcal{Q}(\theta)$, as the restrictions hold with strict inequalities. When $R(\theta)$ is empty, no draw satisfies the indicator in (C.8) and $\kappa(\theta) = 0$. The uniform measure on $\mathcal{Q}(\theta)$ assigns positive mass to every nonempty relatively open subset.

Definition 2 is the analogue of KZ's target (C.4), with Haar measure on $\mathcal{O}(n)$ replaced by the conditional-Haar measure on $\mathcal{Q}(\theta)$. The restrictions truncate only the conditional distribution of the rotation, as in Proposition 3 of KZ, and the marginal posterior of θ remains $\pi(\theta \mid y)$, consistent with Corollary 1. Definition 1 states the target on (θ, X), the coordinates in which the chain runs, and Definition 2 states the target on (θ, Q), the coordinates of the impulse responses.

Proposition 1 (The induced measure on $\mathcal{Q}(\theta)$)**.** *Fix θ satisfying the conditions of Lemma 1 with $\kappa(\theta) > 0$. The distribution over $\mathcal{Q}(\theta)$ that $\pi_X(\cdot \mid \theta, SR)$ induces through γ_z is the null-space construction measure $\mu_z(\cdot\,; \theta)$ of (C.7), restricted to the admissible set $R(\theta)$. The conditional of (6) given θ is the conditional-Haar measure restricted to $R(\theta)$, written $\pi_v(dQ \mid \theta, y, SR)$. The restricted construction measure has two properties.*

(a) When the zeros restrict a single column of Q and the ordering builds that column first, the restricted construction measure coincides with $\pi_v(\cdot \mid \theta, y, SR)$.

(b) Otherwise the restricted construction measure differs from $\pi_v(\cdot \mid \theta, y, SR)$ by the volume element $v(Q; \theta)$ of the construction (ARRW, Theorem 3), and weighting each retained draw by $v(Q; \theta)$ recovers $\pi_v(\cdot \mid \theta, y, SR)$.

Proof. See Appendix D.4. □

Proposition 1 states that the conditional $\pi_X(\cdot \mid \theta, SR)$ induces the construction measure on $Q(\theta)$ through γ_z, not the uniform (conditional-Haar) measure on $Q(\theta)$. When the zeros load on a single column and the ordering builds that column first, the two coincide and need no correction. With zeros on more than one column, I weight the retained rotations by the volume element and normalize the weights within each retained draw of θ.[33] Proposition 1(b) states the correction as a density, known only up to a constant that depends on θ. The next corollary removes the constant. Ratios of weighted expectations under the sampled measure reproduce conditional expectations under (6) exactly.

Corollary 2 (Conditional-Haar expectations from weighted draws)**.** *Fix θ satisfying the conditions of Lemma 1 with $\kappa(\theta) > 0$. Let $E_z[\,\cdot \mid \theta\,]$ denote the expectation under the restricted construction measure of Proposition 1, the distribution of the retained rotations at θ. For every function h integrable under $\pi_v(\cdot \mid \theta, y, SR)$,*

$$\int h(\theta, Q)\, \pi_v(dQ \mid \theta, y, SR) \;=\; \frac{E_z\big[v(Q;\theta)\, h(\theta, Q) \mid \theta\big]}{E_z\big[v(Q;\theta) \mid \theta\big]}. \tag{C.10}$$

Proof. See Appendix D.5. □

The rotation draw. With the target of Definition 1 in place, the rotation draw keeps θ fixed and applies the column-wise ESS update of KZ (Proposition 4) through γ_z. The next proposition establishes its validity on $Q(\theta)$.

Proposition 2 (Column-wise ESS on $Q(\theta)$)**.** *Fix θ satisfying the conditions of Lemma 1 with $\kappa(\theta) > 0$, and let $\pi_X(\cdot \mid \theta, SR)$ denote the conditional distribution of X given θ in (C.8). The*

[33] The weight is the part of the importance weight of ARRW's Algorithm 3 that survives at fixed θ. Their importance weight is the ratio of the target normal-generalized-normal density to the density of their Algorithm 2, a product of two volume elements. The element v_{f_h} converts between the structural and the orthogonal reduced-form parameterizations (their Proposition 1). The element $v_{(g\circ f_h)|Z}$ corrects their Algorithm 2 on the zero-restricted set (their Theorem 3). Only the second element arises here. The Metropolis step targets the structural posterior of θ directly, and the sampled and target distributions differ only in the conditional of Q given θ. The volume element $v(Q;\theta)$ of Proposition 1 is their $v_{(g\circ f_h)|Z}$ at fixed θ.

column-wise ESS update proposes

$$x_j(\vartheta) = x_j \cos\vartheta + \tau_j \sin\vartheta, \qquad \tau_j \sim \mathcal{N}(0, I_n),$$

where x_j denotes the j-th column of the current $X = [x_1, \ldots, x_n]$, and checks admissibility through γ_z. It has the following properties.

(a) Almost surely, it produces proposals whose rotations satisfy every zero restriction exactly along the entire ellipse, and the zeros never reject a proposal.

(b) Each column update terminates almost surely.

(c) It leaves $\pi_X(\cdot \mid \theta, SR)$ invariant.

Proof. See Appendix D.6. □

Lemma 1(ii) permits the column-wise ESS update on $Q(\theta)$. The zeros hold under every elliptical proposal by construction, and the slice check evaluates only sign and narrative admissibility. The angle-zero proposal is the current admissible state. The admissible angular set is nonempty, and the bracket-shrinking rule terminates (Proposition 2(b)).

The validity of Algorithm 1 step by step. The validity of HARS-Z follows step by step from KZ. The posterior draw in Step 1 is the sampler of BPSS, and its invariant distribution is the posterior target of θ. By Proposition 2, the rotation draw in Step 3 leaves the X-block conditional target (C.8) invariant. The retained rotations of Step 3 are serially dependent but are valid draws from the Q-block conditional target (KZ, Definition 1 and Proposition 4), extended to $Q(\theta)$ by Proposition 2. Their distribution at fixed θ is the induced measure of Proposition 1. The global rotation repair in Step 2b targets the same restricted construction measure through independent proposals. By KZ's Theorem 1, the posterior-draw repair in Step 2a redraws a completed parameter candidate and preserves the sign-restricted marginal target of $z_0 = (A_0, \Lambda)$. The sampler never uses the conditional shortcut for z_+, which would have led to the incorrect marginal target of z_0.[34] Because X does not enter the likelihood, by Corollary 1, none of the rotation rebuilds, rotation draws, or repair steps reweights θ. The retained pairs (θ, X) are draws from the target of Definition 1. Where the zeros load on more than one column, the volume weight of Proposition 1(b) restores the conditional $\pi_v(\cdot \mid \theta, y, SR)$.

C.5 Posterior Inference

Let θ_d denote the d-th retained draw of θ, $Q^{(d,r)} = \gamma_z(X^{(d,r)}; \theta_d)$ its r-th retained rotation, and $v_{d,r} = v(Q^{(d,r)}; \theta_d)$ the volume element at that rotation. The estimator of the posterior mean of a function h of the impulse responses is

$$\widehat{E}[h] = \frac{1}{D} \sum_{d=1}^{D} \frac{\sum_{r=1}^{N_{\text{ess}}} v_{d,r}\, h\big(\theta_d, Q^{(d,r)}\big)}{\sum_{r=1}^{N_{\text{ess}}} v_{d,r}}, \tag{C.11}$$

where D is the number of retained draws of θ. The reported quantiles apply the same weights, normalized within each retained draw. The numerator and the denominator of the inner ratio are sample averages of the numerator and the denominator on the right side of (C.10) at θ_d. The outer average runs over the posterior draws of θ, whose marginal the rotation block leaves unchanged by Corollary 1. The estimator is consistent for expectations under (6) as D and N_{ess} grow. Any θ-dependent rescaling of the volume element cancels in the inner ratio. In the single-column case of Proposition 1(a), the weights are constant within each draw and (C.11) reduces to the unweighted average of KZ.

[34] The natural alternative holds $z_0 = (A_0, \Lambda)$ from the stale draw fixed and redraws only $z_+ = A_+$ until the check passes. This targets the wrong distribution. Under the sign-restricted posterior, the marginal of z_0 has the admissibility weight $\omega_0(z_0; s_0, Q)$, the probability that a fresh Gibbs draw of A_+ completes the pair admissibly. Redrawing A_+ until success gives every z_0 a completion. It strips this factor and over-represents structural values whose admissible completions are rare (KZ, Theorem 1).

D Proofs for Section 4 and Appendix C

Throughout, the state convention of Section 4 applies. The chain stores (θ, X) and recomputes $Q = \gamma_z(X; \theta)$ whenever it needs one.

D.1 Proof of Remark 1

Proof of Remark 1. (i) Adding the zeros to the indicator SR forces $R(\theta) \subseteq \mathcal{Q}(\theta)$. Monotonicity gives $\mu(R(\theta)) \leq \mu(\mathcal{Q}(\theta)) = 0$. The condition $\mu(R(\theta)) > 0$ of KZ's Definition 1 fails at every θ with at least one zero restriction.

(ii) The map γ is the QR map that normalizes the R factor to a positive diagonal and returns the orthogonal factor. The rotation $\gamma(X)$ is Haar distributed on $\mathcal{O}(n)$ (KZ, Proposition 4). The equality $\mu(\mathcal{Q}(\theta)) = 0$ gives $\Pr\{\gamma(X) \in \mathcal{Q}(\theta)\} = 0$. The construction projects each Gaussian column only onto the orthogonal complement of the preceding columns, never onto the null spaces of $Z_j(\theta)$.

(iii) The rotation-side steps of KZ's Algorithm 1 fail by (ii). Only Steps 1 and 2a need a separate argument. Both redraw θ rather than Q.

Initialization. Each accepted column of the initialization, Algorithm 4 of Arias et al. (2026), is a projected Gaussian draw. A projected Gaussian draw lies in the null space of $Z_j(\theta)$ with probability zero, by (ii). Almost surely no initial admissible pair exists. The chain never occupies a state whose rotation satisfies the zeros.

Steps 1 and 2a. Both hold the retained rotation fixed and check completed draws. For a fixed unit column $q_j \neq 0$, the condition $Z_j(\theta)\, q_j = 0$ depends on θ only through A_0 and $\Phi_\ell(B)$. The condition is an exact alignment among functions of A_0^{-1} and $\Phi_\ell(B)$. The set of structural parameters at which the alignment holds is of Lebesgue measure zero, following the argument ARRW use for their set U (their Appendix A.3). Every completed draw and candidate comes from a proposal density and the Gibbs conditional density $p_+(A_+ \mid z_0, s_0)$. Draws with a density avoid the alignment set almost surely. When the MH step rejects and sets $z_0^\star = z_0$, the zeros already fail at the current first block, by the initialization argument.

Step 2b. Each independent Haar draw lies off $\mathcal{Q}(\theta^\star)$ almost surely, by (ii).

Step 3. The slice step starts from a current admissible point. No such point is ever available, by the initialization argument. Even from a state on $\mathcal{Q}(\theta)$, the update fails. Consider the ESS update of a column j that carries zero restrictions. Following KZ's Proposition 4, draw an auxiliary vector $\tau_j \sim \mathcal{N}(0, I_n)$ and consider the ellipse

$$x_j(\vartheta) = x_j \cos\vartheta + \tau_j \sin\vartheta, \qquad \vartheta \in [0, 2\pi].$$

Under γ, the proposed column is the projection of $x_j(\vartheta)$ onto the orthogonal complement of the preceding columns, normalized. Write P for that projection. The projection P is fixed along the ellipse. The preceding columns do not move when the ESS step updates column j. The normalization does not affect the zero condition. The zero restrictions hold at the proposal if and only if $Z_j(\theta)\, P\, x_j(\vartheta) = 0$. Linearity of P gives

$$Z_j(\theta)\, P\, x_j(\vartheta) \;=\; \cos\vartheta\; Z_j(\theta) P\, x_j \;+\; \sin\vartheta\; Z_j(\theta) P\, \tau_j.$$

The coefficient $Z_j(\theta)P\,x_j$ is zero at the current state. The current column satisfies its zeros and is proportional to Px_j. The coefficient $Z_j(\theta)P\,\tau_j$ is nonzero almost surely. The auxiliary vector is a fresh Gaussian draw, and $P\tau_j$ is a projected Gaussian column. A projected Gaussian column satisfies its zero restrictions with probability zero, by (ii). The zero restrictions hold on the ellipse only where $\sin\vartheta = 0$, at $\vartheta \in \{0, \pi\}$. At these angles the proposal is the current column up to sign. The update never reaches a new rotation on $\mathcal{Q}(\theta)$. The slice step retains only angles at which the admissibility indicator passes, and no angle off $\{0, \pi\}$ passes. □

D.2 Proof of Lemma 1

Proof of Lemma 1. (i) At step j the subspace $N_j(\theta)$ in (C.5) is, by construction, orthogonal to every preceding column and contained in $\mathrm{null}\big(Z_{\sigma(j)}(\theta)\big)$. The projected and normalized column $q_{\sigma(j)}$ therefore has unit norm, is orthogonal to its predecessors, and satisfies its zero restrictions exactly. The matrix $M_j(\theta)$ in (C.5) has $(j-1)+z_{\sigma(j)}$ rows. The dimension of $N_j(\theta)$ equals $n+1-j-z_{\sigma(j)} \geq 1$ if and only if $M_j(\theta)$ is of full row rank (ARRW). Full row rank holds by hypothesis at every orthonormal collection of preceding columns whose columns satisfy their zero restrictions. The construction completes with Q orthogonal. The projection of a Gaussian column is zero only on a null set. The rotation $\gamma_z(G;\theta)$ lies in $\mathcal{Q}(\theta)$ for almost every G.

(ii) Fix a step k. The subspace $N_k(\theta)$ has two ingredients, the preceding columns and the constraint matrix $Z_{\sigma(k)}(\theta)$. The constraint matrix is fixed at the given θ. The preceding columns come from steps $1,\ldots,k-1$, and step k adds g_k and nothing else. By induction, every construction step k depends on G only through $g_1,\ldots,g_k$. In particular, g_j is not among the inputs of steps $1,\ldots,j-1$ and is among the inputs of steps $j,\ldots,n$. Perturbing g_j leaves steps $1,\ldots,j-1$ untouched. The construction recomputes steps $j,\ldots,n$. Each recomputed column passes through the projection onto its own null space. The zeros hold after the perturbation.

(iii) Let $Q \in \mathcal{Q}(\theta)$ be admissible. Each of its columns already lies in its own subspace $N_j(\theta)$. Orthogonal projection acts as the identity on $N_j(\theta)$. Normalization preserves a unit vector. The construction reproduces Q column by column. The reproduction property is the projection argument ARRW use to establish that their Algorithm 2 can generate every rotation satisfying the zero restrictions. □

D.3 Proof of Corollary 1

Proof of Corollary 1. Fix θ and the regime path. Given (θ, X), the rotated absorbed system evaluated at $Q=\gamma_z(X;\theta)$ generates the data,

$$y_t' A_0 \Xi_{s_t} Q = x_t' A_+ \Xi_{s_t} Q + \tilde{\varepsilon}_t', \qquad E(\tilde{\varepsilon}_t \tilde{\varepsilon}_t' \mid s_t) = I_n,$$

where $\tilde{\varepsilon}_t' = \varepsilon_t' Q = \eta_t' \Xi_{s_t} Q$ collects the labeled shocks of Appendix C.1. The conditional density of y given (θ, X) is $p(y \mid \theta, Q)$ evaluated at $Q=\gamma_z(X;\theta)$. The evaluation gives the first equality of the corollary. By Lemma 1(i), $\gamma_z(X;\theta) \in \mathcal{Q}(\theta) \subseteq \mathcal{O}(n)$ for almost every X. KZ's Proposition 1 gives

$p(y \mid \theta, Q) = p(y \mid \theta)$ for every $Q \in \mathcal{O}(n)$. Evaluating at $Q = \gamma_z(X;\theta)$ yields the second equality of the corollary. The dependence of the map on θ does not affect the evaluation. The invariance $p(y \mid \theta, Q) = p(y \mid \theta)$ holds at each fixed θ for every $Q \in \mathcal{O}(n)$, and $\gamma_z(X;\theta)$ is one such rotation.

For the last claim, let $\kappa(\theta) > 0$. The conditional $\pi_X(\cdot \mid \theta, SR)$ of (C.8) is then well defined. The conditional is the Gaussian distribution of X restricted to the admissible set $\{X : SR(\theta, \gamma_z(X;\theta)) > 0\}$. Specify the prior of the pair as $p(\theta)$ for θ and $\pi_X(\cdot \mid \theta, SR)$ for X given θ. The joint posterior under this prior satisfies

$$\begin{aligned}
\pi(\theta, X \mid y) &\propto p(y \mid \theta, X)\, p(\theta)\, \pi_X(X \mid \theta, SR) \\
&= p(y \mid \theta)\, p(\theta)\, \pi_X(X \mid \theta, SR) \\
&\propto \pi(\theta \mid y)\, \pi_X(X \mid \theta, SR).
\end{aligned}$$

The first line is Bayes' rule. The second line applies the first part. The third line collects $\pi(\theta \mid y) \propto p(y \mid \theta)\, p(\theta)$. The conditional distribution of X is the Gaussian distribution restricted to the admissible set. The posterior density of θ remains $\pi(\theta \mid y)$. The joint posterior under this prior is exactly the target (C.9). Drawing θ from its posterior and then drawing X from $\pi_X(\cdot \mid \theta, SR)$ produces exact draws from the target. The argument is KZ's Proposition 3 with Haar measure restricted to $R(\theta)$ replaced by the Gaussian distribution of X restricted to the admissible set. □

D.4 Proof of Proposition 1

Proof of Proposition 1. Two preliminary steps characterize the induced distribution. Parts (a) and (b) then follow in turn.

Step 1: π_X induces the construction measure restricted to the admissible set. The argument follows the image claim of KZ's Proposition 4, with the QR map γ replaced by γ_z and Haar measure replaced by the construction measure $\mu_z(\cdot\,;\theta)$. Recall the conditional (C.8),

$$\pi_X(dX \mid \theta, SR) = \frac{\mathbf{1}\big\{SR\big(\theta, \gamma_z(X;\theta)\big) > 0\big\}\, \varphi(X)}{\kappa(\theta)}\, dX.$$

Without the indicator the conditional reduces to the Gaussian density φ. The matrix X then has the same distribution as the Gaussian matrix G in (C.7). The relation $\gamma_z(X;\theta) \sim \mu_z(\cdot\,;\theta)$ holds by definition.

Now restore the indicator. The indicator in (C.8) does not test X directly. The indicator builds the rotation $\gamma_z(X;\theta)$ and tests whether that rotation satisfies the sign and narrative restrictions. The input X passes exactly when its rotation lies in the admissible set $R(\theta) = \{Q \in \mathcal{O}(n) : SR(\theta, Q) > 0\}$ of Section 4,

$$\bigl\{X : SR\bigl(\theta, \gamma_z(X;\theta)\bigr) > 0\bigr\} \;=\; \bigl\{X : \gamma_z(X;\theta) \in R(\theta)\bigr\}. \tag{D.1}$$

Take any set C of rotations. Under the conditional $\pi_X(\cdot \mid \theta, SR)$, the probability that the rotation lands in C is

$$\begin{aligned}
\Pr\bigl\{\gamma_z(X;\theta) \in C \,\big|\, SR\bigl(\theta, \gamma_z(X;\theta)\bigr) > 0\bigr\} \;&=\; \Pr\bigl\{\gamma_z(X;\theta) \in C \,\big|\, \gamma_z(X;\theta) \in R(\theta)\bigr\} \\
&=\; \frac{\Pr\bigl\{\gamma_z(X;\theta) \in C \cap R(\theta)\bigr\}}{\Pr\bigl\{\gamma_z(X;\theta) \in R(\theta)\bigr\}} \\
&=\; \frac{\mu_z\bigl(C \cap R(\theta);\theta\bigr)}{\mu_z\bigl(R(\theta);\theta\bigr)}.
\end{aligned}$$

The first line rewrites the conditioning event. Equation (D.1) shows the two events are the same set of X. The second line applies the definition of conditional probability. The probabilities on the second line are unconditional. On the second line $X \sim \varphi$, and the relation $\gamma_z(X;\theta) \sim \mu_z(\cdot\,;\theta)$ from the start of the proof evaluates each probability as a μ_z-value on the third line.

The set C is arbitrary, and the values on every set determine a distribution. The induced distribution is

$$\frac{\mathbf{1}\bigl\{SR(\theta, Q) > 0\bigr\}\; \mu_z(dQ;\theta)}{\mu_z\bigl(R(\theta);\theta\bigr)}. \tag{D.2}$$

The measure (D.2) is the construction measure restricted to the admissible set $R(\theta)$. The restricted construction measure is the counterpart of KZ's Haar measure restricted to $R(\theta)$.

The normalizer of (D.2) coincides with the normalizer $\kappa(\theta)$ of (C.8), $\kappa(\theta) = \mu_z(R(\theta);\theta)$. Both equal the probability of one event, that the rotation is admissible. The normalizer $\kappa(\theta)$ computes this probability in X. The normalizer $\mu_z(R(\theta);\theta)$ computes it on $Q(\theta)$.

Step 2: relation to conditional-Haar. The comparison proceeds column by column. The construction builds the rotation column by column. The construction generates $q_{\sigma(1)},\ldots,q_{\sigma(n)}$ in the order σ. Each step projects a fresh Gaussian column onto the null space $N_j(\theta)$ that the zero restrictions and the previously built columns determine, then normalizes. The projected Gaussian column favors no direction within $N_j(\theta)$. Conditional on the preceding columns,

$$q_{\sigma(j)} \,\big|\, q_{\sigma(1)},\ldots,q_{\sigma(j-1)} \;\sim\; \text{uniform on the unit sphere of } N_j(\theta).$$

The uniform law of the projected column is the second property of the construction in Appendix C.3. Write $\mu_j(\cdot \mid q_{\sigma(1)},\ldots,q_{\sigma(j-1)};\theta)$ for this conditional distribution, the uniform distribution on the unit sphere of $N_j(\theta)$. The chain rule of probability writes the joint law as the product of these conditionals,

$$\mu_z(dQ;\theta) \;=\; \prod_{j=1}^{n} \mu_j\big(dq_{\sigma(j)} \mid q_{\sigma(1)},\ldots,q_{\sigma(j-1)};\theta\big). \tag{D.3}$$

Equation (C.7) defines $\mu_z(\cdot\,;\theta)$ only as the distribution of $\gamma_z(G;\theta)$. The product (D.3) makes the measure explicit. Parts (a) and (b) below use (D.3).

Part (a): the zeros load on a single column. The ordering builds the restricted column first. The first null space $N_1(\theta)$ then equals the null space of the zero restrictions. Every later null space $N_j(\theta)$, $j \geq 2$, is the orthogonal complement of the preceding columns and carries no further constraint.

The comparison involves two distinct measures on $Q(\theta)$. The construction measure $\mu_z(\cdot\,;\theta)$ is the law the sampler induces. The measure $\mu(\cdot \mid Q(\theta))$ is the conditional-Haar (uniform) measure on $Q(\theta)$ and serves as the reference measure of the target. The subscript z denotes the measure that the zero-restricted construction induces. The conditioning on $Q(\theta)$ denotes Haar measure restricted to $Q(\theta)$. Factor both measures at the first column,

$$\mu_z(dQ;\theta) \;=\; \mu_1\big(dq_{\sigma(1)};\theta\big)\,\mu_z\big(dq_{\sigma(2)},\ldots,dq_{\sigma(n)} \mid q_{\sigma(1)};\theta\big), \tag{D.4}$$

$$\mu\big(dQ \mid \mathcal{Q}(\theta)\big) \;=\; \mu\big(dq_{\sigma(1)} \mid \mathcal{Q}(\theta)\big)\, \mu\big(dq_{\sigma(2)}, \ldots, dq_{\sigma(n)} \mid q_{\sigma(1)}, \mathcal{Q}(\theta)\big), \tag{D.5}$$

and compare factor by factor.

The first factors of (D.4) and (D.5) coincide. The first factor of (D.4) is $\mu_1(\cdot\,;\theta)$, the uniform distribution on the unit sphere of $N_1(\theta)$, by the second property of the construction. The first factor of (D.5) is uniform on the same sphere, by the following symmetry argument. Take any orthogonal matrix R that rotates $N_1(\theta)$ within itself and fixes its orthogonal complement. The map $Q \mapsto RQ$ moves the first column along the sphere and leaves every zero restriction intact. The map $Q \mapsto RQ$ therefore sends $\mathcal{Q}(\theta)$ onto itself. Haar measure is invariant under this map. Conditional-Haar inherits the invariance. Every rotation of $N_1(\theta)$ leaves the marginal distribution of $q_{\sigma(1)}$ unchanged. The only probability distribution on a sphere that every rotation leaves unchanged is the uniform distribution. Both first factors are the same uniform law on the same sphere.

The second factors of (D.4) and (D.5) coincide as well. Fix $q_{\sigma(1)}$. No zero restriction remains after the first column. Every later null space is the orthogonal complement of the preceding columns inside $q_{\sigma(1)}^{\perp}$. Each remaining step projects a fresh Gaussian column onto that complement and normalizes. A projected standard Gaussian column is standard Gaussian within the complement. The remaining steps therefore run the QR map γ of KZ on $q_{\sigma(1)}^{\perp}$. The rotation $\gamma(X)$ of a Gaussian matrix is Haar distributed, by the Gaussian-to-Haar construction of Rubio-Ramírez et al. (2010) in KZ's Proposition 4. The second factor of (D.4) is the uniform distribution over the orthonormal completions of $q_{\sigma(1)}$.

The second factor of (D.5) is the same uniform distribution. Apply the symmetry argument of the first factor to rotations that fix $q_{\sigma(1)}$ and rotate its complement. Every such rotation preserves $\mathcal{Q}(\theta)$ and preserves conditional-Haar. The conditional law of the completions is rotation invariant. The uniform distribution is the only rotation-invariant law. ARRW report constant importance weights for their Algorithm 3 without zero restrictions, consistent with the equality of the two factors.

The factorizations (D.4) and (D.5) coincide factor by factor. The two measures are equal,

$$\mu_z(\cdot\,;\theta) \;=\; \mu\big(\cdot \mid Q(\theta)\big).$$

The restrictions of equal measures to $R(\theta)$ are equal. The volume element is constant. This proves part (a).

Part (b): the zeros load on two or more columns. The construction still draws each column uniformly on the unit sphere of its null space, as in (D.3). The measure $\mu_z(\cdot\,;\theta)$ is the law of the construction that ARRW's Theorem 3 describes. Their change-of-variable theorem for maps into lower-dimensional smooth manifolds gives $\mu_z(\cdot\,;\theta)$ a density proportional to $1/v(Q;\theta)$ with respect to the uniform (volume) measure on $Q(\theta)$. The factor $v(Q;\theta)$ is the volume element of the map implied by their Algorithm 2 at fixed θ. The ratio of the two densities gives

$$\frac{d\,\mu\big(\cdot \mid Q(\theta)\big)}{d\,\mu_z(\cdot\,;\theta)}(Q) \;\propto\; v(Q;\theta).$$

The density relation holds on all of $Q(\theta)$ and hence on $R(\theta)$. Weighting each retained draw by $v(Q;\theta)$ recovers the conditional-Haar measure restricted to $R(\theta)$, the conditional $\pi_v(\cdot \mid \theta, y, SR)$ of (6). The weighting is necessary. As ARRW document, the distribution that their Algorithm 2 implies is not invariant to a reordering of the shocks. This proves part (b) and the proposition. □

D.5 Proof of Corollary 2

Proof of Corollary 2. Write $\tilde{\mu}_z(dQ \mid \theta)$ for the restricted construction measure (D.2). By Proposition 1(b), the conditional of (6) has a density proportional to $v(Q;\theta)$,

$$\pi_v(dQ \mid \theta, y, SR) \;=\; c(\theta)\, v(Q;\theta)\, \tilde{\mu}_z(dQ \mid \theta),$$

with a constant $c(\theta) > 0$. For every h integrable under the conditional,

$$\int h(\theta, Q)\, \pi_v(dQ \mid \theta, y, SR) \;=\; c(\theta) \int h(\theta, Q)\, v(Q; \theta)\, \tilde{\mu}_z(dQ \mid \theta) \;=\; c(\theta)\, E_z\big[v(Q; \theta)\, h(\theta, Q) \mid \theta\big].$$

The choice $h = 1$ makes the left side one and pins down the constant, $c(\theta) = 1/E_z[v(Q; \theta) \mid \theta]$. Substituting the constant into the display gives (C.10). □

D.6 Proof of Proposition 2

Proof of Proposition 2. We prove the three parts in turn.

Part (a): the zeros never reject a proposal. Fix a column j, and let $X^\star(\vartheta)$ denote the current X with column j replaced by $x_j(\vartheta)$. The sampler evaluates admissibility only through the candidate rotation $\gamma_z\big(X^\star(\vartheta); \theta\big)$. By Lemma 1(i), every candidate satisfies the zero restrictions exactly wherever the construction is defined. The construction fails only where some projected column in (C.6) equals the zero vector, an exceptional set of inputs with Gaussian probability zero.

Lemma 1(i) concerns a fresh Gaussian input. The candidates along the ellipse are not fresh Gaussian inputs. A Fubini argument extends the statement to the ellipse. Fix an angle with $\sin\vartheta \neq 0$. Given the current state, the candidate column $x_j(\vartheta) = x_j \cos\vartheta + \tau_j \sin\vartheta$ is a full-support Gaussian draw, and the remaining columns arise from draws with densities. The candidate matrix $X^\star(\vartheta)$ lies in the exceptional set with probability zero. By Fubini's theorem over the uniformly drawn angle, with nonnegative integrands, the exceptional angles form a null set almost surely. The zeros hold along the entire ellipse off this null set, and the zeros never reject a proposal. The slice step retains only angles at which the sign and narrative indicator holds, exactly as in KZ's HARS.

Part (b): almost-sure termination. We argue in three steps.

Step 1: the angle $\vartheta = 0$ is admissible. The bracket of the slice search always contains $\vartheta = 0$. At $\vartheta = 0$ the candidate matrix is the current X. By Lemma 1(iii), γ_z maps the current X to the current rotation. The chain retains only admissible states, and the current rotation is admissible.

Step 2: $\vartheta \mapsto \gamma_z\big(X^\star(\vartheta); \theta\big)$ is continuous at $\vartheta = 0$ almost surely. Each construction step depends on the stacked matrix $M_j(\theta)$ in (C.5), abbreviated M_j, only through the orthogonal projector P_j onto its null space $N_j(\theta)$. The projector P_j does not depend on the choice of orthonormal basis. Wherever M_j has full row rank,

$$P_j \;=\; I_n - M_j'\bigl(M_j M_j'\bigr)^{-1} M_j, \tag{D.6}$$

a continuous function of M_j. The proposition fixes θ under the maintained conditions of Lemma 1. Those conditions give M_j full row rank at every orthonormal collection of preceding columns whose columns satisfy their zero restrictions. Along the ellipse, the construction steps $1, \ldots, j-1$ do not move, and only the construction steps from j onward vary, by Lemma 1(ii). Off the exceptional set of part (a), each construction step is the composition of $M_j \mapsto P_j$ through (D.6) and the normalization in (C.6). Both pieces are continuous at almost every angle. The composition of the n steps is continuous at $\vartheta = 0$ almost surely.

Step 3: the bracket-shrinking rule terminates. The sign and narrative restrictions are strict inequalities. The admissible set of rotations is open. By Steps 1 and 2, the admissible angular set contains an open neighborhood of $\vartheta = 0$ and is nonempty. The bracket-shrinking rule reaches the admissible angular set in finitely many steps almost surely. Steps 1 and 3 make explicit the nonemptiness assertion in the proof of KZ's Proposition 4. That assertion suffices in their setting. The QR map γ is continuous at every full-rank X. Step 2 supplies the analogous continuity for γ_z, the only new ingredient.

Part (c): invariance. The target (C.8) is a standard Gaussian density multiplied by a nonnegative likelihood factor, here the admissibility indicator. The Gaussian density factorizes across columns, $\varphi(X) \propto \exp\bigl(-\frac{1}{2}\sum_{j=1}^{n}\|x_j\|^2\bigr)$, and the full conditional of a single column is

$$p(x_j \mid x_{-j}, \theta, SR) \;\propto\; \mathbf{1}\bigl\{SR\bigl(\theta, \gamma_z(X;\theta)\bigr) > 0\bigr\} \exp\bigl(-\tfrac{1}{2}\|x_j\|^2\bigr),$$

a Gaussian density times an indicator, as in KZ's Proposition 4. Elliptical slice sampling is exact for targets of this form (Murray et al., 2010). Part (a) reduces the indicator along every ellipse to a sign-and-narrative indicator. Part (b) supplies the almost-sure termination. The argument of KZ's Proposition 4 applies. Each column update samples from the admissible angular set and satisfies detailed balance for the full conditional $p(x_j \mid x_{-j}, \theta, SR)$. The systematic scan over the columns composes invariant full-conditional updates. The scan leaves $\pi_X(\cdot \mid \theta, SR)$ invariant. The retained states are serially dependent draws from $\pi_X(\cdot \mid \theta, SR)$. This proves the proposition. $\square$

E Estimation Diagnostics

The baseline estimates come from one chain of 6,010 structural sweeps after 1,000 burn-in iterations, retaining $N_{\mathrm{ess}} = 5$ rotations per sweep, 30,000 rotations in total. A second chain with a different random seed and the same settings serves as the convergence check. The repair budgets are $N_\theta = 300$ and $N_Q = 200$.

Sampler convergence. The two chains agree on 23 of the 24 band verdicts of Section 5.1. The verdicts are the 68% and 90% bands of the two shocks' GDP and deflator responses at twelve, twenty-four, and thirty-six months. The one exception is the 90% band of the RFBS deflator response at thirty-six months. Its upper bound is +0.01 percentage points in the first chain and −0.04 in the second, and the text reports the first chain's verdict. The regime variances mix slowly. The two monitored series, $\lambda_{\mathrm{SOMA},5}$ and $\lambda_{\mathrm{GDP},4}$, have lag-one autocorrelations of 0.99 and 0.98 across sweeps, and their effective sample sizes are 5 and 18 in the first chain and 6 and 17 in the second. The variance ratios of the two chains agree to within 5 percent for every entry of Table E.1, and the impulse response medians agree to within 0.04 percentage points.

The volume weights stay near uniform, with a within-block weight ESS share of 0.997 in the first chain and 0.996 in the second. The largest volume element is 3.7 times the median in the first chain and 5.2 times in the second. The Metropolis acceptance rate is 64% in the first chain and 63% in the second. The chains discard 36% and 30% of the proposed transitions, the stale rotations that the joint redraw does not repair within the budgets.

Table E.1 reports the estimated variance ratios $\omega_{i,s} = \lambda_{i,s}/\lambda_{i,1}$ over the pooled draws of the two chains.

The likelihood identifies the level of the ON RRP ratio variances only weakly. The 68% bands of that equation span a factor of about three in every regime, the widest in the table. The two chains place the medians within 5 percent of each other. The reported impulse responses and decompositions agree across the two chains.

Table E.1: Estimated variance ratios by regime, $\omega_{i,s} = \lambda_{i,s}/\lambda_{i,1}$.

Equation	Regime 2	Regime 3	Regime 4	Regime 5
FFR	0.55 [0.32, 0.90]	32.45 [24.50, 49.86]	5.70 [4.44, 8.79]	29.42 [21.90, 45.34]
SOMA	0.21 [0.18, 0.25]	0.07 [0.04, 0.10]	2.87 [2.79, 2.93]	0.05 [0.03, 0.08]
log(R + RRP)	0.62 [0.60, 0.63]	0.57 [0.56, 0.59]	0.68 [0.65, 0.71]	0.51 [0.47, 0.55]
ON RRP ratio	13.13 [8.89, 24.98]	34.42 [23.76, 64.88]	48.27 [33.25, 90.06]	53.02 [36.53, 101.11]
GDP	0.58 [0.56, 0.63]	0.28 [0.26, 0.30]	1.43 [1.41, 1.47]	0.45 [0.34, 0.57]
GDPDEF	1.63 [1.59, 1.69]	1.14 [1.10, 1.19]	1.29 [1.21, 1.34]	2.49 [2.05, 3.25]
Term spread	0.85 [0.81, 0.88]	0.64 [0.58, 0.67]	0.69 [0.67, 0.73]	0.99 [0.94, 1.04]

Notes: Entries are posterior medians with 68% credible bands in brackets over the pooled draws of the two chains, weighted by the volume weights. Rows index the structural equations, labeled by the variable on which each equation is normalized. Regime 1 equals one by normalization. Regime dates follow Table 1.

F Impulse Responses to the Demand, Supply, and MP Shocks

Figure F.1 reports the impulse responses to the demand, supply, and MP shocks under the baseline identification.

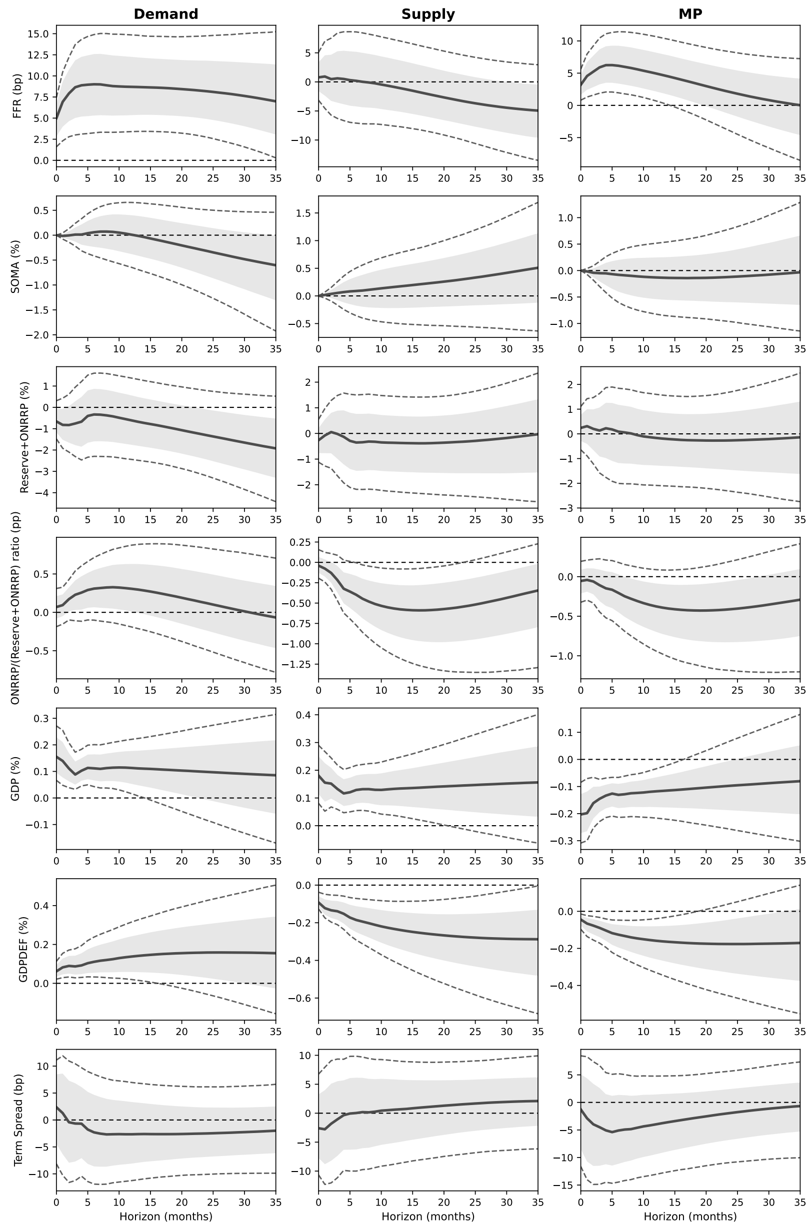
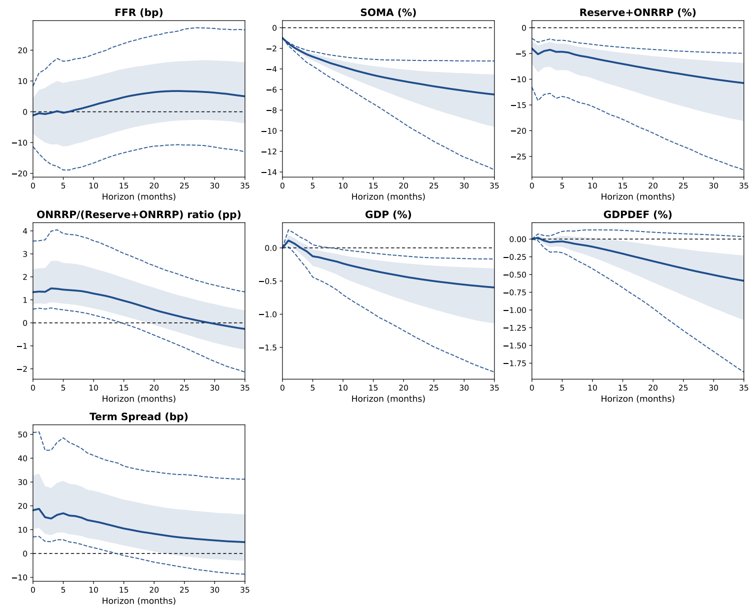


Figure F.1: Impulse responses to the demand, supply, and MP shocks.

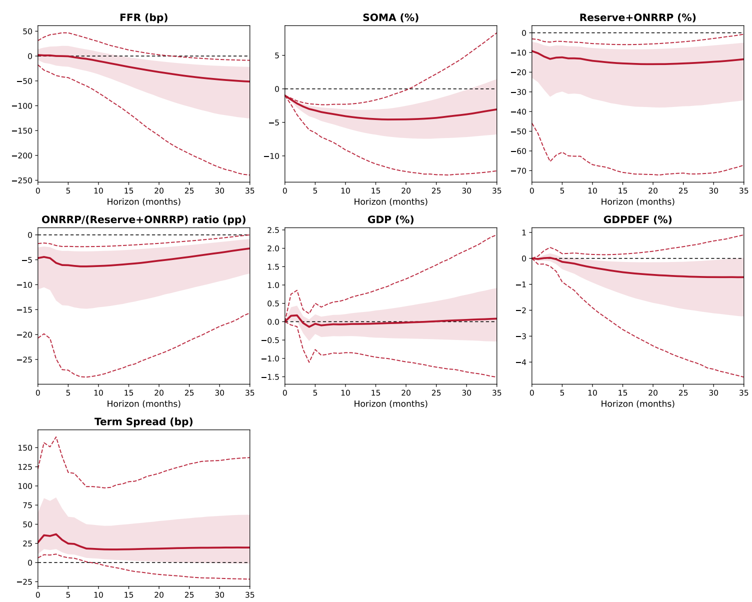
Note. Columns report the demand, supply, and MP shocks, and rows report the seven variables with their units. Each shock is scaled to one standard deviation. Under Assumption 2 the SOMA impact response to all three shocks is zero, and the one-percent-SOMA normalization of the balance-sheet figures does not apply. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws.

G Robustness of the Balance-Sheet Impulse Responses

Figures G.1 through G.12 report the robustness specifications of Section 5.2.



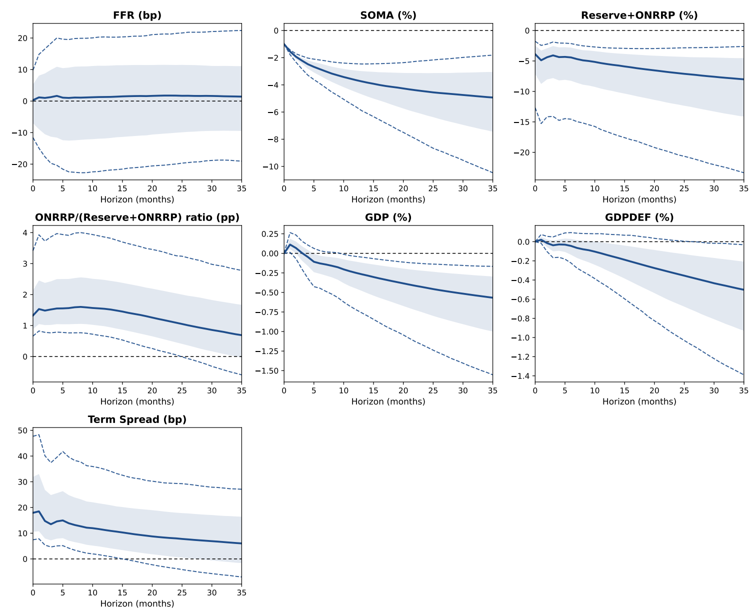
(**a**) RFBS



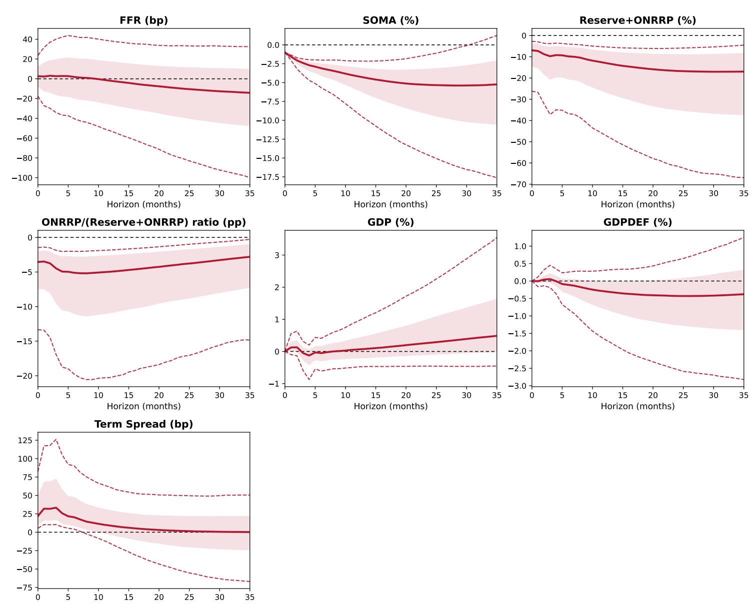
(**b**) OFBS

Figure G.1: Robustness with the constant at \$100 billion.

Note. Panel (a) reports the RFBS shock and panel (b) the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings. Each panel shows its shock in the contractionary direction, and panel (b) is the direction of the QT2 drain. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. The ON RRP ratio is $(\text{ON RRP} + c)/(\text{Reserves} + \text{ON RRP} + c)$ with $c = \$100$ billion in place of the baseline \$250 billion. Regimes and all sign, zero, and narrative restrictions are unchanged from the baseline.



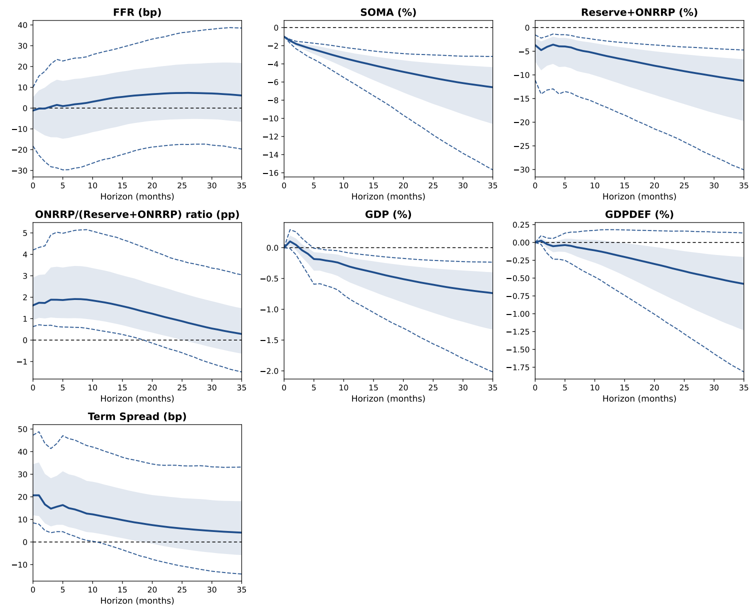
(a) RFBS



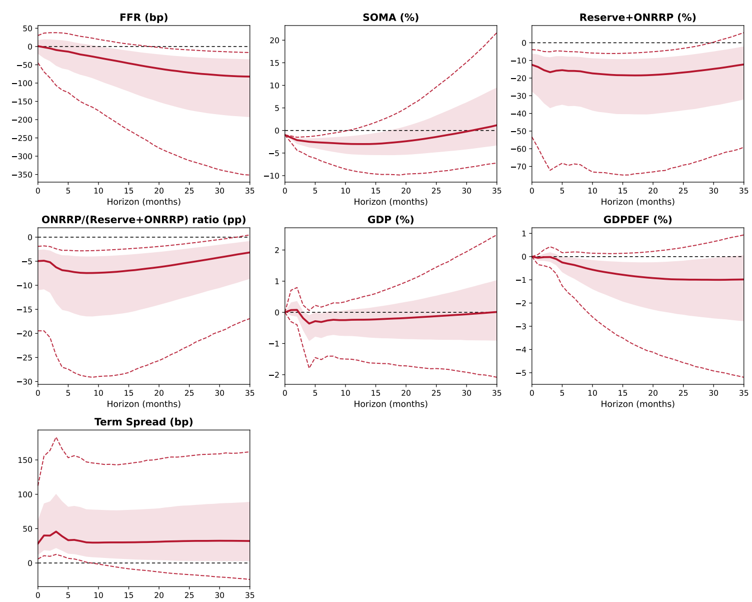
(b) OFBS

Figure G.2: Robustness with the constant at \$500 billion.

Note. Panel (a) reports the RFBS shock and panel (b) the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings. Each panel shows its shock in the contractionary direction, and panel (b) is the direction of the QT2 drain. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. The ON RRP ratio is $(\text{ON RRP} + c)/(\text{Reserves} + \text{ON RRP} + c)$ with $c = \$500$ billion in place of the baseline \$250 billion. Regimes and all sign, zero, and narrative restrictions are unchanged from the baseline.



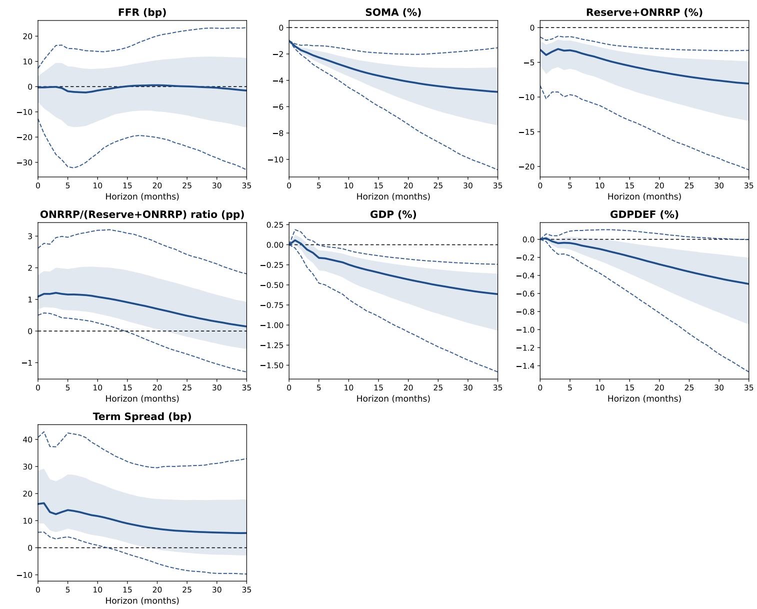
(a) RFBS



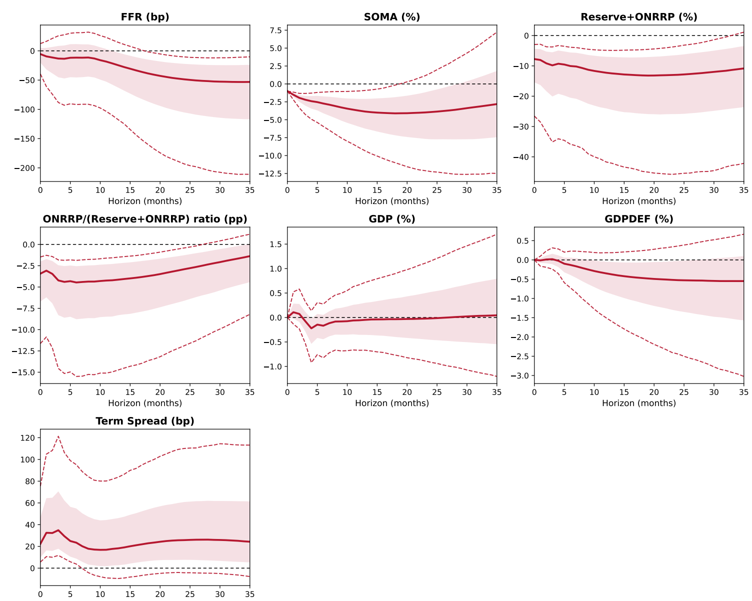
(b) OFBS

Figure G.3: Robustness with three variance regimes, 2015 and 2022 breaks.

Note. Panel (a) reports the RFBS shock and panel (b) the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings. Each panel shows its shock in the contractionary direction, and panel (b) is the direction of the QT2 drain. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. The model uses three variance regimes with breaks at 2015:M7 and 2022:M3. All sign, zero, and narrative restrictions are unchanged from the baseline.



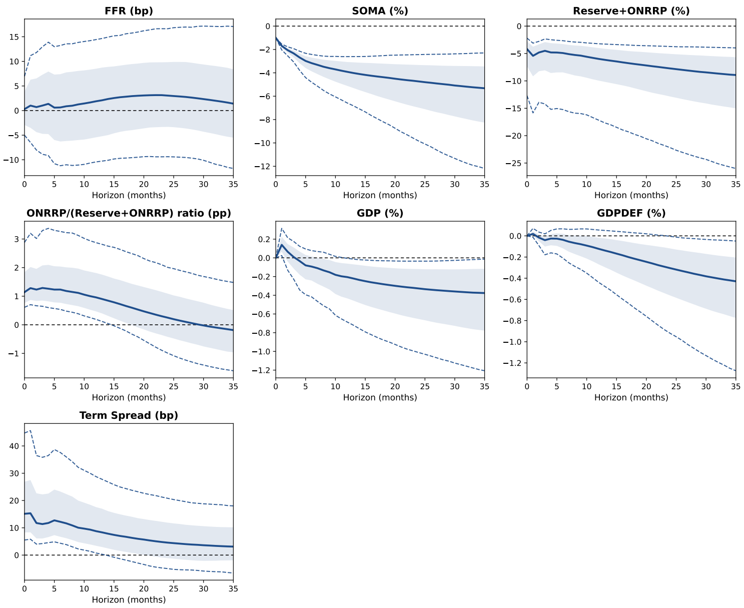
(**a**) RFBS



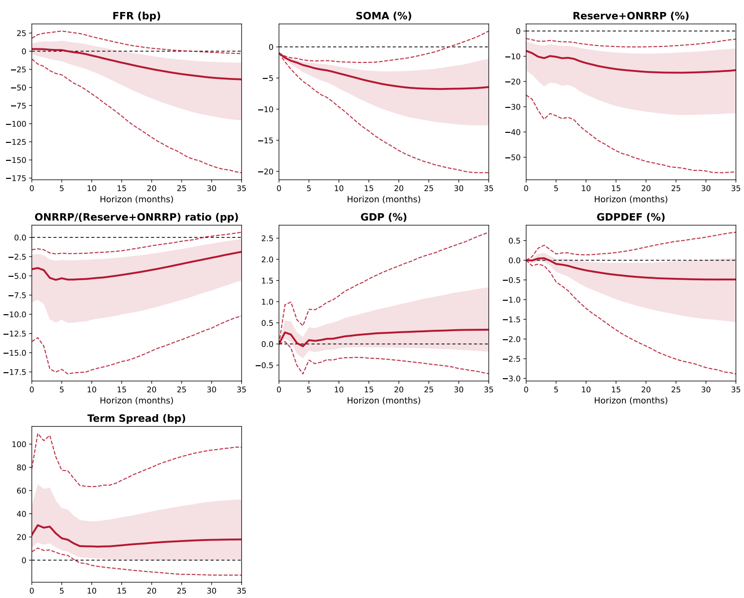
(**b**) OFBS

Figure G.4: Robustness with three variance regimes, 2011 and 2020 breaks.

Note. Panel (a) reports the RFBS shock and panel (b) the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings. Each panel shows its shock in the contractionary direction, and panel (b) is the direction of the QT2 drain. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. The model uses three variance regimes with breaks at 2011:M1 and 2020:M4. All sign, zero, and narrative restrictions are unchanged from the baseline.



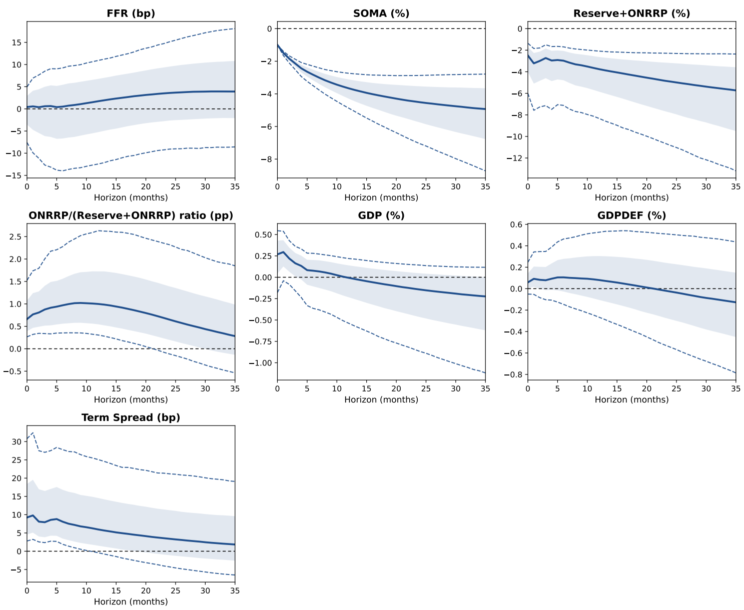
(a) RFBS



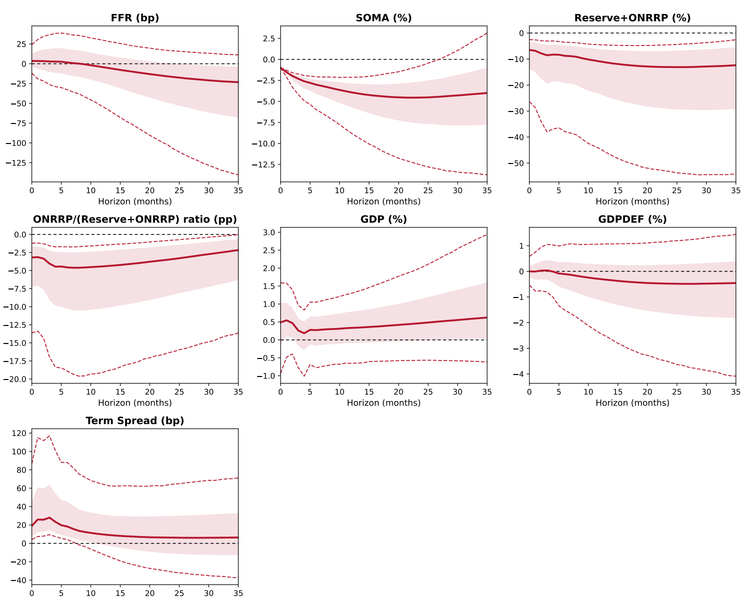
(b) OFBS

Figure G.5: Robustness with Gaussian errors.

Note. Panel (a) reports the RFBS shock and panel (b) the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings. Each panel shows its shock in the contractionary direction, and panel (b) is the direction of the QT2 drain. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. The structural shocks are Gaussian, and the latent Student-t scales of the baseline are removed. Regimes and all sign, zero, and narrative restrictions are unchanged from the baseline.



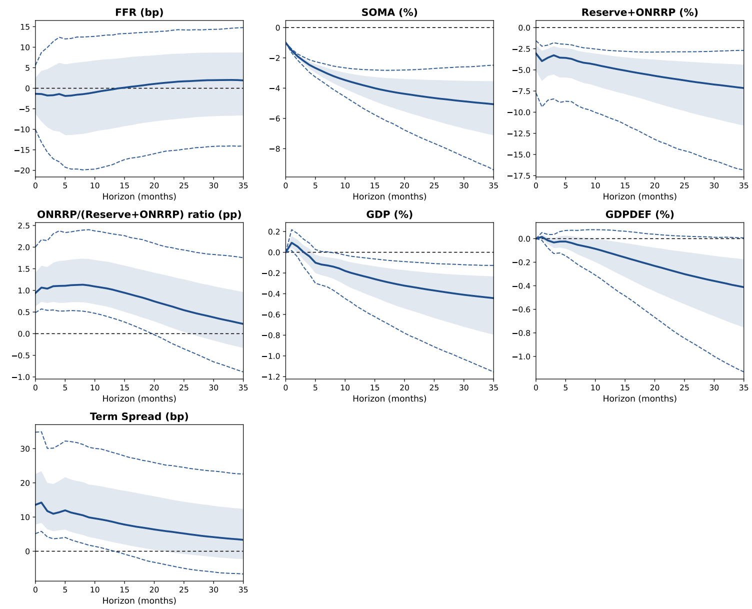
(a) RFBS



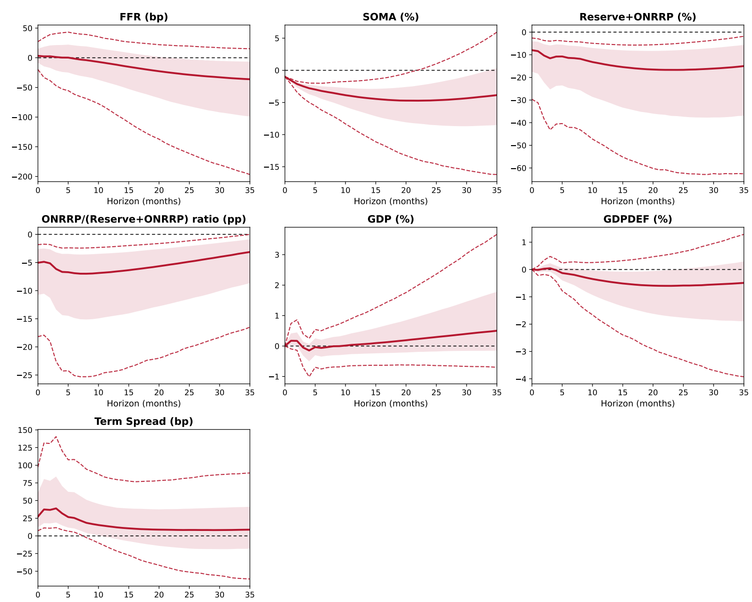
(b) OFBS

Figure G.6: Robustness with Assumption 1 removed.

Note. Panel (a) reports the RFBS shock and panel (b) the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings. Each panel shows its shock in the contractionary direction, and panel (b) is the direction of the QT2 drain. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. The GDP and GDP-deflator zeros of Assumption 1 are removed, and the three SOMA zeros of Assumption 2 remain. Sign and narrative restrictions are unchanged from the baseline.



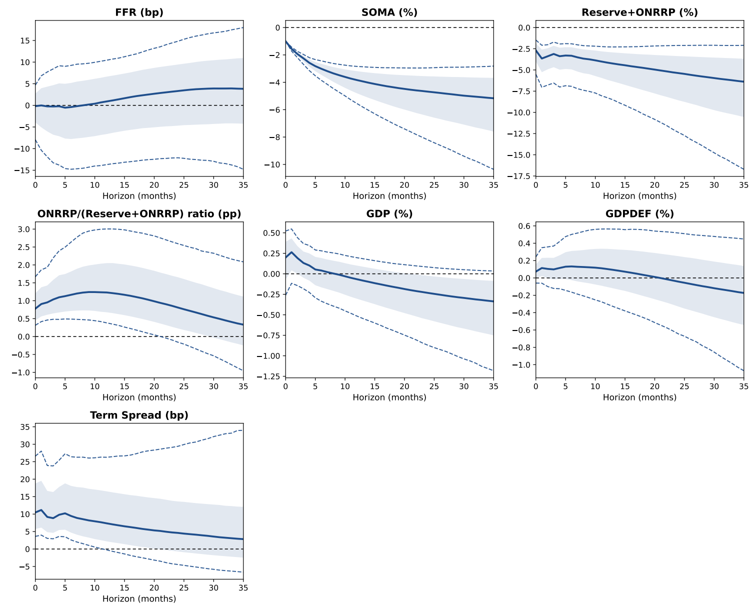
(a) RFBS



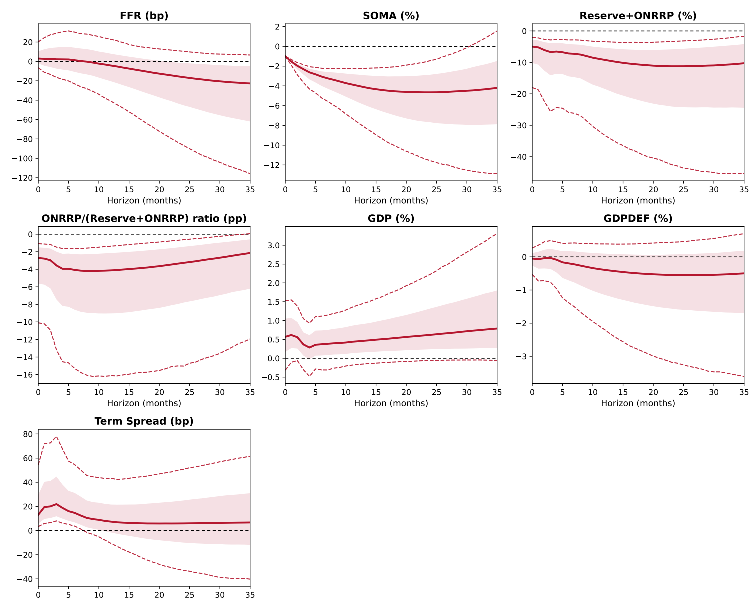
(b) OFBS

Figure G.7: Robustness with Assumption 2 removed.

Note. Panel (a) reports the RFBS shock and panel (b) the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings. Each panel shows its shock in the contractionary direction, and panel (b) is the direction of the QT2 drain. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. The SOMA zeros of Assumption 2 are removed, and the four GDP and GDP-deflator zeros of Assumption 1 remain. Sign and narrative restrictions are unchanged from the baseline.



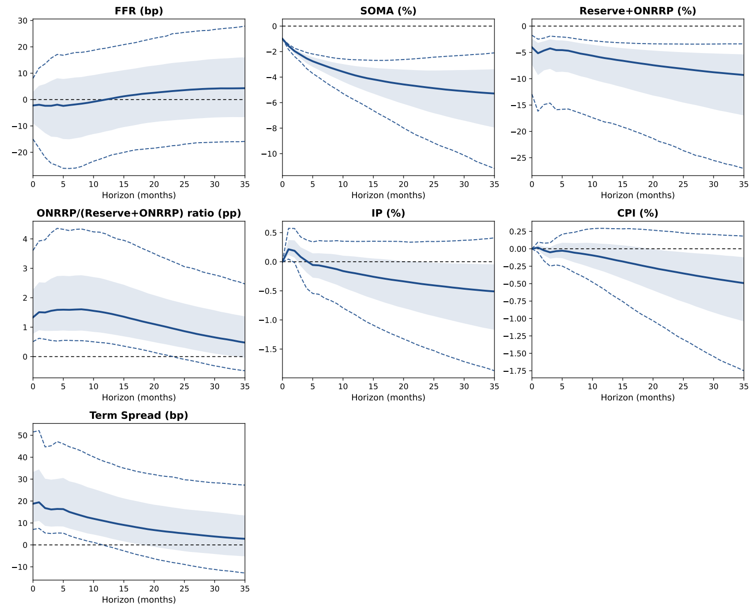
(a) RFBS



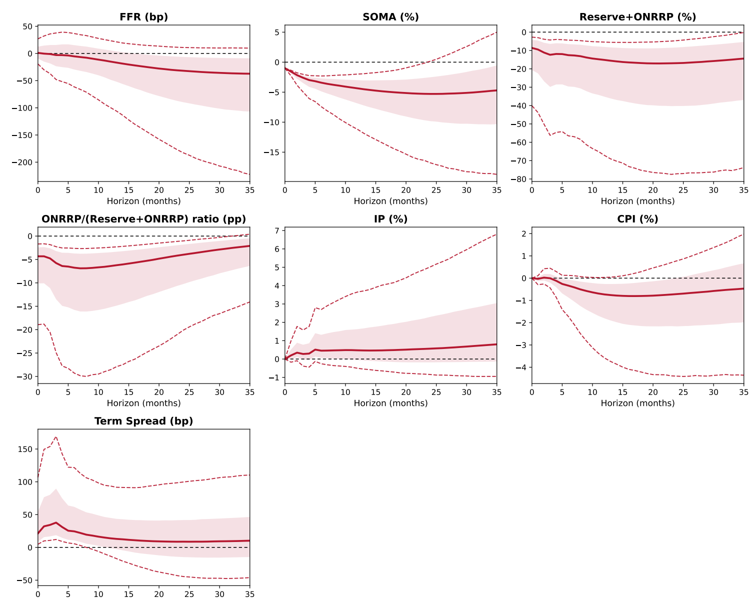
(b) OFBS

Figure G.8: Robustness with the zero restrictions removed.

Note. Panel (a) reports the RFBS shock and panel (b) the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings. Each panel shows its shock in the contractionary direction, and panel (b) is the direction of the QT2 drain. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. All zero restrictions of Assumptions 1 and 2 are removed. Sign and narrative restrictions and the variance regimes are unchanged from the baseline.



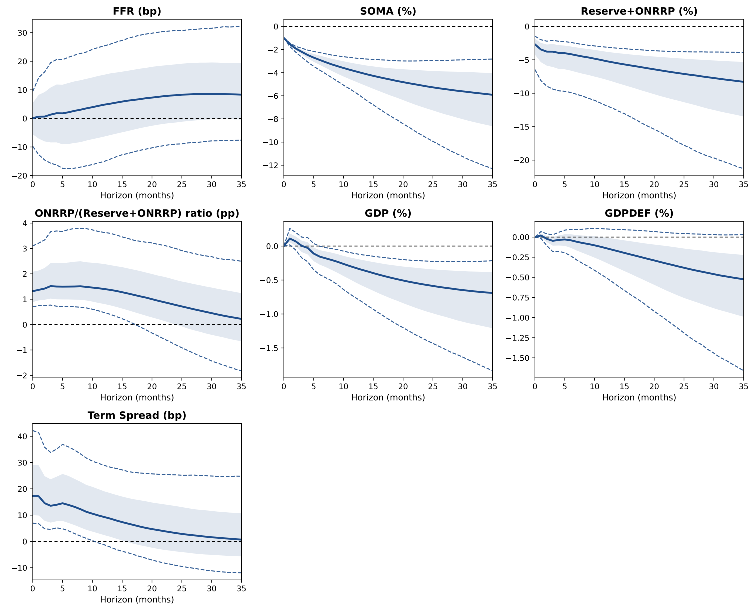
(a) RFBS



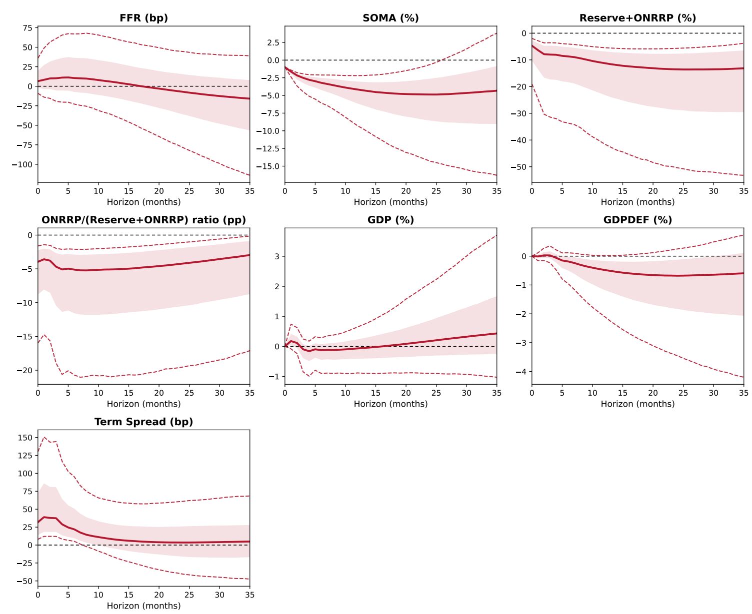
(b) OFBS

Figure G.9: Robustness with monthly outcome variables.

Note. Panel (a) reports the RFBS shock and panel (b) the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings. Each panel shows its shock in the contractionary direction, and panel (b) is the direction of the QT2 drain. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. The log of industrial production and the log of the CPI replace the interpolated GDP and GDP deflator. The variance regimes and the sign, zero, and narrative restrictions are unchanged from the baseline.



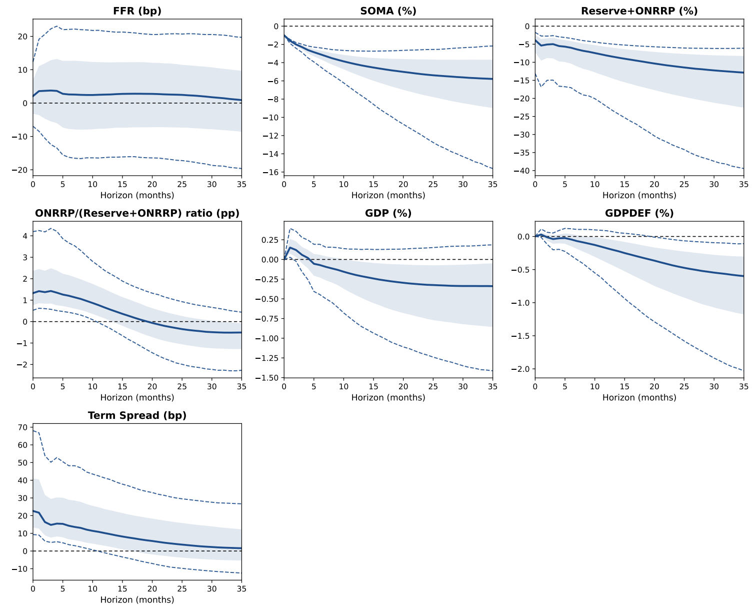
(a) RFBS



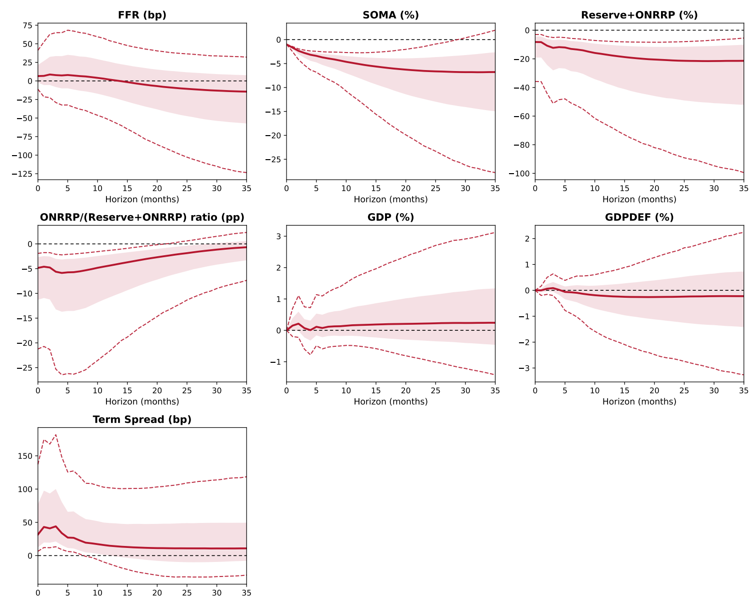
(b) OFBS

Figure G.10: Robustness with the TGA as an exogenous regressor.

Note. Panel (a) reports the RFBS shock and panel (b) the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings. Each panel shows its shock in the contractionary direction, and panel (b) is the direction of the QT2 drain. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. The ratio of the Treasury General Account to the sum of reserves, ON RRP, and the TGA enters every equation at lags zero to two, with unshrunk coefficients. The variance regimes and the sign, zero, and narrative restrictions are unchanged from the baseline.



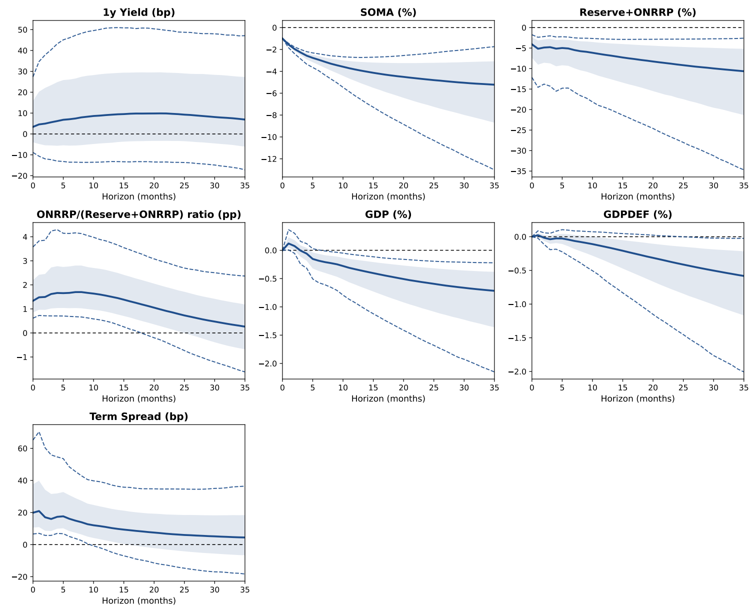
(a) RFBS



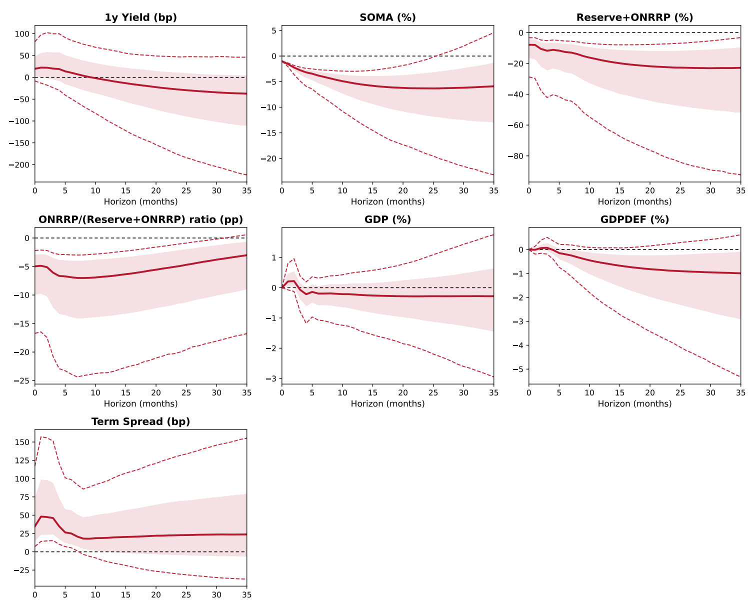
(b) OFBS

Figure G.11: Robustness with Treasury issuance as exogenous regressors.

Note. Panel (a) reports the RFBS shock and panel (b) the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings. Each panel shows its shock in the contractionary direction, and panel (b) is the direction of the QT2 drain. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. The bill share of marketable debt and the log change of coupon securities outstanding enter every equation at lags zero to two, with unshrunk coefficients. The variance regimes and the sign, zero, and narrative restrictions are unchanged from the baseline.



(a) RFBS



(b) OFBS

Figure G.12: Robustness with the one-year yield as the policy rate.

Note. Panel (a) reports the RFBS shock and panel (b) the OFBS shock, each normalized to a one-percent impact decline in SOMA holdings. Each panel shows its shock in the contractionary direction, and panel (b) is the direction of the QT2 drain. Solid lines are posterior medians, shaded regions are 68% credible bands, and dashed lines are 90% credible bands over the retained draws. The one-year Treasury yield replaces the federal funds rate as the first variable. The variance regimes and the sign, zero, and narrative restrictions are unchanged from the baseline.

H Historical Decompositions for All Shocks

This appendix expands Figure 8. Tables H.1 and H.2 report the contributions of all seven shocks to the unexpected changes in GDP and in the GDP deflator over the six windows of the figure. Table H.3 reports the contributions of the two balance-sheet shocks to the unexpected changes in the two liability variables over the same windows. All three tables use the measure of Figure 8, the multi-horizon interval historical decomposition of Antolín-Díaz and Rubio-Ramírez (2018), and report posterior medians with 68% credible intervals over the retained draws.

Table H.1: Historical decomposition by episode, all shocks: GDP.

	Demand	Supply	MP	RFBS	OFBS	Unlabeled 1	Unlabeled 2
QE	+0.21	+0.64	+0.70	+3.96	+0.07	−0.94	−0.94
	[−0.46, +0.99]	[−0.74, +2.07]	[−0.17, +1.92]	[+2.09, +6.44]	[−0.26, +0.42]	[−2.79, +0.25]	[−2.77, +0.25]
QT1	−0.03	+0.43	+0.12	−0.81	+0.18	+0.62	+0.64
	[−0.48, +0.60]	[−0.02, +1.08]	[−0.12, +0.40]	[−1.47, −0.33]	[−0.19, +0.61]	[+0.03, +1.58]	[+0.04, +1.56]
COVID QE	−0.49	−0.50	−0.41	+0.88	−0.02	−1.50	−1.51
	[−1.03, −0.04]	[−1.15, +0.00]	[−1.12, +0.17]	[+0.28, +1.52]	[−0.25, +0.18]	[−3.03, −0.34]	[−2.98, −0.34]
ON RRP buildup	+0.45	−0.73	+1.42	+0.19	−0.28	+0.04	+0.06
	[+0.01, +1.04]	[−1.36, −0.15]	[+0.65, +2.28]	[−0.05, +0.43]	[−0.87, +0.30]	[−0.55, +0.83]	[−0.53, +0.85]
Early QT2	+0.70	+0.19	−0.45	+0.38	+0.00	+0.23	+0.23
	[−0.00, +1.66]	[−0.31, +0.94]	[−0.90, −0.13]	[−0.05, +1.06]	[−0.13, +0.22]	[−0.10, +0.93]	[−0.10, +0.93]
QT2 drain	+0.06	+0.03	+0.01	+0.12	+0.61	−0.07	−0.07
	[−0.65, +0.65]	[−0.77, +0.92]	[−0.46, +0.53]	[−0.67, +0.95]	[−0.51, +1.91]	[−0.88, +0.61]	[−0.88, +0.59]

Notes: Contribution of each shock to the accumulated forecast error of GDP over the window, in percent of the variable's level relative to its within-window forecast. The first row of each pair is the posterior median and the second is the 68% credible interval, both over the retained draws. The two unlabeled shocks carry no identifying restriction. Windows are QE 2009:M5–2014:M11, QT1 2017:M10–2019:M8, COVID QE 2020:M3–2021:M2, ON RRP buildup 2021:M3–2022:M3, Early QT2 2022:M6–2023:M5, and QT2 drain 2023:M6–2025:M12. The QE window opens at 2009:M5, the first estimable month of the episode under the six estimation lags.

Table H.2: Historical decomposition by episode, all shocks: the GDP deflator.

	Demand	Supply	MP	RFBS	OFBS	Unlabeled 1	Unlabeled 2
QE	+0.31	−0.97	+0.45	+3.53	+0.04	−0.46	−0.44
	[−0.67, +1.58]	[−2.80, +0.76]	[−0.59, +2.29]	[+1.50, +6.40]	[−0.40, +0.50]	[−2.25, +0.73]	[−2.14, +0.72]
QT1	+0.04	−0.41	−0.12	−0.38	−0.55	+0.07	+0.06
	[−0.40, +0.59]	[−1.31, +0.32]	[−0.59, +0.29]	[−0.89, −0.01]	[−1.12, −0.08]	[−0.47, +0.60]	[−0.49, +0.59]
COVID QE	−0.32	+0.54	−0.45	+0.46	+0.26	−0.46	−0.45
	[−0.86, +0.10]	[−0.06, +1.19]	[−1.08, +0.00]	[−0.09, +1.07]	[+0.02, +0.60]	[−1.68, +0.47]	[−1.63, +0.48]
ON RRP buildup	+0.40	+0.60	+0.82	+0.03	+0.85	+0.13	+0.13
	[−0.01, +1.21]	[−0.05, +1.52]	[+0.26, +1.71]	[−0.07, +0.15]	[+0.22, +1.46]	[−0.09, +0.69]	[−0.08, +0.69]
Early QT2	+0.66	−0.33	−0.82	+0.09	+0.09	−0.01	−0.01
	[+0.04, +1.56]	[−1.68, +0.80]	[−1.76, −0.25]	[−0.05, +0.34]	[−0.06, +0.33]	[−0.58, +0.35]	[−0.57, +0.34]
QT2 drain	+0.00	−0.02	+0.38	+0.03	−1.01	+0.02	+0.03
	[−0.80, +0.83]	[−1.75, +1.57]	[−0.31, +1.82]	[−0.41, +0.56]	[−2.73, +0.33]	[−0.69, +0.70]	[−0.65, +0.70]

Notes: Contribution of each shock to the accumulated forecast error of the GDP deflator over the window, in percent of the variable's level relative to its within-window forecast. The first row of each pair is the posterior median and the second is the 68% credible interval, both over the retained draws. The two unlabeled shocks carry no identifying restriction. Windows are QE 2009:M5–2014:M11, QT1 2017:M10–2019:M8, COVID QE 2020:M3–2021:M2, ON RRP buildup 2021:M3–2022:M3, Early QT2 2022:M6–2023:M5, and QT2 drain 2023:M6–2025:M12. The QE window opens at 2009:M5, the first estimable month of the episode under the six estimation lags.

Table H.3: Historical decomposition by episode, the two liabilities.

	log(Reserves + ON RRP)		ON RRP ratio	
	RFBS	OFBS	RFBS	OFBS
QE	+61.13	+0.18	−0.56	−0.19
	[+33.32, +95.60]	[−9.15, +9.67]	[−5.49, +4.32]	[−2.54, +2.49]
QT1	−11.14	−10.10	+2.74	−4.79
	[−20.54, −4.43]	[−17.12, −4.72]	[+0.72, +5.30]	[−7.84, −2.15]
COVID QE	+27.91	+6.53	−6.12	+3.74
	[+20.01, +37.29]	[+2.22, +12.07]	[−9.26, −3.57]	[+1.20, +6.61]
ON RRP buildup	−1.90	+25.26	+0.60	+13.31
	[−9.68, +6.06]	[+17.96, +33.81]	[−1.07, +2.52]	[+10.14, +16.79]
Early QT2	+3.94	+6.31	−1.98	+5.18
	[+0.02, +10.26]	[+1.19, +11.86]	[−4.36, −0.10]	[+0.83, +9.42]
QT2 drain	+1.35	−24.78	−0.07	−11.17
	[−6.62, +9.22]	[−43.88, −10.60]	[−2.06, +1.75]	[−19.90, −4.72]

Notes: Contribution of each balance-sheet shock to the accumulated forecast error of the liability variable over the window. Size entries are in percent of the level of log(Reserves + ON RRP) and ratio entries are in percentage points of the ON RRP ratio, both relative to the within-window forecast. The first row of each pair is the posterior median and the second is the 68% credible interval, both over the retained draws. Windows are those of Tables H.1 and H.2.